



WHO INFLUENZA CENTRE LONDON

**Report prepared for the WHO annual
consultation on the composition of influenza
vaccine for the Southern Hemisphere**

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Southern Hemisphere Vaccine Recommendation Meeting for the 2012 season.

Summary of NIMR Results

Final Report for WHO meeting 26-28 September 2011

Influenza Activity, February 2011- September 2011.

Influenza activity in Europe peaked in January and February 2011 and had declined by the end of March. In many countries after a peak in influenza A activity influenza B activity followed, for example in Germany, Latvia, Romania and Georgia. Since then influenza activity has been low (**Figure 1** and **Figure 2, pages 9 & 10**). The influenza season was timed similarly to those seen prior to the emergence of the pandemic A(H1N1) virus in 2009 (**Figure 3, page 11**). Peak RSV activity preceded influenza activity and was at higher levels than seen prior to the influenza pandemic of 2009 (**Figure 4, page 12**), whether this is due to increased sampling and capacity is not known.

Cumulative influenza virus typing and subtyping showed that influenza A(H1N1) predominated with influenza B viruses representing fewer than 30% of the total viruses identified while A(H3N2) viruses represented only 2% of identified viruses. Despite both being in the minority, influenza B and A(H3N2) activity increased relative to A(H1N1) activity over the previous season. There were no reports of former A(H1N1) seasonal influenza viruses indicating that any erroneous reporting during the 2009-10 season had been eliminated (**Figure 1** and **Figure 2, pages 9 & 10**).

Antigenic and genetic characteristics of influenza viruses collected between February and August 2011

A summary of the clinical samples and virus isolates collected between 2011-02-01 and 2011-08-31 received by the Collaborating Centre is shown in **Table 1, page 13**. Viruses have been received from Western, Central and Eastern Europe; the Middle East; North, Central and South Africa; Asia and South America. Batches of specimens for which work is ongoing is described as in progress. In cases where a large number of clinical samples were received a proportion was chosen for propagation based on real-time PCR results (Ct values).

The recovery efficiency from clinical samples and virus isolates is shown in **Table 2, page 14**. Recovery of viruses from clinical samples varied between 40% (H1N1 viruses) and 74% (B/Victoria-lineage viruses); recoveries from virus isolates were around 80 to 90% irrespective of type/subtype. **Table 3, page 15** summarises the numbers of viruses of the A(H1N1) and A(H3N2) subtypes, and the two influenza B-lineages that showed ≥ 8 -fold reductions in HI titres compared to the titres with homologous reference/vaccine viruses.

Influenza A(H1N1) Virus analysis.

Table 4, page 16, summarises the antigenic results for the H1N1 viruses received. All H1N1 viruses were of the 2009 pandemic lineage. No former seasonal influenza viruses were received by the Collaborating Centre and no reports of former-seasonal H1N1 viruses were made to WHO Euro.

The majority of H1N1 viruses were antigenically similar to the vaccine virus (**Table 4, page 16**) with 13% of viruses showing a >4-fold reduction in HI titre compared with the titre of the vaccine virus (A/California/7/2009) with its homologous post-infection ferret antiserum. The results are shown in **Tables 5 to 10, pages 17 to 22**. The majority of viruses that showed low reactivity in HI assays were chosen for sequence analysis of the HA and NA genes along with other representative viruses. Viruses for which HA and NA sequence has been included for phylogenetic analysis are highlighted in green in **Tables 5 to 10, pages 17-22**.

Figure 5, page 23 shows a phylogenetic tree for the HA gene of selected H1N1 viruses. The polymorphisms and substitutions K154X, G155X/E and N156S/D are marked on the tree. No viruses with sequence polymorphisms at 153 and 157 were included in the analysis. The sequence changes D222G/N/E and the polymorphism D222X are also marked, as is the substitution Q223R associated with the isolation and passage of virus in hens' eggs. Single amino acid substitutions or amino acid polymorphisms between residues 154 to 156 continue to be observed, but not exclusively associated with the propagation of virus in MDCK-SIAT cells, and occur in all genetic groups. Viruses carrying these substitutions frequently showed reduced titres in HI assays with several sera but reductions with the antiserum raised against the vaccine virus, A/California/7/2009, were frequently ≤ 4 -fold compared with the homologous titre.

HA genes of recent H1N1 viruses clustered into seven main genetic groups that can be defined by encoded amino acid substitutions:

- (2) **N31D, S162N** (resulting in the gain of a glycosylation site) & **A186T**, e.g. A/Czech Republic/32/2011;
- (3) **A134T & S183P**, e.g. A/Hong Kong/3934/2011;
- (4) **N125D**, e.g. A/Christchurch/16/2010;
- (5) **D97N, R205K, I216V & V249L**, e.g. A/Astrakhan/1/2011;
- (6) **D97N & S185T**, e.g. A/St Petersburg/27/2011;
- (7) **D97N, S143G, S185T & A197T**, e.g. A/St Petersburg/100/2011;
- (8) **A186T & V272A**, e.g. A/Ghana/601/2011.

Viruses collected in June and July 2011 fall into groups 6 and 7.

Viruses showing low reactivity (> 4-fold reductions) fall into each genetic group. In genetic group 3, defined by the HA1 amino acid substitutions A134T and S183P and represented by five viruses here, three, which also carry A141S and I295V in HA1 and V152A in HA2, show reduced reactivity with antiserum raised against the vaccine virus. Two of the three viruses also carry polymorphism at 155 of HA1. This group is not new and does not have a strong presence among viruses circulating recently, but previously it had not been identified as being antigenically distinct from the vaccine virus.

Viruses that show low reactivity with the antiserum raised against A/California/7/2009 and not shown on the phylogenetic tree fall into genetic group 7 (N=2), group 5 (N=1) and one aligns with A/Stockholm/13/2011 lying in an undefined minor genetic group.

Amino acid alignments of HA for a selected group of viruses is shown in **Figure 6, page 24**: the vaccine virus is shown in red and glycosylation motifs are highlighted in yellow.

Figure 7, page 27, illustrates the location of amino acid residues on the structure of the H1 HA. Single amino acid substitutions are located in the vicinity of the receptor binding site in genetic groups 2, 3, 4 and 5. Notably an additional substitution in genetic group 3, A141S, which was observed in 3 of the 5 viruses in this group, is located adjacent to amino acid residue 134 of the HA and is not shown in **Figure 7, page 27**.

Figure 8, page 28, shows a phylogenetic tree for the NA gene of selected H1N1 viruses and **Figure 9, page 29**, shows an amino acid alignment for a selected group of viruses. A large genetic clade of NA genes is defined by the amino acid substitutions V241I and N369K: this group included viruses from HA genetic groups 5, 6 and 7. Viruses in HA genetic group 3 cluster into a group defined by the substitutions Q313R and V394I in the NA, with additional substitutions of R220K and I389K or I46T and I467V.

Antiviral analysis. From a total of nearly 600 H1N1 viruses tested since 2011-02-01, two were resistant to oseltamivir but sensitive to zanamivir; these two viruses were collected prior to 2011-02-01. None of the close to 100 viruses collected since 2011-02-01 were resistant to either inhibitor (**Table 30, page 73**).

Of the 18 M gene sequences analysed, all retain the amino acid substitution S31N in the M2 ion channel protein indicative of resistance to adamantanes (**Table 31, page 74**).

Conclusion: The majority of viruses have remained antigenically similar to the vaccine virus. Viruses that showed low reactivity with antisera raised against the vaccine virus fell into each genetic group of the HA gene. Some viruses with substitutions or polymorphisms between residues 154 and 156 of the HA show a reduced reactivity with antiserum raised against A/California/7/2009. Viruses that carry HA genes of genetic group 3, albeit represented in small numbers, show evidence of antigenic divergence from the vaccine virus although this group also contains viruses with polymorphism at residue 155 of the HA. This divergence was not observed in the analyses carried out for the previous influenza vaccine selection meeting.

Influenza A(H3N2) Virus analysis.

Influenza A(H3N2) viruses were received from Europe, Africa, Hong Kong and Argentina. These viruses have continued to be difficult to characterise antigenically by HI assay due to variable agglutination of red blood cells from guinea pig, turkey and humans. Those viruses with sufficient titre in HA assays using guinea pig red blood cells in the presence of 20nM oseltamivir, to circumvent the NA mediated binding of the H3N2 viruses to red blood cells, were analysed by HI assay. Such viruses represented in the order of 65% of the H3N2 samples received. A summary is presented in **Table 11, page 32**, and the results of the HI assays are shown in **Tables 12 to 20, pages 33 to 38**. Viruses for which HA and NA gene sequences have been determined and included in phylogenetic analysis are highlighted in green. The results of antigenic analysis as assayed by plaque reduction assay (virus neutralisation) is shown in **Table 21, page 39**.

Approximately one third (33%) of the viruses showed 8-fold or lower reactivity in the HI assay with antiserum raised against the vaccine virus A/Perth/16/2009 (**Table 11, page 32**). The majority of viruses that showed reduced activity with this antiserum showed improved titres with some of the other antisera, relative to that with the homologous viruses against which the antisera were raised: notably with antisera raised against A/Wisconsin/15/2009, A/Alabama/5/2010 and A/Perth/10/2010 - representatives of both the genetic clades of H3N2 viruses currently circulating. Generally there was low reactivity, compared with the reactivity of the homologous virus, with antisera raised against A/Victoria/208/2009 and A/Victoria/210/2009, two viruses that often show high homologous titres in HI assays carried out at the London CC.

To supplement the HI assays, virus neutralisation assays were carried out on a set of viruses that were able to be stably propagated and grew to sufficient titre (**Table 21, page 39**). These viruses were analysed with a small panel of antisera and were neutralised efficiently by the antiserum raised against the vaccine virus A/Perth/16/2009. Three of the six viruses had shown low reactivity in HI assays - A/Stockholm/16/2011, A/Stockholm/17/2011 and A/Ireland/M27357/2011 but only A/Stockholm/16/2011 showed slightly reduced activity in the virus neutralisation test when using the ferret antiserum against the vaccine virus while showing reduced neutralisation with antisera raised against A/Perth/10/2010 and A/Alabama/5/2010.

Phylogenetic analysis of the HA1 coding-region of the HA genes of H3N2 viruses is shown in **Figure 10, page 40**. Viruses continue to fall into both the Perth/16 and the Victoria/208 clades but with the majority falling into the Victoria/208 genetic clade. Amino acid alignments of a selected group of viruses is shown in **Figure 11, page 41**, with the vaccine virus shown in red and glycosylation motifs highlighted in yellow.

Six genetic groups for the HA gene can be seen in circulating H3N2 viruses. Within the A/Perth/16/2009 genetic clade there are two genetic groups defined by amino acid substitutions:

- (1) **P162S, I260M & R261Q**, e.g. A/Victoria/210/2009 with some viruses also carrying the substitutions **E50K**, e.g. A/Norway/1186/2011, or **N81D**, e.g. A/Niigata/510/2011;

- (2) **N133D** (resulting in the loss of a glycosylation site), **R142G**, **T212A** & **V213A**, this group is not represented in our recently collected viruses and is not shown in **Figure 11, page X**.

Four genetic groups, in two recognizable clusters, can be defined in the A/Victoria/208/2009 genetic clade of H3N2 viruses by amino acid substitutions additional to those that distinguish viruses of the A/Victoria/208/2009 clade from those in the A/Perth/16/2009 clade (**K62E**, **K144N** & **T212A**).

- (3) **V223I** often with **N145S**, e.g. A/Johannesburg/73/2011; this group can be divided into two subgroups of viruses, with one sub-group having the substitution **N144D** (resulting in the loss of a glycosylation site), e.g. A/Stockholm/18/2011, the other sub-group having **A198S** often with **S45N** (resulting in the gain of a glycosylation site), **T48I** and **N312S**, e.g. A/Hong Kong/3969/2011;
- (4) **N312S**, e.g. A/Brisbane/11/2010, with many viruses also carrying **T48A** & **K92R**, e.g. A/Rhode Island/01/2010 and A/Massachusetts/08/2011;
- (5) **D53N**, **Y94H**, **I230V** & **E280A**, e.g. the reference virus A/Perth/10/2010, with some viruses also carrying the substitution **I192T**, e.g. A/Athens/108/2011;
- (6) **D53N**, **Y94H**, **I230V** & **E280A**, as in genetic group 5 but with **S199A**, e.g. the reference virus A/Iowa/19/2010, and some with **S45N** (resulting in the gain of a glycosylation site), e.g. A/Hong Kong/3951/2011;

A further cluster of viruses, defined by **S45N** (resulting in the gain of a glycosylation site), is represented particularly by recently circulating South African viruses, e.g. A/Johannesburg/114/2011; some viruses also carry **L3F** & **L59I** substitutions, e.g. A/Norway/685/2011; this cluster has not yet been defined as a distinct genetic group.

Recent viruses fell into each genetic group; those that showed low reactivity in the HI tables distributed into:

- group 1 with the substitution **E50K**,
- group 3 with the loss of glycosylation at **144**, or the substitution **N145S**
- group 5
- group 6
- the cluster defined by the substitution **S45N**, above.

The virus neutralisation assay had examples of viruses from genetic groups 1, 3 (N=3), 5 and 6 (N=2). Viruses from each group reacted well with the antisera raised against reference viruses in genetic groups 1 and 5.

Figure 12, page 44, shows the location on the H3 HA structure of amino acid substitutions that define the genetic groups observed among recent H3N2 viruses. Many of the substitutions that define genetic groups are located away from the HA receptor binding site. Substitutions are seen in the 140-loop, others are located in the lower side of the HA head in the vestigial esterase domain and the hinge region of the HA.

Figure 13, page 45, shows a NA gene phylogenetic tree of selected viruses with the HA clades and genetic groups indicated. **Figure 14, page 46** shows an amino acid alignment for a selected group of these viruses. The NA genes remain in two genetic clades and the main subgroups within the Victoria/208 clade are distinguishable. Viruses in HA group 3 have alternative NA genes corresponding to the two subgroups

described above and based on NA substitutions of **N144D** or **A198S**. Viruses in the HA gene cluster defined by **S45N**, above, also carry NA genes that cluster together. Only limited phylogenetic signal is seen in the NA genes corresponding to HA genes in genetic groups 5 and 6.

Antiviral analysis. Of the 75 H3N2 viruses tested since 2011-02-01, including 39 viruses collected after 2011-02-01, all were sensitive to oseltamivir and zanamivir (**Table 30, page 73**).

Of the 11 M gene sequences analysed, all retain the amino acid substitution S31N in the M2 ion channel protein indicative of resistance to adamantanes (**Table 31, page 74**).

Conclusion. Antigenic analysis by HI carried out in the presence of oseltamivir showed that one third of viruses had reduced titre with the antiserum raised against the vaccine virus. Virus neutralisation assays of a small number of viruses, that included viruses of different genetic groups some of which had low reactivity in HI assays, showed good reactivity with the antiserum raised against the vaccine virus. There is no evidence from our results that viruses showing low reactivity with antisera raised against the vaccine virus, or viruses genetically similar to the vaccine virus, predominate in any of the emerging genetic groups.

Influenza B viruses.

Influenza viruses of the B/Victoria-lineage predominated over those of the B/Yamagata-lineage in most regions of the world from which we have received influenza B viruses (**Table 1, page 13**).

B/Victoria-lineage.

Influenza B/Victoria-lineage viruses were received from Europe, Africa and Hong Kong. **Table 22, page 49**, lists the centres from which influenza B/Victoria-lineage viruses were received and summarises the antigenic results for these viruses.

The results of the HI assays are presented in **Tables 23 to 27, pages 50 to 54**. Viruses shown in the phylogenetic trees are highlighted in green. Cell-propagated viruses generally displayed low HI titres using antisera raised against egg-grown viruses as expected. The vast majority reacted well with antisera raised against cell-grown viruses belonging to the Brisbane/60/2008 genetic clade (B/Paris/1762/2010, B/Hong Kong/514/2009, B/Odessa/3886/2010). Some of the test viruses propagated in cells showed raised titres with the antiserum raised against the egg-propagated vaccine virus B/Brisbane/60/2008. Two test viruses, from a total of 124 tested, showed unusually low reactivity: one from Sweden (A/Malmö/2/2011 in **Table 25, page 52**) and one from Ghana (A/Ghana/850/2011 in **Table 26, page 53**). B/Cambodia/30/2011- received from WHO CC Melbourne as a serology antigen - also showed reduced titres in our HI assays (**Table 27, page 54**).

Phylogenetic analysis of the HA1 coding-region of the HA gene of B/Victoria-lineage viruses is shown in **Figure 15, page 55**, and a line-up of selected amino acid sequences is presented in **Figure 16, page 56**. All recently collected viruses clustered into the B/Brisbane/60/2008 clade including A/Malmö/2/2011, which showed low HI reactivity and carried the amino acid substitution **K165E** in HA1, and B/Ghana/850/2011, which carried substitutions **T121N**, **I137V** and **K257R** - residue **137** lying on the ridge of the receptor binding site.

Figure 17, page 59, shows a phylogenetic analysis of the NA gene of B/Victoria-lineage viruses and a line-up of selected amino acid sequences is shown in **Figure 18, page 60**. The majority of viruses cluster similarly to the clustering observed in the HA but a small group of viruses from China can be seen to carry an NA gene distinct from the origin of the HA gene, this group is highlighted in **Figure 17, page 59**.

Antiviral analysis. From a total of 373 influenza B viruses tested since 2011-02-01, of which 110 were collected after 2011-02-01, all but three viruses were sensitive to zanamivir and oseltamivir (**Table 30, page 73**). The three B/Victoria-lineage viruses resistant to oseltamivir and zanamivir were collected from a single French patient over the course of their infection (B/Lyon-chu/15216/2011, B/Lyon-chu/16167/2011 and B/Lyon-chu/16432/2011) and carried the NA substitution I221L - **Figures 15 & 17, pages 55 & 59**.

B/Yamagata-lineage.

Influenza B/Yamagata-lineage viruses were received from Northern Europe (Estonia, Latvia, Sweden and Norway), South-Eastern Europe (Turkey and Greece) and Hong Kong. **Table 28, page 63**, lists the centres from which influenza B/Yamagata-lineage viruses were received and summarises the antigenic results for these viruses.

The antigenic results are shown in **Table 29, page 64**. All 20 test viruses reacted well with antisera raised against the reference viruses B/Bangladesh/3333/2007 and B/Wisconsin/1/2010; none showed greater than a 4-fold reduction in HI titre with these antisera compared with the titres observed with the homologous reference viruses.

Phylogenetic analysis of the HA1 coding-region of the HA gene of B/Yamagata-lineage viruses is shown in **Figure 19, page 65**, and a line-up of selected amino acid sequences is presented in **Figure 20, page 66**. Most of the recently collected viruses fell into the B/Bangladesh/3333/2007 genetic clade with four exceptions. The viruses analysed genetically from Estonia and one of the viruses from Sweden fell into a distinct genetic clade along with previously recognised viruses from Finland collected at the end of 2010, and a virus collected in June 2011 from Hong Kong.

Phylogenetic analysis of the NA gene of B/Yamagata-lineage viruses is shown in **Figure 21, page 69**, and a line-up of selected amino acid sequences is presented in **Figure 22, page 70**. In the NA gene, like the HA gene, the recently collected viruses from Estonia, Sweden and Hong Kong cluster together outside the B/Bangladesh/3333/2007 genetic clade.

Conclusion: Influenza B viruses of the B/Victoria-lineage have predominated and the vast majority of these viruses were antigenically similar to influenza B viruses closely related to the B/Brisbane/60/2008 vaccine virus but propagated in cells. For the B/Yamagata-lineage, viruses were antigenically and genetically similar to B/Bangladesh/3333/2007 and B/Wisconsin/1/2010.

Antiviral Resistance

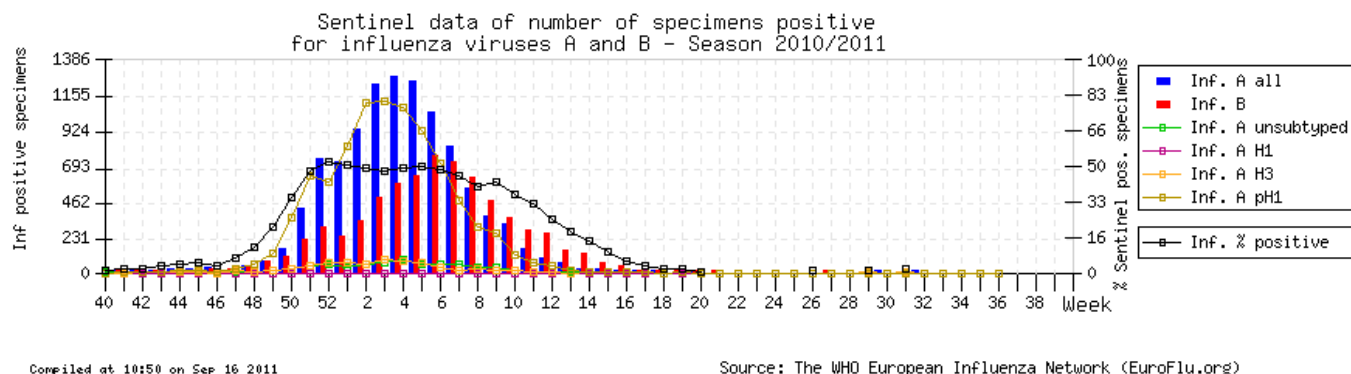
Results of antiviral susceptibility testing are shown in **Table 30, page 73** which summarises both for viruses collected from 2011-02-01 and for viruses received from 2011-02-01 but with earlier collection dates. Only five viruses showed resistance in MUNANA-based fluorescence assays. Two H1N1 viruses from Parma, Italy, both collected in January 2011, showed resistance to oseltamivir and three B/Victoria-lineage viruses from France showed resistance to oseltamivir and zanamivir. These influenza B/Victoria-lineage viruses are under further investigation.

Antiviral resistance studies from the ECDC for the 2010-11 influenza season up to week 35/2011 showed that 117 of 2592 H1N1 viruses (4.5%) were resistant to oseltamivir but none to zanamivir; no H3N2 viruses were resistant to either drug out of 112 tested. There were no influenza B viruses shown to be resistant to either drug by the end of week 35/2011 out of 521 viruses tested - the resistant B/Victoria-lineage viruses from France not being included in the database at the time of this update.

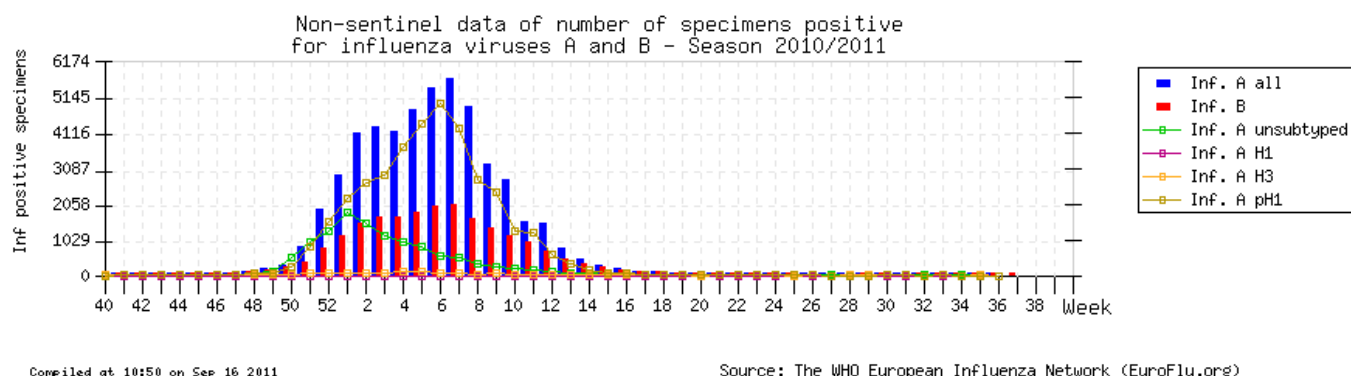
Influenza A M-gene sequencing has shown H1N1 and H3N2 viruses to retain resistance to adamantanes due to S31N substitution in the M2 protein (**Table 31, page 74**). The latter result is supported by ECDC figures for 263 H1N1 and 71 H3N2 viruses analysed.

Figure 1: Current European Situation (EuroFlu.org: 16.09.2011)

Sentinel Data



Non-sentinel Data



Virus typing/subtyping.

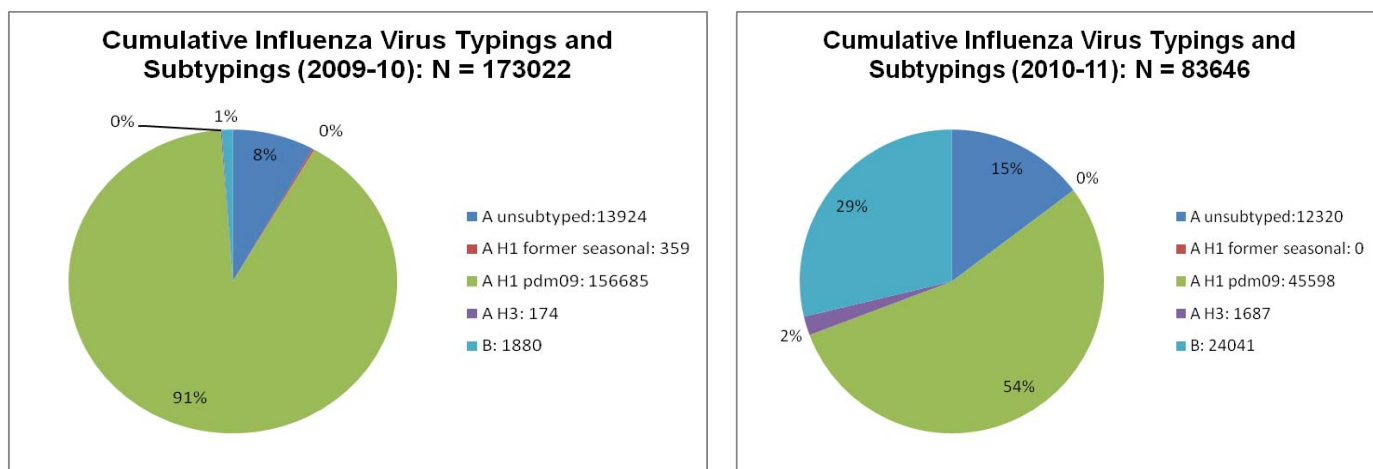
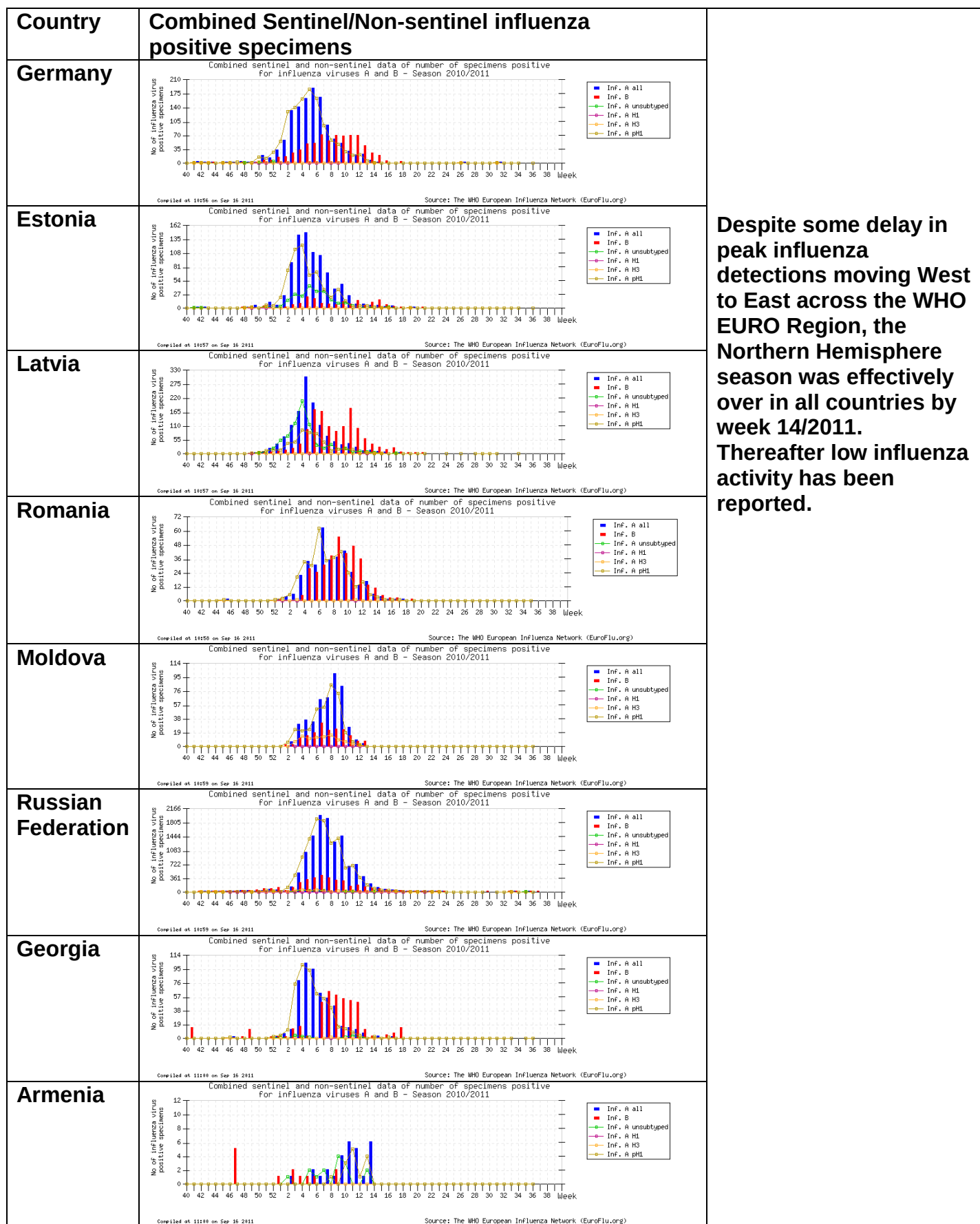


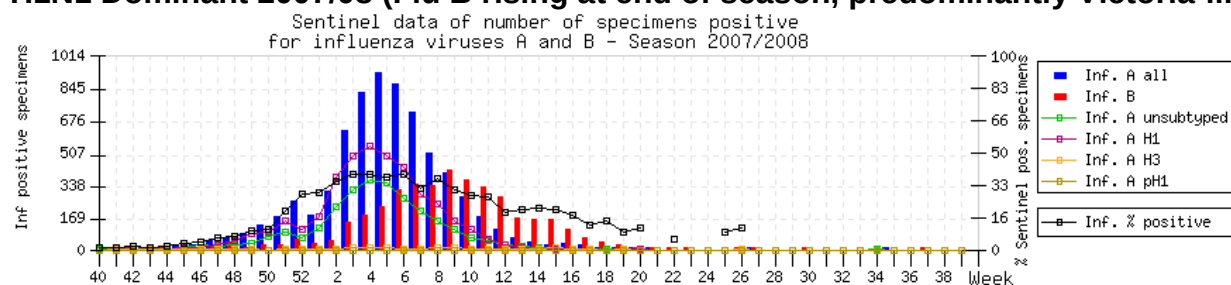
Figure 2: Eastern WHO Euro Country situations – EuroFlu.org (week 36/2011)



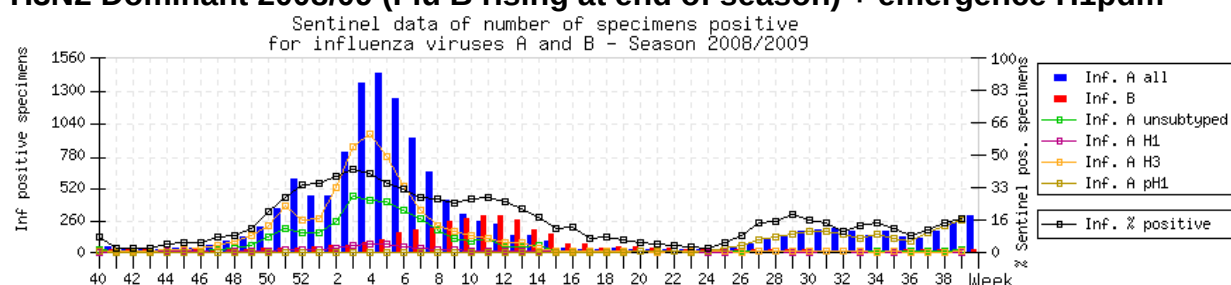
Despite some delay in peak influenza detections moving West to East across the WHO EURO Region, the Northern Hemisphere season was effectively over in all countries by week 14/2011. Thereafter low influenza activity has been reported.

**Figure 3: Influenza Season 'timing' by year (EuroFlu.org: 16.09.2011)
Sentinel Data**

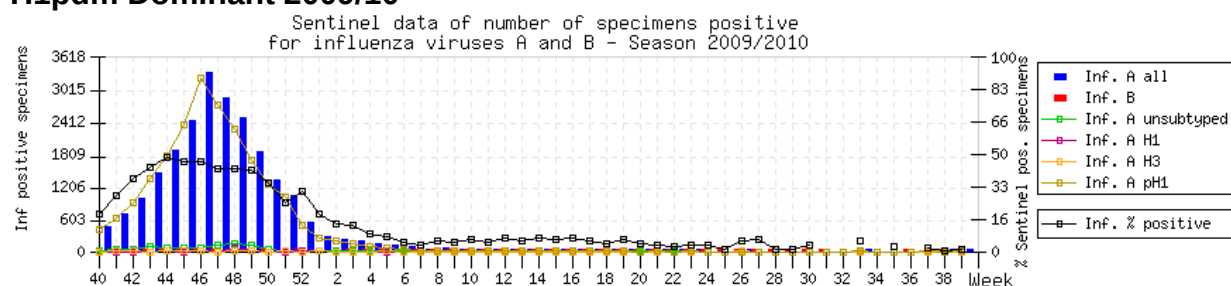
H1N1 Dominant 2007/08 (Flu B rising at end of season, predominantly Victoria-lineage)



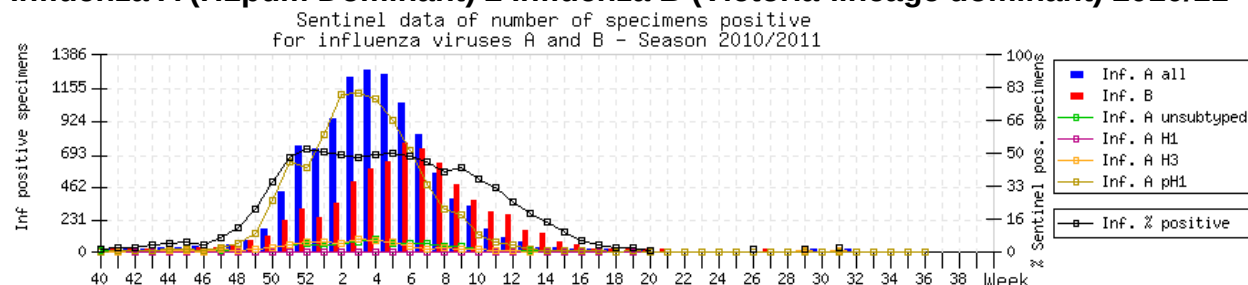
H3N2 Dominant 2008/09 (Flu B rising at end of season) + emergence H1pdm



H1pdm Dominant 2009/10



Influenza A (H1pdm Dominant) ≥ Influenza B (Victoria-lineage dominant) 2010/11

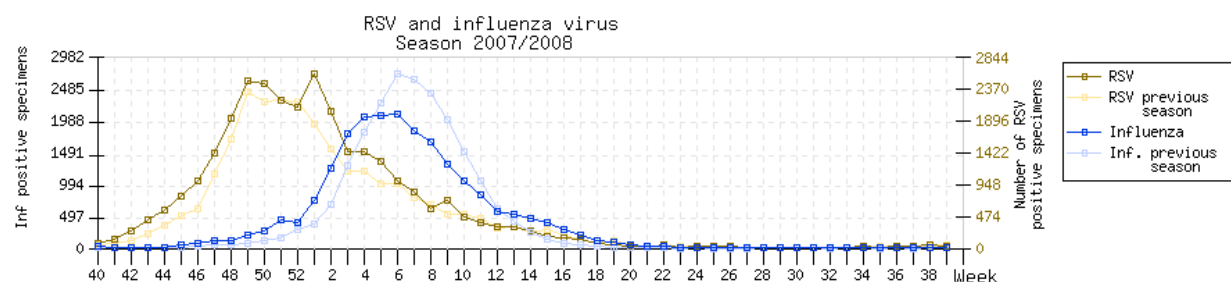


In 2010/11, numbers of positive specimens have decreased to levels seen in seasons prior to the 2009/10 H1pdm season.

The timing of the season has also returned to the timeframes seen prior to the 2009/10 H1pdm season.

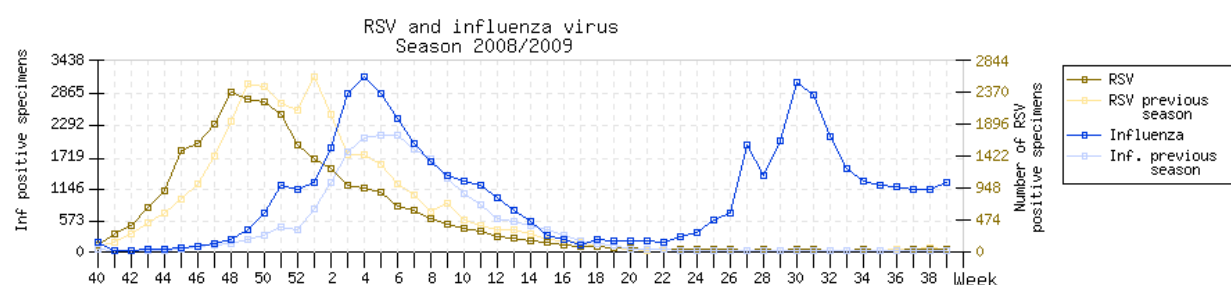
Figure 4: RSV and influenza – activity timings (EuroFlu.org: 16.09.2011)

Combined sentinel and non-sentinel data for numbers of positive specimens.



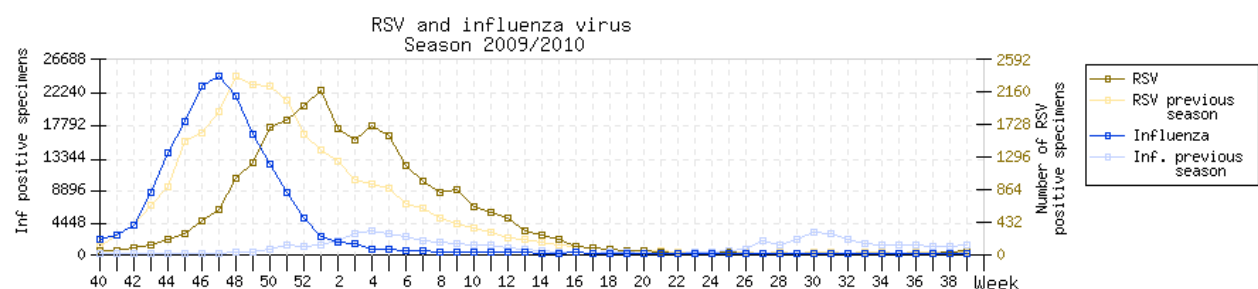
Compiled at 11:09 on Sep 16 2011

Source: The WHO European Influenza Network (EuroFlu.org)



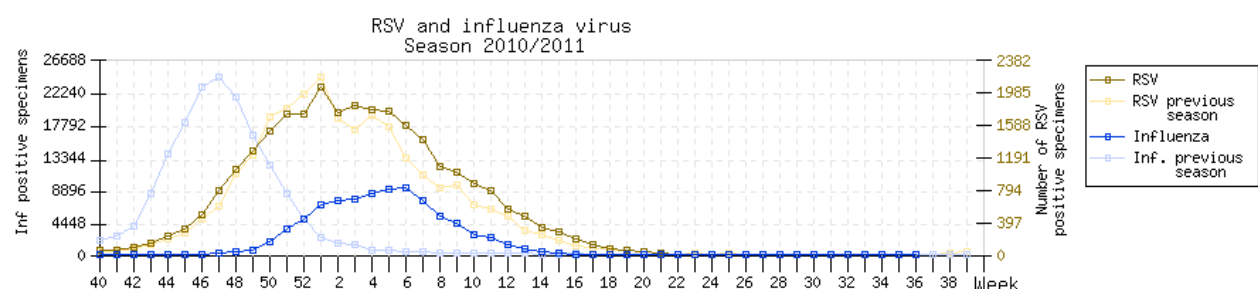
Compiled at 11:09 on Sep 16 2011

Source: The WHO European Influenza Network (EuroFlu.org)



Compiled at 11:07 on Sep 16 2011

Source: The WHO European Influenza Network (EuroFlu.org)



Compiled at 11:05 on Sep 16 2011

Source: The WHO European Influenza Network (EuroFlu.org)

In 2010/11 the pattern of RSV emerging and peaking before influenza was re-established. The approximate ten-fold increase in RSV detections observed in 2009/10 was maintained in 2010/11.

Table 1. Summary of clinical samples and isolates received, with collection dates since 2011-02-01*

MONTH Country	A Untyped	H1N1pdm		H3N2		B untyped	B Yamagata lineage		B Victoria lineage	
		Number received	Number recovered	Number received	Number recovered		Number received	Number recovered	Number received	Number recovered
FEBRUARY										
Albania		2	0	1	0				2	2
Belarus		5	5						2	in process
Cameroon		4	in process						3	3
Czech Republic		1	1							
Estonia		31	28			1	1	1		
France		11	in process	2	2			in process	5	in process
Ghana		2	1	1	0		3		6	6
Greece	1	9	7							
Hungary									2	in process
Iceland									11	in process
Ireland		2	2	2	2				1	1
Italy		1	1							
Kazakstan	3	4	in process	6	in process	8				
Latvia		1	1				1	1	3	2
Madagascar				10	9				9	8
Malta		5	1							
Moldova		17	6	15	6	22			1	1
Norway		4	3	5	5					
Portugal		6	2							
Russia		8	8	1	0	1			7	7
Senegal		1	in process							
Serbia		60	3	7	6				3	3
Slovakia		1	1						2	2
Slovenia		6	6						6	6
Spain		5	4	2	1				5	5
Sweden		2	2	4	4		1	1	3	3
Switzerland				2	in process					
Turkey		12	9			2	2	2		
Ukraine		20	in process	1	1	18			3	3
MARCH										
Albania		1	0							
Belarus									1	1
Cameroon		13	in process							
Estonia		6	4				5	5	1	1
France		2	in process	1	1		3	in process	10	in process
Ghana		5	1						4	3
Greece	4	5	4			5	1	1	2	2
Hong Kong		1	1							
Hungary										
Iceland		4	in process				1	in process	3	in process
Ireland									9	in process
Kazakstan		1	in process						1	1
Latvia									2	2
Madagascar	1	1	1	3	3				7	7
Malta									5	5
Norway		5	3	3	2		1	0	6	6
Romania				1	1					
Russia		3	3							
Senegal		7	in process							
Serbia		5	0	2	2		3	3	7	7
Slovenia		1	1							
Spain									1	1
Sweden		5	5	1	1		2	2	9	9
Turkey	2	5	0							
Ukraine	1	25	5			6				
APRIL										
Albania		2	0							
Cameroon		2	in process							
France									7	7
Ghana		6	4	4	2				6	5
Hong Kong		1	1	2	2		2	0	2	1
Iceland									2	in process
Ireland				1	1					
La Reunion				1	1					
Madagascar				3	in process				4	in process
Malta									1	1
Senegal		4	in process							
Serbia				1	0				1	1
Slovenia		1	1							
Sweden		2	2	3	3				2	2
MAY										
Albania				1	0					
Argentina		3	3	1	1					
Cameroon									4	in process
France				1	1				1	0
Ghana		3	3	2	1				4	3
Hong Kong				2	2					
La Reunion				4	4					
Madagascar		2	in process	3	in process				5	in process
Senegal		3	in process							
South Africa		5	4	3	3					
Sweden		1	1	2	2				1	1
JUNE										
Argentina		2	2	3	3					
Cameroon		1	in process						10	in process
Ghana				3	3				1	1
Hong Kong		2	0				2	2	2	0
La Reunion				1	1					
Madagascar		7	in process	4	in process				3	2
South Africa		15	in process	7	7				1	in process
United Kingdom							5	in process	2	in process
JULY										
Argentina				4	4					
Cameroon		1	in process						10	in process
Madagascar		8	in process	5	in process				5	in process
Norway				2	2					
Senegal		1	in process							
South Africa		8	in process	1	1					
AUGUST										
Madagascar		4	in process	3	in process				1	in process
Norway				1	1					
Total Received = 862	12	399	140	138	91	63	33	18	217	121
% of total	1%	46%		16%		8%	4%		25%	
* Total receipts since 2011-02-01 were 1940										
	23	894		311		202 (17)	79		414	

(17) - Seventeen specimens were reported as Influenza B + A(H1)/A(H3)

Table 2. Isolation rates for samples collected after 2011-02-01

	Clinical sample			Virus isolate		
	number positive	number negative	percent recovered	number positive	number negative	percent recovered
H1	80	124	40%	300	64	82%
H3	39	36	52%	123	17	88%
B Victoria	20	7	74%	204	12	94%
B Yamagata	4	1	80%	31	8	79%

Table 3. Summary of isolates collected since 2011-02-01 showing ≥ 8 -fold reduction in reactivity with antisera raised against reference viruses

	Number tested	Number showing ≥ 8 -fold reduction	% showing ≥ 8 -fold reduction
H1¹	140	18	13
H3²	91	30	33
B/Victoria³	121	3	3
B/Yamagata⁴	18	0	0

1. compared to homologous titre for ferret antiserum raised against vaccine virus A/California/7/2009

2. compared to homologous titre for ferret antiserum raised against vaccine virus A/Perth/16/2009

3. compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents

(B/Paris/1762/2008, B/Hong Kong/514/2009 and B/Odessa/3886/2010)

4. compared to homologous titre for ferret antiserum raised against prototype virus B/Florida/4/2006

Table 4. Summary of influenza A (H1N1)pdm09 samples received, collected between 01 February 2011 and present

MONTH Country	H1N1			
	Number propagated	≤2 fold*	4 fold*	≥8 fold*
FEBRUARY				
Albania	0			
Belarus	5	4		1
Czech Republic	1	1		
Estonia	28	21	5	2
Ghana	1	1		
Greece	7	7		
Ireland	2		2	
Italy	1	1		
Latvia	1		1	
Malta	1	1		
Moldova	6	5	1	
Norway	3	2	1	
Portugal	2	2		
Russia	8	7	1	
Serbia	3	2	1	
Slovakia	1	1		
Slovenia	6	4		2
Spain	4	4		
Sweden	2		2	
Turkey	9	2	4	3
MARCH				
Albania	0			
Estonia	4	2	2	
Ghana	1	1		
Greece	4		4	
Hong Kong	1			1
Madagascar	1	1		
Norway	3	3		
Russia	3	3		
Serbia	0			
Slovenia	1	1		
Sweden	5	4	1	
Turkey	0			
Ukraine	5	1	2	2
APRIL				
Albania	0			
Ghana	4	3	1	
Hong Kong	1	1		
Slovenia	1	1		
Sweden	2			2
MAY				
Argentina	3			3
Ghana	3	2	1	
South Africa	4	4		
Sweden	1			1
JUNE				
Argentina	2		1	1
Hong Kong	0			
Total	140	92 66%	30 21%	18 13%

* Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

Table 5. Antigenic analyses of pandemic influenza A(H1N1)pdm09 viruses

Viruses	Collection date	Passage History	Haemagglutination inhibition titre						
			Post infection ferret sera						
			A/Cal 7/09 F05/10	A/Eng 195/09 F06/10	A/Auck 3/09 F17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/09 C4/34/09	A/HK 2212/10 F21/10 Egg	A/C'church 16/10 F30/10
REFERENCE VIRUSES									
A/California/7/2009	2009-04-09	E2/E4	2560	2560	5120	1280	2560	2560	2560
A/England/195/2009	2009-04-28	MDCK1/MDCK2	2560	2560	2560	1280	2560	2560	2560
A/Auckland/3/2009	2009-04-25	Ex/E3	2560	2560	5120	1280	2560	2560	2560
A/Bayern/69/2009	2009-07-01	MDCK4/MDCK1	320	160	80	640	320	160	160
A/Lviv/N6/2009	2009-10-27	MDCK5	1280	160	160	2560	2560	640	640
A/Hong Kong/2212/2010	2010-07-16	E3	5120	5120	5120	2560	5120	5120	5120
A/Christchurch/16/2010	2010-07-12	E2/E1	5120	5120	5120	2560	5120	5120	5120
TEST VIRUSES									
A/Soria/33/2011	2011-02-01	MDCK1/MDCK2	1280	1280	1280	640	640	1280	640
A/Estonia/53652/2011	2011-02-01	MDCK1/MDCK1	1280	1280	1280	320	320	1280	640
A/Czech Republic/26/2011	2011-02-02	MDCK3/MDCK1	2560	2560	2560	640	1280	2560	1280
A/Moldova/291/2011	2011-02-02	MDCK3	2560	2560	1280	640	1280	2560	1280
A/Estonia/53730/2011	2011-02-02	MDCK2/MDCK1	1280	2560	2560	640	1280	2560	1280
A/Estonia/53731/2011	2011-02-02	MDCK2/MDCK1	1280	1280	1280	640	1280	2560	5120
A/Moldova/338/2011	2011-02-03	MDCK3	2560	2560	1280	640	1280	2560	1280
A/Estonia/53904/2011	2011-02-04	MDCK2/MDCK1	640	640	1280	320	640	640	640
A/Estonia/53902/2011	2011-02-04	MDCK2/MDCK1	1280	1280	2560	640	1280	1280	640
A/Moldova/366/2011	2011-02-07	MDCK3	1280	1280	1280	640	1280	1280	640
A/Moldova/414/2011	2011-02-07	MDCK2	1280	1280	1280	320	1280	1280	640
A/Moldova/448/2011	2011-02-07	MDCK2	640	1280	320	640	1280	1280	640
A/Burgos/36/2011	2011-02-07	MDCK1/MDCK2	1280	1280	1280	640	1280	1280	640
A/St. Petersburg/6/2011	2011-02-07	E1/E1	1280	2560	1280	640	1280	1280	1280
A/St. Petersburg/15/2011	2011-02-07	E1/E1	2560	2560	2560	1280	2560	2560	5120
A/St. Petersburg/8/2011	2011-02-07	E1/E2	2560	2560	2560	1280	2560	1280	1280
A/St. Petersburg/10/2011	2011-02-07	E1/E2	1280	1280	1280	640	1280	1280	1280
A/Estonia/54094/2011	2011-02-08	MDCK2/MDCK1	640	640	640	320	320	640	640
A/Estonia/54133/2011	2011-02-08	MDCK2/MDCK1	1280	640	1280	320	320	640	640
A/Moldova/399/2011	2011-02-09	MDCK3	1280	1280	1280	320	1280	1280	640
A/Palencia/37/2011	2011-02-10	MDCK1/MDCK2	1280	1280	1280	640	1280	1280	640
A/Estonia/54273/2011	2011-02-10	MDCK1/MDCK1	1280	2560	2560	1280	1280	1280	1280
A/Estonia/54494/2011	2011-02-11	MDCK1/MDCK1	1280	1280	640	320	640	640	640
A/Estonia/54372/2011	2011-02-11	MDCK1/MDCK1	1280	1280	1280	320	640	640	320
A/St. Petersburg/27/2011	2011-02-14	E1/E1	2560	1280	1280	640	1280	1280	640
A/St. Petersburg/37/2011	2011-02-14	E1/E1	1280	1280	1280	640	1280	1280	640
A/Estonia/54532/2011	2011-02-15	MDCK2/MDCK1	1280	640	1280	320	640	640	640
A/Estonia/54553/2011	2011-02-15	MDCK2/MDCK1	1280	1280	1280	640	1280	1280	1280
A/Ghana/FS-11-206/2011	2011-02-16	C1/MDCK3	5120	5120	5120	2560	5120	5120	5120
A/Segovia/41/2011	2011-02-17	MDCK1/MDCK1	1280	1280	1280	640	1280	1280	640
A/Crete/GR6779/2011	2011-02-17	MDCK3	1280	1280	1280	640	1280	1280	1280
A/Malta/2/2011	2011-02-19	MDCK2	2560	2560	2560	640	2560	2560	1280
A/Arta/GR6666/2011	2011-02-21	MDCK2	5120	2560	5120	1280	2560	2560	2560
A/Ioannina/GR6664/2011	2011-02-21	MDCK3	2560	2560	2560	640	1280	2560	1280
A/Estonia/54922/2011	2011-02-22	MDCK2/MDCK1	1280	2560	2560	640	1280	1280	1280
A/Estonia/55236/2011	2011-03-02	MDCK1/MDCK1	2560	2560	2560	640	2560	2560	1280
A/Kaluga/1/2011	2011-03-14	E1/E1	2560	2560	2560	1280	2560	2560	1280
A/St. Petersburg/100/2011	2011-03-14	E1/E1	1280	1280	1280	640	1280	640	640
A/Ghana/FS391/2011	2011-03-21	MDCK2	5120	5120	2560	1280	2560	2560	2560

Sequences included in HA phylogeny

Vaccine virus

Table 6. Antigenic analyses of pandemic influenza A(H1N1)pdm09 viruses

Viruses	Collection date	Passage History	Haemagglutination inhibition titre						
			Post infection ferret sera						
			A/Cal 7/09 F05/10	A/Eng 195/09 NIB F18/09	A/Auck 3/09 F17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/09 C4/34/09	A/HK 2212/10 F21/10 Egg	A/C'church 16/10 F30/10
REFERENCE VIRUSES									
A/California/7/2009	2009-04-09	E2/E4	2560	5120	5120	1280	5120	2560	2560
A/England/195/2009	2009-04-28	MDCK1/MDCK4	2560	2560	2560	1280	1280	2560	1280
A/Auckland/3/2009	2009-04-25	Ex/E3	5120	5120	5120	2560	5120	5120	2560
A/Bayern/69/2009	2009-07-01	MDCK4/MDCK2	320	80	80	320	320	160	160
A/Lviv/N6/2009	2009-10-27	MDCK5	1280	320	160	1280	1280	640	320
A/Hong Kong/2212/2010	2010-07-16	E4	5120	5120	5120	2560	5120	5120	5120
A/Christchurch/16/2010	2010-07-12	E2/E1	5120	2560	2560	2560	5120	5120	5120
TEST VIRUSES									
A/Estonia/53545/2011	2011-02-01	MDCK1/MDCK2	1280	640	640	320	640	1280	640
A/Estonia/53646/2011	2011-02-01	MDCK1/MDCK2	1280	2560	2560	640	1280	1280	1280
A/Estonia/53649/2011	2011-02-01	MDCK2/MDCK3	1280	1280	1280	640	640	1280	640
A/Ireland/11542/2011	2011-02-01	MDCKP2/MDCK1	640	1280	1280	640	1280	1280	1280
A/Ukraine/60/2011	2011-02-01	MDCK1/MDCK1	640	1280	2560	640	1280	1280	1280
A/Estonia/53727/2011	2011-02-02	MDCK2/MDCK3	1280	1280	2560	640	1280	1280	1280
A/Ireland/12263/2011	2011-02-02	MDCKP3/MDCK1	640	640	640	640	640	640	640
A/Latvia/2-32584p/2011	2011-02-04	MDCK1/MDCK3	640	640	640	160	320	640	320
A/Estonia/54025/2011	2011-02-07	MDCK2/MDCK3	1280	1280	1280	640	640	1280	640
A/Thessaloniki/15/2011	2011-02-07	MDCK3/MDCK1	1280	1280	1280	640	1280	2560	1280
A/Thessaloniki/80/2011	2011-02-07	MDCK3/MDCK1	1280	2560	2560	1280	1280	2560	5120
A/Veria/111/2011	2011-02-07	MDCK2/MDCK1	5120	5120	2560	2560	5120	5120	5120
A/Estonia/54132/2011	2011-02-08	MDCK1/MDCK3	320	160	160	320	160	320	160
A/Estonia/54131/2011	2011-02-08	MDCK1/MDCK3	1280	1280	1280	640	1280	1280	640
A/Estonia/54086/2011	2011-02-08	MDCK2/MDCK3	320	160	80	320	160	320	160
A/Estonia/54122/2011	2011-02-08	MDCK1/MDCK3	640	640	640	320	320	640	320
A/Kavala/19/2011	2011-02-10	E2/E1	1280	1280	640	640	640	640	640
A/Estonia/54427/2011	2011-02-11	MDCK1/MDCK2	2560	1280	2560	640	1280	2560	1280
A/Estonia/54534/2011	2011-02-15	MDCK1/MDCK3	1280	1280	1280	640	640	640	640
A/Estonia/54626/2011	2011-02-16	MDCK1/MDCK3	1280	1280	2560	640	640	1280	640
A/Estonia/54582/2011	2011-02-16	MDCK2/MDCK3	1280	1280	1280	640	640	640	640
A/Estonia/54849/2011	2011-02-21	MDCK1/MDCK2	640	320	640	160	160	640	320
A/Estonia/55082/2011	2011-02-28	MDCK2/MDCK2	640	320	640	160	320	320	320
A/Kaliningrad/13/2011	2011-02-28	C1/MDCK1	1280	640	640	320	640	1280	640
A/Astrakhan/1/2011	2011-02-28	C1/MDCK1	640	1280	1280	640	640	640	640
A/Estonia/55247/2011	2011-03-02	MDCK1/MDCK3	640	640	640	320	640	640	320
A/Estonia/55306/2011	2011-03-03	MDCK1/MDCK2	640	640	640	320	640	640	320
A/Estonia/55305/2011	2011-03-03	MDCK1/MDCK3	1280	640	1280	320	640	640	640
A/Thessaloniki/1099/2011	2011-03-08	E2/E1	640	320	640	640	640	640	2560
A/Thessaloniki/1078/2011	2011-03-08	E2/E1	640	320	640	640	640	640	1280
A/Thessaloniki/1376/2011	2011-03-08	E1/E1	640	80	80	1280	1280	320	160
A/Thessaloniki/1151/2011	2011-03-08	E1/E1	640	640	640	640	640	640	320
A/Madagascar/0822/2011	2011-03-10	MDCK3	1280	2560	2560	1280	1280	1280	1280
A/Voronezh/21/2011	2011-03-14	C1/MDCK1	2560	2560	2560	1280	2560	2560	5120
Sequences included in HA phylogeny			Vaccine virus						

Table 7. Antigenic analyses of pandemic influenza A(H1N1)pdm09 viruses

Viruses	Collection date	Passage History	Haemagglutination inhibition titre						
			Post infection ferret sera						
			A/Cal 7/09 F05/10	A/Eng 195/09 F06/10	A/Auck 3/09 F29/10	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/HK 2212/10 F21/10 Egg	A/C'church 16/10 F30/10
REFERENCE VIRUSES									
A/California/7/2009	2009-04-09	E2/E2	2560	1280	1280	1280	1280	1280	1280
A/England/195/2009	2009-04-28	MDCK7	1280	1280	1280	640	640	1280	1280
A/Auckland/3/2009	2009-04-25	Ex/E4	1280	1280	2560	640	1280	1280	640
A/Bayern/69/2009	2009-07-01	MDCK5	160	40	40	640	160	80	80
A/Lviv/N6/2009	2009-10-27	M4/S2	320	80	80	1280	640	320	160
A/Hong Kong/2212/2010	2010-07-16	E3	2560	2560	2560	1280	2560	2560	2560
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	1280	2560	2560	2560	1280	5120
TEST VIRUSES									
A/Slovenia/772/2011	2011-02-03	MDCK1/MDCK1	1280	1280	1280	640	640	1280	320
A/Minsk/970/2011	2011-02-07	MDCK2/MDCK1	1280	1280	1280	640	640	1280	640
A/Slovenia/892/2011	2011-02-07	MDCK1/MDCK2	1280	1280	1280	640	640	1280	640
A/Stockholm/3/2011	2011-02-08	MDCK1/MDCK2	640	320	320	640	640	320	320
A/Minsk/972/2011	2011-02-08	MDCK2/MDCK1	320	640	640	320	320	320	320
A/Lisbon/SU429/2011	2011-02-10	MDCK2	1280	5120	1280	320	640	640	640
A/Lisbon/SU468/2011	2011-02-11	MDCK4	1280	1280	1280	640	1280	1280	640
A/Minsk/1127/2011	2011-02-14	MDCK2/MDCK1	1280	1280	1280	640	1280	1280	640
A/Minsk/1130/2011	2011-02-15	MDCK2/MDCK1	1280	1280	1280	640	640	640	640
A/Minsk/1131/2011	2011-02-15	MDCK2/MDCK1	640	640	320	320	320	320	320
A/Slovenia/1039/2011	2011-02-16	MDCK1/MDCK2	1280	1280	1280	1280	1280	1280	640
A/Lyon/1067/2011	2011-02-21	MDCK2/MDCK1	1280	1280	2560	640	1280	1280	640
A/Toulon/1173/2011	2011-02-22	MDCK2/MDCK1	1280	1280	1280	320	640	1280	640
A/Slovenia/1273/2011	2011-03-11	MDCK1/MDCK2	1280	1280	1280	640	1280	1280	1280
A/Ghana/570/2011	2011-04-12	SIAT1/MDCK1	2560	2560	2560	640	1280	2560	2560
A/Ghana/601/2011	2011-04-14	SIAT2/MDCK1	640	1280	1280	320	640	640	640
A/Ghana/688/2011	2011-04-27	SIAT1/MDCK1	1280	1280	1280	320	640	1280	640
A/Ghana/FS-11-687/2011	2011-04-27	SIAT3	1280	640	640	2560	1280	640	640
A/Slovenia/1552/2011	2011-04-30	MDCKx/MDCK1	1280	1280	1280	640	1280	1280	1280
A/Johannesburg/32/2011	2011-05-04	MDCK1/MDCK2	1280	1280	1280	640	1280	1280	1280
A/Ghana/701/2011	2011-05-06	SIAT1/MDCK1	1280	1280	1280	640	640	1280	640
A/Johannesburg/28/2011	2011-05-07	MDCK2/MDCK2	1280	1280	1280	640	640	1280	1280
A/Johannesburg/29/2011	2011-05-09	MDCK2/MDCK2	1280	1280	1280	640	640	1280	1280
A/Ghana/763/2011	2011-05-13	SIAT1/MDCK1	640	1280	1280	320	640	640	640
A/Johannesburg/24/2011	2011-05-16	MDCK2/MDCK1	1280	1280	1280	640	1280	1280	1280
A/Neuquén/50677674/2011	2011-05-17	SIAT1/MDCK1	320	320	320	160	320	320	320
A/Neuquén/2452/2011	2011-05-24	MDCK2/MDCK1	320	320	320	160	320	320	160
A/Mendoza/2587/2011	2011-05-26	MDCK2/MDCK1	320	320	320	160	320	320	160
A/Ghana/812/2011	2011-05-31	SIAT1/MDCK2	2560	2560	2560	640	1280	5120	1280
A/Cape Town/60/2011	2011-06-02	E1/E1	1280	1280	1280	640	640	640	640
A/Cape Town/61/2011	2011-06-02	E1/E1	1280	1280	1280	640	640	1280	640
A/Mendoza/2672/2011	2011-06-02	MDCK2/MDCK1	640	320	640	320	320	320	320
A/Johannesburg/40/2011	2011-06-03	E1/E1	2560	2560	1280	1280	1280	1280	1280
A/Johannesburg/43/2011	2011-06-03	E1/E1	1280	1280	640	640	640	640	640
A/Johannesburg/45/2011	2011-06-03	E1/E1	1280	1280	1280	640	1280	1280	1280
A/Johannesburg/47/2011	2011-06-06	MDCK5	1280	1280	2560	640	1280	1280	1280
A/Johannesburg/49/2011	2011-06-08	MDCK4	640	640	1280	320	320	640	640
A/Mendoza/2685/2011	2011-06-12	MDCK1/MDCK1	320	160	320	160	160	160	80

Sequences included in HA phylogeny

Vaccine virus

Table 8. Antigenic analyses of pandemic influenza A(H1N1)pdm09 viruses

Viruses	Collection date	Passage History	Haemagglutination inhibition titre						
			Post infection ferret sera						
			A/Cal 7/09 F05/10	A/Eng 195/09 F06/10	A/Auck 3/09 F29/10	A/Bayern 69/09 F33/09	A/Lviv N6/09 C4/34/09	A/HK 2212/10 F21/10 Egg	A/C'church 16/10 F30/10
REFERENCE VIRUSES									
A/California/7/2009	2009-04-09	E2/E2	2560	2560	2560	640	1280	1280	1280
A/England/195/2009	2009-04-28	MDCK6	5120	2560	2560	1280	1280	1280	2560
A/Auckland/3/2009	2009-04-25	Ex/E4	5120	2560	2560	1280	2560	2560	1280
A/Bayern/69/2009	2009-07-01	MDCK5	320	160	80	640	160	160	80
A/Lviv/N6/2009	2009-10-27	M4/S2	640	160	160	1280	1280	320	320
A/Hong Kong/2212/2010	2010-07-16	E3	5120	2560	5120	2560	2560	5120	2560
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	2560	2560	1280	2560	2560	5120
TEST VIRUSES									
A/Slovakia/226/2011	2011-02-01	MDCKx/MDCK1	2560	1280	2560	1280	1280	1280	1280
A/Serbia/840/2011	2011-02-02	MDCK3	1280	640	1280	320	640	640	640
A/Ukraine/52/2011	2011-02-02	MDCK1/MDCK2	1280	1280	2560	1280	1280	1280	1280
A/Trieste/31/2011	2011-02-02	MDCK1/MDCK1	2560	2560	2560	640	2560	2560	1280
A/Serbia/857/2011	2011-02-03	MDCK4	1280	1280	2560	320	640	1280	640
A/Serbia/902/2011	2011-02-05	MDCK1	640	320	320	160	320	320	160
A/Stockholm/4/2011	2011-02-08	MDCK1/MDCK2	640	160	320	1280	640	320	320
A/Norway/949/2011	2011-02-17	MDCK2/MDCK4	1280	640	640	640	640	640	320
A/Norway/950/2011	2011-02-21	MDCK2/MDCK3	1280	640	640	320	320	640	320
A/Norway/924/2011	2011-02-22	MDCK1/MDCK2	640	80	160	1280	640	160	160
A/Ukraine/84/2011	2011-02-27	MDCK1/MDCK2	640	1280	1280	640	640	640	640
A/Norway/1047/2011	2011-03-01	MDCK2/MDCK3	2560	2560	2560	1280	1280	1280	1280
A/Norway/1043/2011	2011-03-01	MDCK2/MDCK2	2560	2560	2560	1280	1280	2560	1280
A/Norway/1058/2011	2011-03-03	MDCK2/MDCK2	1280	1280	1280	640	640	1280	640
A/Ukraine/132/2011	2011-03-10	MDCK1/MDCK2	640	1280	1280	640	1280	1280	640
A/Stockholm/12/2011	2011-03-14	MDCK1/MDCK1	2560	1280	1280	640	1280	1280	1280
A/Stockholm/10/2011	2011-03-14	MDCK2/MDCK1	2560	1280	1280	640	1280	1280	1280
A/Stockholm/11/2011	2011-03-14	MDCK3/MDCK2	1280	640	1280	640	640	640	640
A/Ukraine/101/2011	2011-03-16	MDCK1/MDCK1	640	640	1280	640	640	640	640
A/Ukraine/102/2011	2011-03-16	MDCK1/MDCK1	80	640	640	320	640	640	160
A/Ukraine/182/2011	2011-03-21	MDCK1/MDCK2	80	320	160	160	160	160	160
A/Ukraine/104/2011	2011-03-22	MDCK1/MDCK1	1280	1280	1280	640	1280	1280	640
A/Stockholm/13/2011	2011-03-28	MDCK2/MDCK1	640	160	160	640	640	320	160
A/Malmoe/4/2011	2011-03-28	MDCK1/MDCK1	1280	1280	1280	640	1280	1280	5120
A/Hong Kong/3934/2011	2011-03-29	MDCK3/MDCK1	320	640	640	160	320	320	160
A/Stockholm/14/2011	2011-04-12	MDCK1/MDCK1	320	160	320	640	320	160	160
A/Göteborg/1/2011	2011-04-12	MDCK1/MDCK1	320	80	160	320	320	160	160
A/Hong Kong/3960/2011	2011-04-21	MDCK3/MDCK1	1280	2560	5120	640	1280	2560	2560
A/Umea/1/2011	2011-05-19	MDCK3/MDCK2	320	160	80	640	320	320	160
Sequences included in HA phylogeny			Vaccine virus						

Table 9. Antigenic analyses of pandemic influenza A(H1N1)pdm09 viruses (2011-08-09)

Haemagglutination inhibition titre ¹													
Post infection ferret sera													
Viruses	Collection date	Passage History	A/Cal 7/09 F5/10	A/Eng 195/09 F6/10	A/Auck 3/09 F29/10	A/Bay 69/09 F11/11	A/Lviv N6/2009 C4/34/09	A/HK 2212/2010 F21/10 Egg	A/Chch 16/2010 F30/10	A/HK 3934/11 F21/11	A/Astr 1/11 F22/11	A/St P 27/11 F23/11	A/St P 100/11 F24/11
REFERENCE VIRUSES													
A/California/7/2009	2009-04-09	E2/E2	2560	2560	2560	2560	2560	1280	1280	2560	2560	5120	5120
A/England/195/2009	2009-04-28	MDCK1/MDCK6	1280	1280	2560	640	1280	1280	1280	1280	1280	1280	2560
A/Auckland/3/2009	2009-04-25	Ex/E4	2560	2560	2560	1280	2560	1280	1280	2560	2560	2560	5120
A/Bayern/69/2009	2009-07-01	MDCK4/MDCK2	160	40	80	640	320	80	80	40	80	160	80
A/Lviv/N6/2009	2009-10-27	MDCK4/SIAT2	640	80	80	1280	640	320	160	80	160	160	160
A/Hong Kong/2212/2010	2010-07-16	E4	2560	2560	2560	1280	2560	2560	1280	2560	2560	2560	2560
A/Christchurch/16/2010	2010-07-12	E2/E1	1280	1280	1280	2560	2560	1280	5120	1280	2560	1280	2560
A/Hong Kong/3934/2011	2011-03-29	MDCK2/MDCK2	640	640	1280	160	320	640	320	1280	640	640	1280
A/Astrakhan/1/2011	2011-02-28	MDCK1/MDCK2	1280	1280	2560	640	1280	1280	1280	2560	1280	1280	2560
A/St Petersburg/27/2011	2011-02-14	E1/E1	5120	5120	5120	2560	5120	2560	2560	5120	2560	2560	5120
A/St Petersburg/100/2011	2011-03-14	E1/E1	1280	1280	1280	640	1280	1280	640	1280	1280	1280	2560
TEST VIRUSES													
A/Tekirdag/TR-34/2011	2011-02-01	SIAT1/MDCK1	640	640	640	160	320	640	320	640	640	640	1280
A/Slovenia/808/2011	2011-02-03	MDCKx/MDCK4	320	160	160	80	160	320	160	320	320	320	640
A/Slovenia/861/2011	2011-02-05	MDCK1/MDCK4	320	160	160	80	160	320	160	320	320	320	1280
A/Konya/TR-35/2011	2011-02-02	SIAT1/MDCK1	640	320	640	320	320	640	320	640	640	640	1280
A/Kibris/TR-38/2011	2011-02-03	SIAT1/MDCK2	640	160	160	320	320	160	160	160	320	320	320
A/Samsun/TR-39/2011	2011-02-08	SIAT1/MDCK1	1280	1280	1280	640	640	1280	2560	1280	1280	1280	2560
A/Ankara/TR-40/2011	2011-02-09	SIAT1/MDCK2	320	640	80	80	160	320	160	320	320	160	640
A/Burdur/TR-42/2011	2011-02-09	SIAT1/MDCK1	160	160	80	80	80	160	80	160	320	160	320
A/Karaman/TR-43/2011	2011-02-11	SIAT1/MDCK3	640	160	80	80	160	320	160	640	640	640	1280
A/Kastamonu/TR-45/2011	2011-02-12	SIAT1/MDCK1	320	80	80	320	320	320	160	80	160	160	160
A/Slovenia/1168/2011	2011-02-24	MDCKx/MDCK3	1280	1280	1280	1280	1280	1280	640	1280	1280	1280	2560
A/Antalya/TR-41/2011	2011-02-08	SIAT1/MDCK4	1280	1280	1280	1280	1280	1280	640	1280	1280	1280	2560
Sequences included in HA phylogeny			Vaccine virus										

1. < = <40

Table 10. Antigenic analyses of pandemic influenza A(H1N1)pdm09 viruses (2011-09-14)

Haemagglutination inhibition titre ¹													
Viruses	Collection date	Passage History	Post infection ferret sera										
			A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/2009	A/C'ch 16/2010	A/HK 3934/2011	A/Astrak 1/2011	A/St. P 27/11	A/St. P 100/11	A/St. P 100/11	A/H-X SWL191/11	A/Voro 01/11
			F31/09	F11/11	C4/34/09	F30/10	F21/11	F22/11	F23/11	F24/11	CDC 2011768870	CNIC 2011-02	CDC 2011768868
REFERENCE VIRUSES													
A/California/7/2009	2009-04-09	E2/E4	2560	1280	1280	640	640	640	640	1280	640	2560	640
A/Bayern/69/2009	2009-07-01	MDCK4/MDCK2	320	640	320	80	80	160	160	80	80	320	<
A/Lviv/N6/2009	2009-10-27	MDCK5	640	2560	1280	320	160	320	320	160	320	640	160
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	2560	2560	5120	2560	2560	2560	5120	5120	5120	2560
A/Hong Kong/3934/2011	2011-03-29	M2/M2	1280	320	640	640	1280	1280	1280	2560	640	2560	640
A/Astrakhan/1/2011	2011-02-28	M1/M4	1280	640	640	640	1280	1280	1280	2560	640	2560	640
A/St. Petersburg/27/2011	2011-02-14	E1/E2	1280	640	640	640	1280	1280	1280	2560	ND	ND	ND
A/St. Petersburg/100/2011	2011-03-14	E1/E1	1280	640	640	640	1280	1280	1280	2560	640	5120	640
TEST VIRUSES													
A/Heilongjiang-Xiangfang/191/2011	2011-04-21	E3	2560	1280	2560	2560	2560	2560	2560	5120	1280	5120	2560
A/Voronezh/01/2011	2011-03-14	E1/E2/E1	2560	1280	1280	1280	1280	2560	2560	5120	1280	5120	1280
IVR-159 (A/Victoria/502/2010)	2011-01-24	E3/D7/E1	1280	1280	1280	2560	1280	1280	640	2560	2560	2560	640
A/Brisbane/70/2011	2011-02-18	E4/E1	640	320	640	320	640	640	640	1280	320	1280	320
A/Minnesota/03/2011	2011-02-18	E4/E1	2560	1280	2560	1280	2560	2560	2560	5120	1280	5120	1280
A/Mexico/2208/2011	2011-03-15	E3/E2/E1	2560	1280	2560	2560	2560	2560	2560	5120	2560	5120	2560
A/New York/09/2011	2011-04-13	E4/E1	640	640	640	640	640	640	640	2560	640	1280	640
A/Tasmania/14/2011	2011-05-07	E3/E1	1280	640	640	640	1280	1280	2560	2560	640	2560	640
A/Wellington/1/2011	2011-05-11	E3/E1	2560	640	1280	1280	1280	2560	1280	5120	1280	5120	1280
Sequences included in HA phylogeny			Vaccine virus										

1. < = <40

Figure 5. Phylogenetic comparison of influenza A(H1N1)pdm09 HA genes

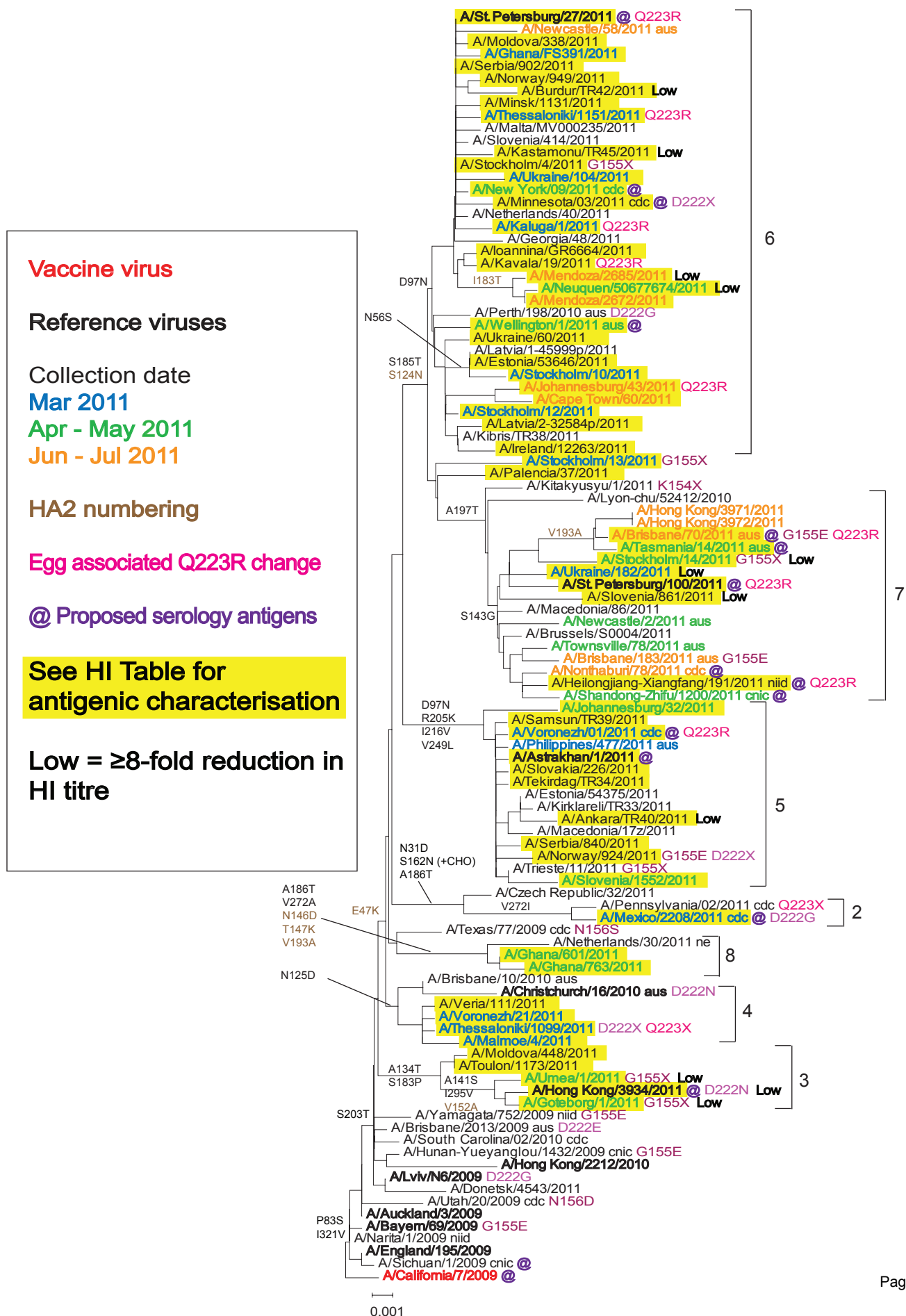


Figure 6.
HA protein alignment for pandemic A(H1N1)pdm09 viruses.

Potential N-linked glycosylation sites

	10	20	30	40	50	60	70	80	90	100
A/California/7/2009	DTLCIGYHANNST	DTVDTVLEKNVT	VTHSVNLLEDKHNGKLC	KLGRVAPLHLGKC	NIAGWILGNPECESL	STASSWSYIVETPSSD	NGTCYPGDFIDYEE			
A/Sichuan/1/2009 CNIC			I			S				
A/England/195/2009			I			S				
A/Bayern/69/2009						S				
A/Auckland/3/2009						S				
A/Lviv/N6/2009						S				
A/Hong Kong/2212/2010						S				
A/Goteburg/1/2011						S				
A/Hong Kong/3934/2011						S				
A/Thessaloniki/1099/2011						S				
A/Christchurch/16/2010 AUS						S			N	
A/Ghana/763/2011						S				
A/Mexico/2208/2011 CDC			D			S				
A/Slovenia/1552/2011						S			N	
A/Astrakhan/1/2011						S			N	
A/Voronezh/01/2011 CDC						S			N	
A/Shandong-Zhifu/1200/2011 CNIC						S				
A/Hjiang-Xiangfang/191/2011 NIID						S				
A/Nonthaburi/78/2011 CDC						S				
A/Newcastle/2/2011 AUS						S				
A/St. Petersburg/100/2011						S				
A/Tasmania/14/2011 AUS				I		S				
A/Brisbane/70/2011 AUS						S				
A/Hong Kong/3972/2011						S				
A/Johannesburg/43/2011						S			N	
A/Stockholm/10/2011					S				N	
A/Wellington/1/2011 AUS					T				N	
A/Mendoza/2685/2011						S			N	
A/Minnesota/03/2011 CDC						S			N	
A/New York/09/2011 CDC						S			N	
A/St. Petersburg/27/2011						S			N	
	110	120	130	140	150	160	170	180	190	200
A/California/7/2009	LREQLSSVSSFERFE	IFPKTSSWPNHDS	SNKGVTAAACPHAGAKS	FYKNLIWLVKKGNSY	PKLSKSYINDKGKEVL	VLWGIHPST	SADQ	QSLYQ	NADAYVF	
A/Sichuan/1/2009 CNIC										
A/England/195/2009										
A/Bayern/69/2009					E					
A/Auckland/3/2009										
A/Lviv/N6/2009										
A/Hong Kong/2212/2010			P				T			A
A/Goteburg/1/2011				T	S			X		
A/Hong Kong/3934/2011									P	
A/Thessaloniki/1099/2011										
A/Christchurch/16/2010 AUS			D							
A/Ghana/763/2011										
A/Mexico/2208/2011 CDC					N			N		
A/Slovenia/1552/2011									T	
A/Astrakhan/1/2011										
A/Voronezh/01/2011 CDC										
A/Shandong-Zhifu/1200/2011 CNIC				G					T	
A/Hjiang-Xiangfang/191/2011 NIID				G					T	
A/Nonthaburi/78/2011 CDC					G				T	
A/Newcastle/2/2011 AUS		K			G				T	
A/St. Petersburg/100/2011					G				T	
A/Tasmania/14/2011 AUS					G				T	
A/Brisbane/70/2011 AUS				T	G		E		T	
A/Hong Kong/3972/2011					G				T	
A/Johannesburg/43/2011									T	
A/Stockholm/10/2011								I		
A/Wellington/1/2011 AUS				R					T	
A/Mendoza/2685/2011									T	
A/Minnesota/03/2011 CDC									XT	
A/New York/09/2011 CDC			X							
A/St. Petersburg/27/2011									T	

HA protein alignment for pandemic A(H1N1) viruses.

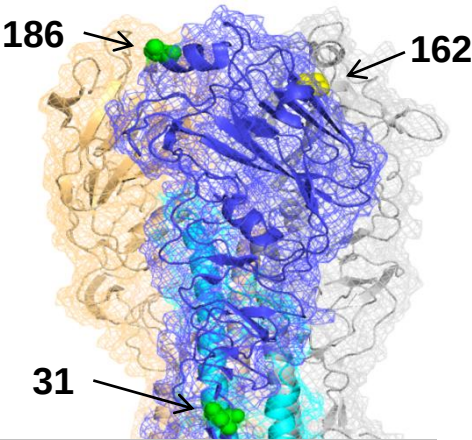
	310	320	330	340	350	360	370	380	390	400
A/California/7/2009	GKCPKYVKSTKLRLATGLRNI	PSIQSRGLFGA	IAGFIEGGWTGMVDGWY	YHHQNEQSGYAADLKSTQNA	IDEITNKVNSVIEKMNTQFTAVGKEF	NHL				
A/Sichuan/1/2009 CNIC		V								
A/England/195/2009		V								
A/Bayern/69/2009		V								
A/Auckland/3/2009		V								
A/Lviv/N6/2009		V								
A/Hong Kong/2212/2010		V					F			
A/Goteburg/1/2011		V						K		
A/Hong Kong/3934/2011		V								
A/Thessaloniki/1099/2011		V						K		
A/Christchurch/16/2010 AUS		V						K		
A/Ghana/763/2011		V						K		
A/Mexico/2208/2011 CDC		V								
A/Slovenia/1552/2011		V						K		
A/Astrakhan/1/2011		V						K		
A/Voronezh/01/2011 CDC		V						K		
A/Shandong-Zhifu/1200/2011 CNIC		V						K		
A/Hjiang-Xiangfang/191/2011 NIID		V						K		
A/Nonthaburi/78/2011 CDC		V						V.K		
A/Newcastle/2/2011 AUS		V						K		
A/St. Petersburg/100/2011		V						K		
A/Tasmania/14/2011 AUS		V						K		
A/Brisbane/70/2011 AUS		V						K		
A/Hong Kong/3972/2011		V						K		
A/Johannesburg/43/2011		V						K		
A/Stockholm/10/2011		V						K		
A/Wellington/1/2011 AUS		V						K		
A/Mendoza/2685/2011		V						K		
A/Minnesota/03/2011 CDC		V						K		
A/New York/09/2011 CDC		V						K		
A/St. Petersburg/27/2011		V						K		

Figure 6 continued.

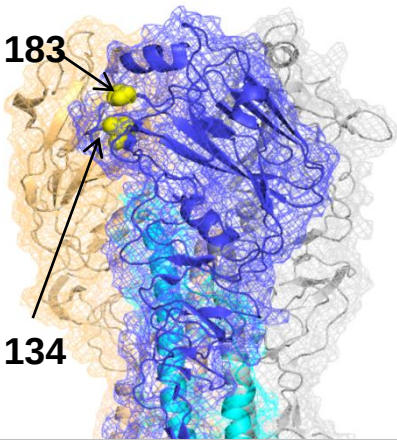
HA protein alignment for pandemic A(H1N1) viruses.

	410	420	430	440	450	460	470	480	490	500
A/California/7/2009	ERRIENLNKKVDDGFLDIWTYN	AEALLVLENER	TLDYHDSNVKNLYEKVRSQ	LKNNAKEIGNGCFEFYHKDNT	CMESVKNGTYDYPKYSEEAKLNREEI					
A/Sichuan/1/2009 CNIC										
A/England/195/2009										
A/Bayern/69/2009										
A/Auckland/3/2009										
A/Lviv/N6/2009										
A/Hong Kong/2212/2010										
A/Goteburg/1/2011								I		
A/Hong Kong/3934/2011								I		
A/Thessaloniki/1099/2011										
A/Christchurch/16/2010 AUS										
A/Ghana/763/2011							DK			
A/Mexico/2208/2011 CDC										
A/Slovenia/1552/2011										
A/Astrakhan/1/2011										
A/Voronezh/01/2011 CDC										
A/Shandong-Zhifu/1200/2011 CNIC					N					
A/Hjiang-Xiangfang/191/2011 NIID					N					
A/Nonthaburi/78/2011 CDC					N					
A/Newcastle/2/2011 AUS					N					
A/St. Petersburg/100/2011					N					
A/Tasmania/14/2011 AUS					N					
A/Brisbane/70/2011 AUS					N					
A/Hong Kong/3972/2011					N					
A/Johannesburg/43/2011					N					
A/Stockholm/10/2011					N					
A/Wellington/1/2011 AUS					N					
A/Mendoza/2685/2011					N					
A/Minnesota/03/2011 CDC					N					
A/New York/09/2011 CDC					N					
A/St. Petersburg/27/2011					N					
	510	520	530	540	550					
A/California/7/2009	DGVKLESTRIYQILAIYSTVASSLV	LVVSLGAISFWMCSNGSLQCRICI*								
A/Sichuan/1/2009 CNIC					*					
A/England/195/2009					*					
A/Bayern/69/2009					*					
A/Auckland/3/2009					*					
A/Lviv/N6/2009					*					
A/Hong Kong/2212/2010					*					
A/Goteburg/1/2011					*					
A/Hong Kong/3934/2011					*					
A/Thessaloniki/1099/2011					*					
A/Christchurch/16/2010 AUS					*					
A/Ghana/763/2011		A			*					
A/Mexico/2208/2011 CDC					*					
A/Slovenia/1552/2011					*					
A/Astrakhan/1/2011					*					
A/Voronezh/01/2011 CDC					*					
A/Shandong-Zhifu/1200/2011 CNIC					*					
A/Hjiang-Xiangfang/191/2011 NIID					*					
A/Nonthaburi/78/2011 CDC					*					
A/Newcastle/2/2011 AUS					*					
A/St. Petersburg/100/2011					*					
A/Tasmania/14/2011 AUS		A			*					
A/Brisbane/70/2011 AUS		A			*					
A/Hong Kong/3972/2011		A			*					
A/Johannesburg/43/2011					*					
A/Stockholm/10/2011					*					
A/Wellington/1/2011 AUS					*					
A/Mendoza/2685/2011		T			*					
A/Minnesota/03/2011 CDC					*					
A/New York/09/2011 CDC					*					
A/St. Petersburg/27/2011					*					

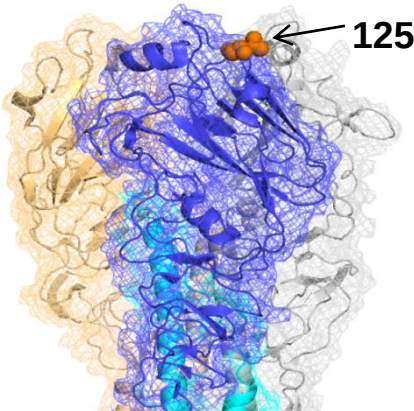
Figure 7. H1-HA locations of genetic cluster defining amino acid substitutions.



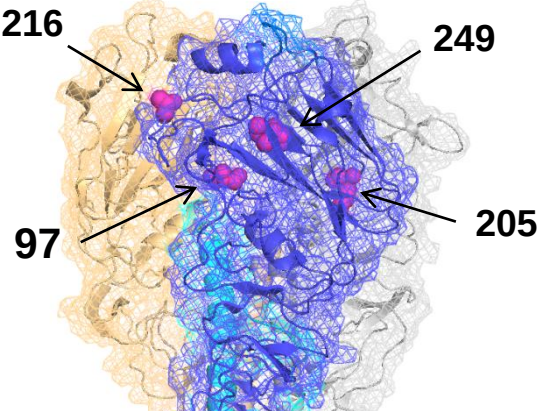
Group 2
N31D, S162N (+CHO), A186T



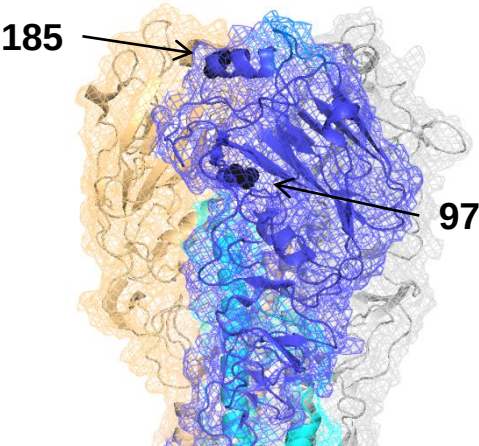
Group 3
A134T, S183P



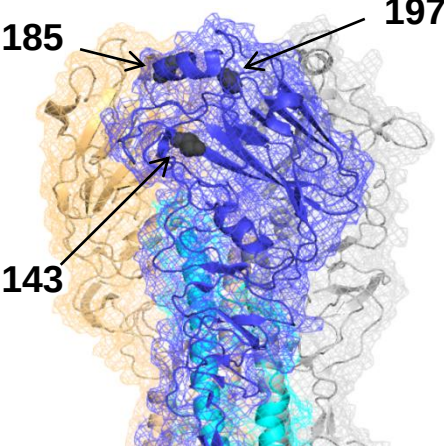
Group 4
N125D



Group 5
D97N, R205K, I216V, V249L



Group 6
D97N, S185T



Group 7
S143G, S185T, A197T

Figure 8. Phylogenetic comparison of influenza A(H1N1)pdm09 NA genes

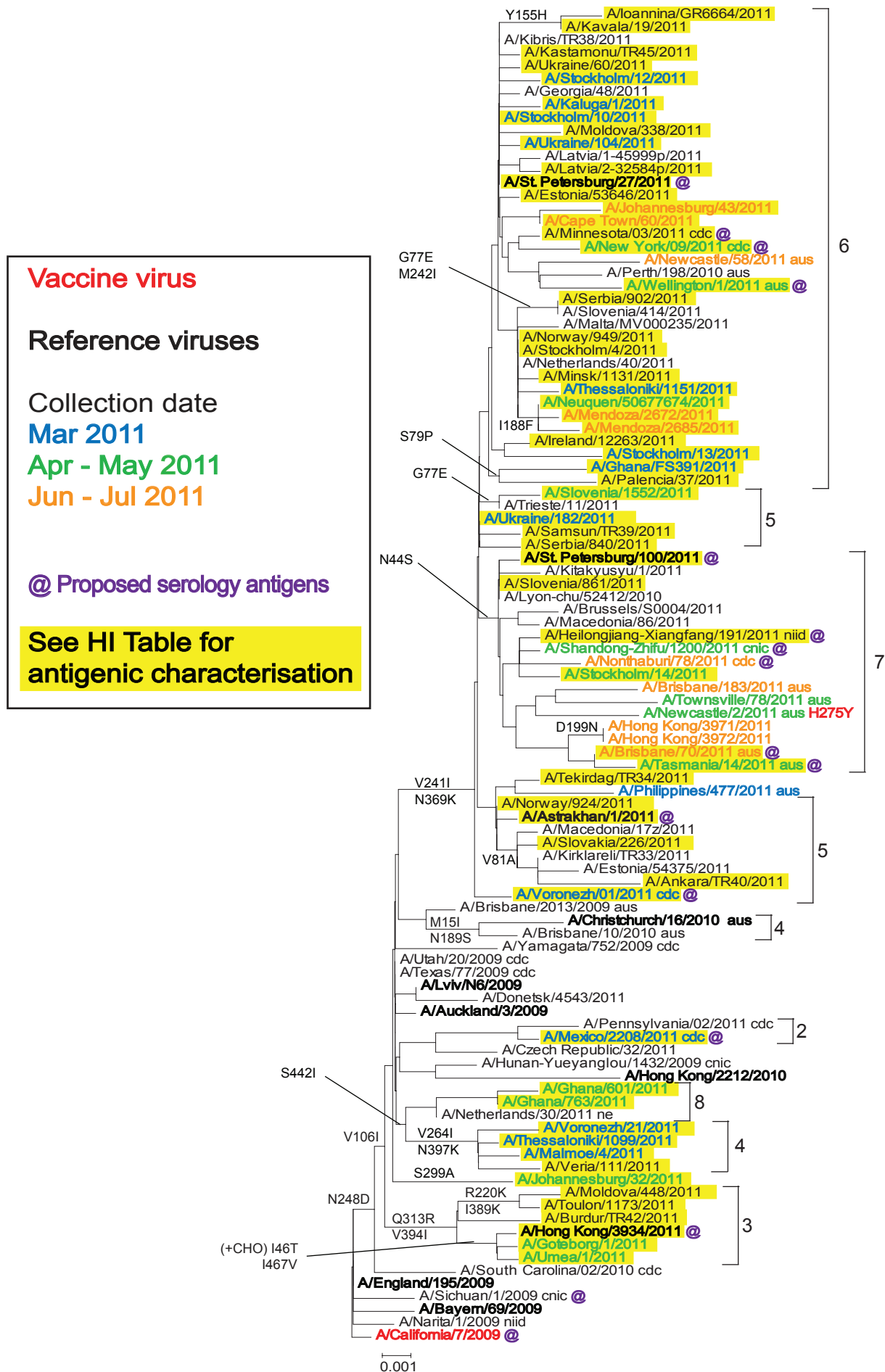
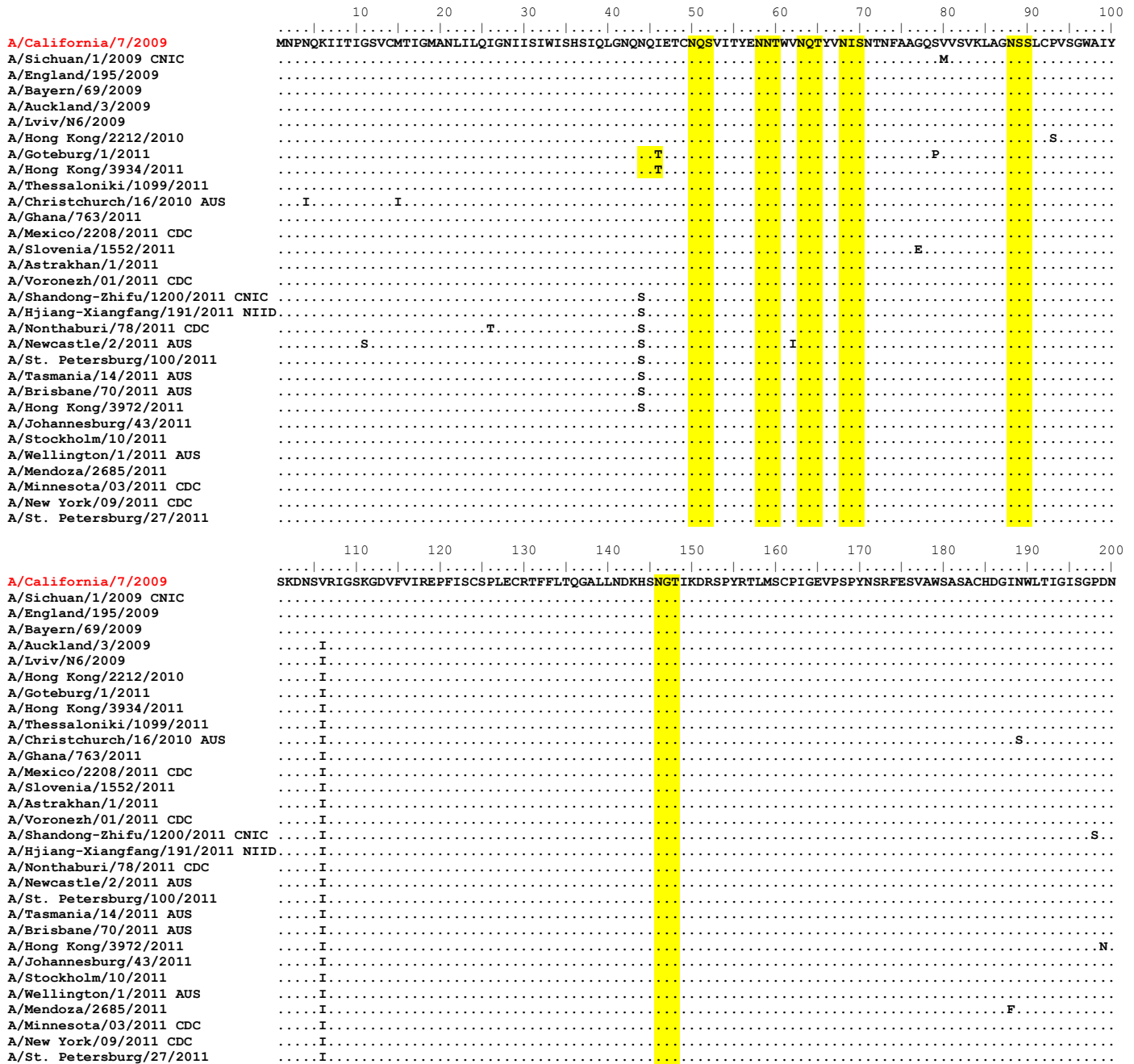


Figure 9.
NA protein alignment for pandemic A(H1N1)pdm09 viruses.

Potential N-linked glycosylation sites



NA protein alignment for pandemic A(H1N1) viruses.



Figure 9 continued.

NA protein alignment for pandemic A(H1N1) viruses.

	410	420	430	440	450	460	470	
A/California/7/2009		GYSGSFVQHP	ELTGLDCIR	PCFWVELIR	GRPKENTIW	TSGSSISFC	GVNSDTV	GWSWPDGAELPFTIDK*
A/Sichuan/1/2009 CNIC								*
A/England/195/2009								*
A/Bayern/69/2009								*
A/Auckland/3/2009								*
A/Lviv/N6/2009								*
A/Hong Kong/2212/2010								*
A/Goteburg/1/2011								*
A/Hong Kong/3934/2011								*
A/Thessaloniki/1099/2011								*
A/Christchurch/16/2010 AUS								*
A/Ghana/763/2011								*
A/Mexico/2208/2011 CDC								*
A/Slovenia/1552/2011								*
A/Astrakhan/1/2011								*
A/Voronezh/01/2011 CDC								*
A/Shandong-Zhifu/1200/2011 CNIC								*
A/Hjiang-Xiangfang/191/2011 NIID								*
A/Nonthaburi/78/2011 CDC								*
A/Newcastle/2/2011 AUS								*
A/St. Petersburg/100/2011								*
A/Tasmania/14/2011 AUS								*
A/Brisbane/70/2011 AUS								*
A/Hong Kong/3972/2011								*
A/Johannesburg/43/2011								*
A/Stockholm/10/2011								*
A/Wellington/1/2011 AUS								*
A/Mendoza/2685/2011								*
A/Minnesota/03/2011 CDC								*
A/New York/09/2011 CDC								*
A/St. Petersburg/27/2011								*

Table 11. Summary of influenza A(H3N2) samples received, collected between 2011-02-01 and present

MONTH Country	H3N2			
	Number propagated	≤2 fold*	4 fold*	≥8 fold*
FEBRUARY				
Albania	0			
France	2		1	1
Ghana	0			
Ireland	2	2		
Madagascar	9	2	7	
Moldova	6	1	5	
Norway	5			5
Russia	0			
Serbia	6	1	1	4
Spain	1		1	
Sweden	4		1	3
Ukraine	1		1	
MARCH				
France	1		1	
Madagascar	3	2		1
Norway	2		1	1
Romania	1		1	
Serbia	2		2	
Sweden	1			1
APRIL				
Ghana	2		2	
Hong Kong	2		2	
Ireland	1			1
La Reunion	1			1
Serbia	0			
Sweden	3			3
MAY				
Albania	0			
Argentina	1	1		
France	1		1	
Ghana	1		1	
Hong Kong	2		2	
La Reunion	4		2	2
South Africa	3	2	1	
Sweden	2			2
JUNE				
Argentina	3		3	
Ghana	3	1	2	
La Reunion	1		1	
South Africa	7		4	3
JULY				
Argentina	4		3	1
Norway	2		1	1
South Africa	1		1	
AUGUST				
Norway	1	1		
Total	91	13 14%	48 53%	30 33%

* Reduction in HI titre with ferret antiserum raised against A/Perth/16/2009 (egg grown vaccine virus) - results only for those viruses that could be tested in the presence of oseltamivir are presented

Table 12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2011-04-12

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			A/Wis 67/05 F01/06	A/Bris 10/07 F29/09	A/Perth 16/09 F25/09	A/Wis 15/09 F24/09	A/Vic 208/09 F7/10	A/Vic 210/09 F11/10	A/Ala 5/10 F27/10	A/Perth 10/10 F03/11
REFERENCE VIRUSES										
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E11	1280	640	<	<	160	<	<	80
A/Brisbane/10/2007	2007-02-06	E2/E4	5120	5120	40	40	320	160	40	160
A/Perth/16/2009	2009-07-04	E3/E3	40	80	1280	640	1280	1280	640	2560
A/Wisconsin/15/2009	2009-07-06	E2/E3	40	<	640	320	40	160	640	320
A/Victoria/208/2009	2009-06-02	E3/E2	160	160	640	640	5120	5120	1280	2560
A/Victoria/210/2009	2009-06-02	E2/E3	160	320	1280	1280	2560	5120	320	2560
A/Alabama/5/2010	2010-07-13	MK1/M2/S3	<	<	80	40	40	40	160	320
A/Perth/10/2010	2010-05-25	E2/E2	<	<	160	40	40	40	160	320
TEST VIRUSES										
A/Moldova/278/2011	2011-02-02	SIAT4	40	80	640	640	320	320	640	1280
A/Moldova/317/2011	2011-02-04	SIAT3	40	80	320	320	320	160	640	1280
A/Moldova/375/2011	2011-02-07	SIAT4	40	80	320	320	320	160	640	1280
A/Moldova/378/2011	2011-02-08	SIAT3	40	80	320	320	320	320	640	1280
A/Valladolid/40/2011	2011-02-16	MDCK1/SIAT1	40	40	320	320	320	160	320	640
A/Moldova/594/2011	2011-02-18	SIAT4	40	40	320	320	320	160	320	640
A/Moldova/601/2011	2011-02-18	SIAT3	40	40	320	640	320	320	1280	1280
A/Romania/55656/2011	2011-03-10	SIAT2	40	40	320	640	320	320	1280	1280

1. < = <40

Vaccine virus

Sequences included in the HA phylogeny

Table 13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2011-05-13

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			A/Wis 67/05 F18/08	A/Bris 10/07 F09/11	A/Perth 16/09 F30/09	A/Wis 15/09 F24/09	A/Vic 208/09 F7/10	A/Vic 210/09 F10/11	A/Ala 5/10 F27/10	A/Perth 10/10 F03/11
REFERENCE VIRUSES										
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E9	640	320	<	<	80	<	<	40
A/Brisbane/10/2007	2007-02-06	E2/E4	2560	2560	<	<	320	160	<	160
A/Perth/16/2009	2009-07-04	E3/E4	<	40	640	320	160	320	320	640
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	<	320	160	40	80	640	320
A/Victoria/208/2009	2009-06-02	E3/E1	160	160	640	640	1280	2560	640	1280
A/Victoria/210/2009	2009-06-02	E2/E3	160	160	1280	640	1280	2560	320	2560
A/Alabama/5/2010	2010-07-13	MK1/M2/SIAT4	<	<	80	80	40	40	160	320
A/Perth/10/2010	2010-05-25	E2/E1	<	<	320	160	40	80	320	640
TEST VIRUSES										
A/Madagascar/0277/2011	2011-02-01	MDCK1/SIAT1	<	80	320	320	320	320	640	1280
A/Ireland/13527/2011	2011-02-09	MDCK2/SIAT1	<	40	320	320	320	320	640	1280
A/Madagascar/0577/2011	2011-02-10	MDCK2/SIAT1	<	80	160	80	160	160	320	640
A/Ireland/13334/2011	2011-02-10	MDCK2/SIAT1	<	40	320	320	320	160	640	1280
A/Madagascar/0638/2011	2011-02-22	MDCK1/SIAT1	<	40	160	320	320	160	640	1280
A/Ukraine/175/2011	2011-02-22	MDCK1/SIAT1	<	40	160	160	320	160	320	640
A/Madagascar/0817/2011	2011-03-09	MDCK1/SIAT1	<	80	320	320	640	320	640	1280
A/Madagascar/1121/2011	2011-03-22	MDCK1/SIAT1	<	80	320	160	640	160	320	1280

1. < = <40

Vaccine virus

Sequences included in the HA phylogeny

Table 14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2011-05-18

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			A/Wis 67/05 F18/08	A/Bris 10/07 F09/11	A/Perth 16/09 F30/09	A/Wis 15/09 F24/09	A/Vic 208/09 F7/10	A/Vic 210/09 F10/11	A/Ala 5/10 F27/10	A/Perth 10/2010 F03/11
REFERENCE VIRUSES										
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E9	640	640	<	<	80	80	<	40
A/Brisbane/10/2007	2007-02-06	E2/E4	2560	5120	40	40	320	1280	<	80
A/Perth/16/2009	2009-07-04	E3/E4	80	160	640	1280	1280	5120	640	2560
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	80	640	320	40	320	1280	320
A/Victoria/208/2009	2009-06-02	E3/E1	160	640	640	1280	2560	5120	1280	5120
A/Victoria/210/2009	2009-06-02	E2/E3	160	320	1280	1280	1280	5120	320	1280
A/Alabama/5/2010	2010-07-13	MK1/M2/SIAT4	<	40	80	80	40	80	320	640
A/Perth/10/2010	2010-05-25	E2/E1	<	40	320	160	80	320	640	640
TEST VIRUSES										
A/Madagascar/0472/2011	2011-02-03	SIAT2	<	80	160	320	160	320	640	1280
A/Madagascar/0481/2011	2011-02-04	MDCK3/SIAT3	<	160	320	320	320	320	640	1280
A/Madagascar/0501/2011	2011-02-10	SIAT2	<	80	160	320	160	320	640	640
A/Madagascar/0575/2011	2011-02-16	SIAT2	<	80	160	320	160	320	640	640
A/Madagascar/0606/2011	2011-02-21	SIAT2	<	80	160	160	160	160	320	640
A/Madagascar/0648/2011	2011-02-21	SIAT2	<	80	160	160	160	160	320	640
A/Madagascar/0751/2011	2011-03-01	SIAT2	<	80	80	160	80	160	320	640
A/Ireland/M27357/2011	2011-04-06	SIAT2	<	40	80	320	80	80	320	320
A/Hong Kong/3951/2011	2011-04-17	MDCK2/SIAT1	<	40	160	320	160	160	640	640
A/Hong Kong/3961/2011	2011-04-23	MDCK2/SIAT1	<	80	160	320	160	160	640	640

1. < = <40

Vaccine virus

Sequences included in the HA phylogeny

Table 15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2011-06-24

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			A/Wis 67/05 F18/08	A/Bris 10/07 F09/11	A/Perth 16/09 F30/09	A/Wis 15/09 F24/09	A/Vic 208/09 F7/10	A/Vic 210/09 F10/11	A/Ala 5/10 F27/10	A/Perth 10/10 F03/11
REFERENCE VIRUSES										
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E8	640	640	<	<	80	80	<	40
A/Brisbane/10/2007	2007-02-06	E2/E4	2560	2560	40	<	160	640	<	80
A/Perth/16/2009	2009-07-04	E3/E4	40	80	1280	640	640	2560	320	1280
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	<	320	160	<	320	640	320
A/Victoria/208/2009	2009-06-02	E3/E1	160	321	640	640	2560	2560	1280	2560
A/Victoria/210/2009	2009-06-02	E2/E3	160	320	1280	1280	1280	5120	320	2560
A/Alabama/5/2010	2010-07-13	MK1/M2/SIAT4	<	40	80	40	<	40	160	320
A/Perth/10/2010	2010-05-25	E2/E1	<	40	320	160	80	160	640	640
TEST VIRUSES										
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT1	<	40	80	40	40	40	160	160
A/Stockholm/16/2011	2011-04-06	MDCK2/SIAT1	40	80	80	80	80	80	160	320
A/Stockholm/15/2011	2011-04-06	MDCK0/SIAT1	<	80	160	320	160	160	320	640
A/Stockholm/17/2011	2011-04-12	MDCK1/SIAT1	<	<	80	80	<	40	160	160
A/Umea/2/2011	2011-05-05	MDCK2/SIAT1	<	40	80	80	80	80	160	320
A/Umea/3/2011	2011-05-23	MDCK3/SIAT1	<	<	80	80	40	80	160	320

1. < = <40

Vaccine virus

Sequences included in the HA phylogeny

Table 16. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2011-06-14

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹											
			Post infection ferret sera											
			A/Wis 67/05 F18/08	A/Bris 10/07 F09/11	A/Perth 16/09 F30/09	A/Wis 15/09 F24/09	A/Vic 208/09 F7/10	A/Vic 210/09 F10/11	A/Ala 5/10 F27/10	A/Perth 10/10 F03/11	A/Alaska 5/10 F14/11	A/Guad 202/10 F16/11	A/Iowa 19/10 F15/11	
REFERENCE VIRUSES														
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E9	640	640	<	<	80	40	<	40	<	<	<	
A/Brisbane/10/2007	2007-02-06	E2/E4	2560	2560	80	40	160	1280	<	80	80	<	80	
A/Perth/16/2009	2009-07-04	E3/E4	40	80	1280	640	640	2560	320	1280	1280	640	1280	
A/Wisconsin/15/2009	2009-07-06	E2/E3	40	<	320	160	<	320	640	320	160	320	640	
A/Victoria/208/2009	2009-06-02	E3/E1	320	640	1280	1280	5120	5120	2560	5120	5120	1280	5120	
A/Victoria/210/2009	2009-06-02	E2/E3	320	320	1280	1280	1280	5120	320	2560	2560	640	1280	
A/Alabama/5/2010	2010-07-13	MK1/M2/SIAT4	<	40	160	80	40	160	320	640	160	320	320	
A/Perth/10/2010	2010-05-25	E2/E2	<	<	80	40	<	<	160	160	40	160	160	
A/Alaska/5/2010	2010-08-12	E3/E1	160	80	320	160	640	1280	320	1280	1280	320	1280	
A/Guadeloupe/202/2010	2010-11-30	E3/E1	<	<	160	80	<	80	320	320	160	640	320	
A/Iowa/19/2010	2010-12-30	E3/E1	80	160	640	640	1280	640	1280	2560	1280	1280	5120	
TEST VIRUSES														
A/Serbia/824/2011	2011-02-01	SIAT4	40	40	160	160	80	160	320	640	160	320	320	
A/Serbia/812/2011	2011-02-02	SIAT3	40	40	80	160	80	160	320	640	160	320	320	
A/Norway/685/2011	2011-02-04	MDCK2/SIAT1	40	80	160	320	160	160	640	640	320	640	640	
A/Norway/842/2011	2011-02-06	MDCK2/SIAT3	<	<	160	160	160	160	320	640	160	320	320	
A/Norway/845/2011	2011-02-07	MDCK2/SIAT1	<	80	40	40	40	40	80	160	40	80	160	
A/Stockholm/6/2011	2011-02-09	C2/SIAT1	40	80	80	80	80	80	160	320	160	320	320	
A/Stockholm/5/2011	2011-02-09	C1/SIAT1	<	40	80	160	80	160	320	640	320	640	640	
A/Serbia/1181/2011	2011-02-12	SIAT4	40	80	80	160	80	80	160	640	320	640	640	
A/Norway/865/2011	2011-02-22	MDCK2/SIAT3	40	40	80	80	80	80	160	320	160	160	320	
A/Norway/884/2011	2011-02-22	MDCK2/SIAT1	40	40	160	80	40	320	80	160	80	160	160	
A/Stockholm/7/2011	2011-02-24	C2/SIAT1	40	40	320	640	160	640	320	640	160	640	640	
A/Uppsala/3/2011	2011-02-24	C2/SIAT1	<	40	160	80	80	40	160	320	160	160	160	
A/Norway/1138/2011	2011-03-09	MDCK2/SIAT2	<	40	320	320	160	640	160	320	160	320	320	
A/Norway/1186/2011	2011-03-16	MDCK2/SIAT1	<	<	160	320	160	160	320	320	320	640	640	

1. < = <40

Vaccine virus

Sequences included in the HA phylogeny

Table 17. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2011-07-14

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			A/Wis	A/Bris	A/Perth	A/Wis	A/Vic	A/Vic	A/Ala	A/Perth
			67/05 F18/08	10/07 F09/11	16/09 F17/11	15/09 F24/09	208/09 F7/10	210/09 F10/11	5/10 F27/10	10/10 F03/11
REFERENCE VIRUSES										
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E8	1280	640	40	<	80	80	<	40
A/Brisbane/10/2007	2007-02-06	E2/E4	2560	2560	160	40	320	640	40	160
A/Perth/16/2009	2009-07-04	E3/E4	40	160	2560	640	640	2560	320	2560
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	40	2560	160	40	320	640	320
A/Victoria/208/2009	2009-06-02	E3/E1	160	640	2560	640	2560	2560	1280	2560
A/Victoria/210/2009	2009-06-02	E2/E3	160	160	2560	640	1280	2560	160	1280
A/Alabama/5/2010	2010-07-13	MK1/M2/SIAT4	<	<	160	40	<	40	160	160
A/Perth/10/2010	2010-05-25	E2/E1	<	<	640	80	80	160	320	640
TEST VIRUSES										
A/Lyon-CHU/6.614/2011	2011-02-09	MDCK2/MDCK2	<	40	320	160	80	160	320	320
A/La Réunion/1499/2011	2011-04-11	MDCK2/MDCK2	<	40	320	320	160	80	320	1280
A/La Réunion/1457/2011	2011-05-04	MDCK2/MDCK2	40	80	320	160	160	160	320	640
A/La Réunion/1523/2011	2011-05-23	MDCK2/MDCK2	40	40	320	320	160	160	320	320
A/La Réunion/1552/2011	2011-06-08	MDCK2/MDCK2	40	80	640	320	80	160	160	320

1. < = <40

Vaccine virus

Table 18. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2011-07-19

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			A/Wis 67/05 F18/08	A/Bris 10/07 F29/09	A/Perth 16/09 F30/09	A/Wis 15/09 F24/09	A/Vic 208/09 F7/10	A/Vic 210/09 F11/10	A/Ala 5/10 F27/10	A/Perth 10/10 F03/11
REFERENCE VIRUSES										
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E9	640	640	<	<	80	<	<	40
A/Brisbane/10/2007	2007-02-06	E2/E4	640	2560	40	<	160	160	<	80
A/Perth/16/2009	2009-07-04	E3/E4	<	80	640	640	640	1280	320	1280
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	<	320	160	40	160	1280	320
A/Victoria/208/2009	2009-06-02	E3/E1	80	160	640	640	1280	2560	320	1280
A/Victoria/210/2009	2009-06-02	E2/E3	160	320	640	640	2560	5120	1280	5120
A/Alabama/5/2010	2010-07-13	MK1/M2/SIAT4	<	<	160	80	40	40	160	320
A/Perth/10/2010	2010-05-25	E2/E1	<	<	320	160	160	320	320	640
TEST VIRUSES										
A/Serbia/822/2011	2011-02-01	MDCK2/SIAT1	<	40	320	160	160	160	320	320
A/Lyon-CHU/7.496/2011	2011-02-16	MDCK2/SIAT2	<	80	160	160	80	160	320	320
A/Serbia/1580/2011	2011-02-24	MDCK2/SIAT1	<	<	80	80	40	80	80	160
A/Serbia/1605/2011	2011-02-26	MDCK2/SIAT1	<	<	160	160	160	160	320	640
A/Serbia/1731/2011	2011-03-01	MDCK2/SIAT1	<	<	160	80	80	80	160	320
A/St Etienne/1130/2011	2011-03-01	MDCK2/SIAT2	<	40	160	160	160	160	320	320
A/Serbia/1891/2011	2011-03-09	MDCK2/SIAT1	<	<	160	160	80	160	320	320
A/La Réunion/1456/2011	2011-05-04	MDCK2/SIAT2	<	40	160	160	80	160	320	320
A/La Réunion/1459/2011	2011-05-04	MDCK2/SIAT2	<	40	160	160	80	160	320	320
A/Clermont-Ferrand/1472/2011	2011-05-18	MDCK2/SIAT2	<	40	160	160	160	320	640	640

1. < = <40

Vaccine virus

Sequences included in the HA phylogeny

Table 19. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2011-07-28

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			A/Wis 67/05 F18/08	A/Bris 10/07 F29/09	A/Perth 16/09 F17/11	A/Wis 15/09 F24/09	A/Vic 208/09 F7/10	A/Vic 210/09 F11/10	A/Ala 5/10 F27/10	A/Perth 10/10 F03/11
REFERENCE VIRUSES										
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E9	1280	640	40	<	80	<	<	40
A/Brisbane/10/2007	2007-02-06	E2/E3	2560	2560	80	40	320	160	<	80
A/Perth/16/2009	2009-07-04	E3/E4	40	80	1280	640	640	5120	320	1280
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	<	1280	160	80	160	640	320
A/Victoria/208/2009	2009-06-02	E3/E2	160	160	1280	640	1280	2560	640	1280
A/Victoria/210/2009	2009-06-02	E2/E3	160	320	1280	1280	2560	5120	640	2560
A/Alabama/5/2010	2010-07-13	MK1/M2/SIAT5	<	<	320	80	80	80	320	320
A/Perth/10/2010	2010-05-25	E2/E2	40	40	1280	160	640	1280	640	1280
TEST VIRUSES										
A/Ghana/FS591/2011	2011-04-07	SIAT2	40	40	320	160	160	160	160	320
A/Ghana/FS648/2011	2011-04-15	SIAT2	<	40	320	80	80	80	160	320
A/Johannesburg/27/2011	2011-05-06	MDCK2/SIAT1	<	40	320	80	80	80	160	320
A/Ghana/731/2011	2011-05-10	SIAT1/MDCK1	40	40	320	80	160	160	160	320
A/Hong Kong/3968/2011	2011-05-17	MDCK2/SIAT1	<	40	320	80	80	80	160	320
A/Hong Kong/3969/2011	2011-05-19	MDCK2/SIAT1	<	80	320	80	80	80	160	160
A/La Plata/65507/2011	2011-05-23	SIAT1/SIAT1	<	40	640	320	160	320	640	640
A/Buenos Aires/10140261/2011	2011-06-13	SIAT1/SIAT1	<	<	320	320	160	160	320	320
A/Santa Fé/1157-04/2011	2011-06-15	SIAT1/SIAT1	<	<	320	160	160	160	320	640
A/Ghana/FS-11-1014/2011	2011-06-21	SIAT2	40	40	640	160	160	160	320	640
A/Buenos Aires/66090/11	2011-06-21	SIAT2/SIAT1	<	40	320	160	160	160	160	320
A/Ghana/FS1037/2011	2011-06-22	SIAT3	40	40	320	160	160	80	320	320
A/Ghana/FS1045/2011	2011-06-27	SIAT3	40	40	320	160	160	160	320	640
A/Santa Fé/1431/2011	2011-07-04	SIAT1/SIAT1	<	40	320	320	160	160	640	640
A/Entre Rios/755282/2011	2011-07-05	SIAT2/SIAT1	<	<	320	160	160	160	160	320
A/Buenos Aires/R230/2011	2011-07-08	SIAT1/SIAT1	<	<	320	160	160	160	160	320
A/Buenos Aires/49987/2011	2011-07-15	SIAT1/SIAT1	<	<	<	80	40	40	40	160

1. < = <40

Vaccine virus

Sequences included in the HA phylogeny

Table 20. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBC with 20nM Oseltamivir) 2011-09-13

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹												
			Post infection ferret sera												
			A/Wis 67/05 F18/08	A/Bris 10/07 F29/09	A/Perth 16/09 F30/09	A/Vic 208/09 F7/10	A/Vic 210/09 F11/10	A/Ala 5/10 F27/10	A/Perth 10/2010 F03/11	A/HK 3969/11 F27/11	A/Stock 18/2011 F28/11	A/Iowa 19/2010 F15/11	A/Mont 05/11 CDC	A/Iowa 19/10 CDC	A/S Aus 3/11 Aust F2018
REFERENCE VIRUSES															
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E2	640	320	<	80	40	<	40	40	<	<	<	<	<
A/Brisbane/10/2007	2007-02-06	E2/E1	1280	1280	80	80	80	<	80	320	40	40	160	40	160
A/Perth/16/2009	2009-07-04	E3/E1	<	40	1280	80	320	320	320	640	320	640	1280	160	640
A/Victoria/208/2009	2009-06-02	E3/E1	160	320	640	2560	2560	1280	2560	2560	2560	2560	1280	2560	2560
A/Victoria/210/2009	2009-06-02	E2/E2	160	320	1280	1280	2560	640	1280	2560	1280	2560	1280	1280	1280
A/Alabama/5/2010	2010-07-13	MK1/M2/SIAT5	<	<	160	40	40	160	160	640	320	160	320	320	640
A/Perth/10/2010	2010-05-25	E2/E1	160	160	1280	1280	1280	640	1280	2560	1280	1280	1280	640	1280
A/Hong Kong/3969/2011	2011-05-19	MDCK3	<	40	640	160	320	320	640	1280	640	640	640	640	1280
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT2	<	<	320	40	80	80	160	640	320	160	160	320	320
A/Iowa/19/2010	2010-12-30	E3/E1	80	160	1280	2560	2560	1280	2560	2560	2560	2560	2560	2560	2560
A/Montana/05/2011	2011-03-02	E2/E1	<	<	320	<	40	80	80	160	80	80	160	80	80
A/South Australia/3/2011	2011-06-10	MDCK2/SIAT1	<	40	320	80	160	160	320	640	320	320	640	640	640
TEST VIRUSES															
A/Sakai/20/2011	2011-02-15	MDCK2+2/SIAT1	<	<	160	40	80	80	160	320	80	160	160	160	320
A/Kumamoto-C/36/2011	2011-04-15	Caco2 2+MDCK2/SIAT1	<	<	160	40	80	80	80	160	80	160	160	80	160
A/Bangladesh/5071/2011	2011-04-15	Cx/C2/SIAT1	<	<	160	40	80	80	160	320	160	160	160	320	320
A/Johannesburg/113/2011	2011-05-17	MDCK1/SIAT1	40	80	640	320	320	640	640	1280	640	640	1280	1280	1280
A/Johannesburg/114/2011	2011-05-26	MDCK1/SIAT1	<	<	640	160	160	320	640	1280	320	640	1280	640	1280
A/Johannesburg/73/2011	2011-06-15	MDCK2/SIAT1	<	<	160	80	160	160	160	320	320	160	160	320	320
A/Durban/92/2011	2011-06-15	MDCK1/SIAT1	40	40	320	160	160	160	320	640	640	320	320	640	320
A/Johannesburg/78/2011	2011-06-20	MDCK1/SIAT1	<	<	160	80	160	80	160	320	320	320	160	320	320
A/Johannesburg/79/2011	2011-06-21	MDCK2/SIAT1	40	40	160	80	160	160	320	640	320	320	320	320	640
A/Johannesburg/94/2011	2011-06-24	MDCK2/SIAT1	40	40	320	160	160	320	640	1280	640	640	640	640	1280
A/Johannesburg/99/2011	2011-06-29	MDCK2/SIAT1	<	<	320	80	160	160	320	640	640	320	320	320	640
A/Johannesburg/107/2011	2011-06-29	MDCK1/SIAT1	<	<	320	80	160	160	320	640	320	320	640	640	640
A/Johannesburg/153/2011	2011-07-05	MDCK1/SIAT1	<	<	320	80	160	320	320	640	320	320	640	640	640
A/Norway/1762/2011	2011-07-21	MDCK2/SIAT1	<	<	160	40	80	80	160	320	160	80	80	320	320
A/Norway/1775/2011	2011-07-29	MDCK1/SIAT1	<	40	320	80	160	160	320	640	640	320	320	640	640
A/Norway/1789/2011	2011-08-02	MDCK1/SIAT1	<	40	640	160	160	320	320	1280	640	320	640	640	1280

1. < = <40

Vaccine virus

Sequences included in the HA phylogeny

Table 21. Antigenic analysis of influenza A H3N2 viruses - Plaque Reduction Neutralisation - MDCK-SIAT

Viruses	Collection Date	Passage History	Neutralisation titre ¹					Genetic Group	
			Post infection ferret sera						
			A/Bris 10/07 F29/09	A/Per 16/09 F30/09	A/Per 10/10 F1777	A/Ala 5/10 F27/10	A/Wis 15/10 F27/10		
REFERENCE VIRUSES									
A/Brisbane/10/2007	2007-02-06	E2/E4	>5120	160	160	80	80	1	
A/Perth/16/2009	2009-07-04	E3/E6	<	320	80	80	80		
A/Perth/10/2010	2010-05-25	E2/E2	80	640	1280	640	160		
A/Alabama/5/2010	2010-07-13	MK1/MDCK2/SIAT5	80	320	640	640	ND		
TEST VIRUSES									
A/Ireland/909/2011	2011-01-27	MDCK2/SIAT2	80	320	160	320	160	1	
A/Stockholm/16/2011	2011-04-06	MDCK2/SIAT2	<	160	160	40	40	3	
A/Ireland/M27357/2011	2011-04-06	SIAT3	160	1280	2560	2560	1280	6	
A/Stockholm/17/2011	2011-04-12	C0/SIAT1	<	1280	1280	640	640	6	
A/La Reunion/1459/2011	2011-05-04	MDCK2/SIAT3	<	320	320	320	160	5	
A/Ghana/731/2011	2011-05-10	SIAT1/MDCK1/SIAT1	40	320	320	160	160	3	
A/Ghana/FS1037/2011	2011-06-22	SIAT4	40	640	640	320	320	3	

Vaccine virus

¹ Based on 50% plaque reduction compared to serum negative controls

Analysis by HI shown in tables

Analysis by HI shown in tables and represented in phylogenies

Figure 10.
Phylogenetic comparison of influenza A(H3N2) HA genes (HA1 region)

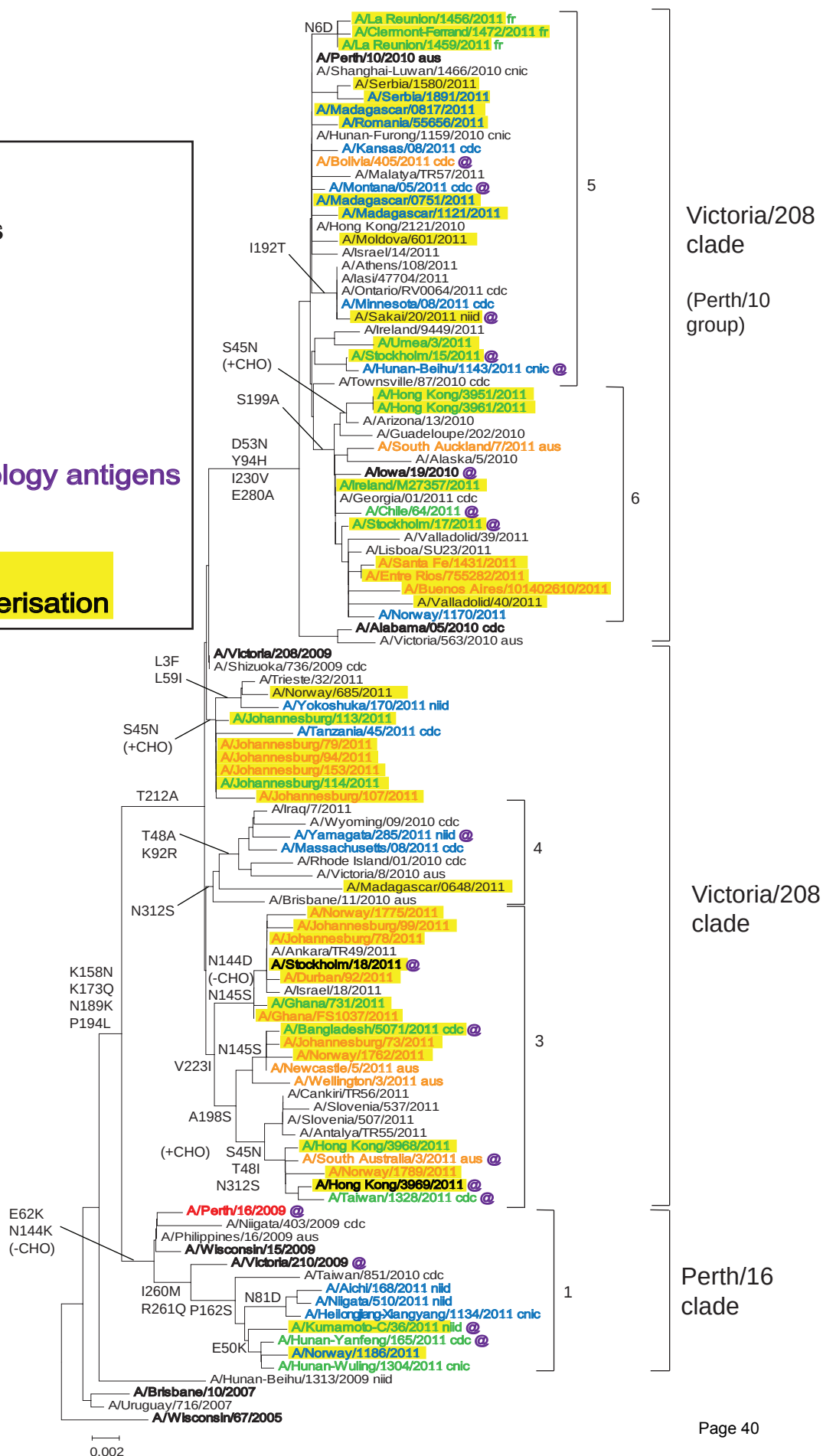
Vaccine strain

Reference strains

Collection date
Mar 2011
Apr - May 2011
Jun - Aug 2011

@ Proposed serology antigens

See HI Table for
 antigenic characterisation



Potential N-linked glycosylation sites

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Figure 11 continued. HA protein alignment for A(H3N2) viruses.

	210	220	230	240	250	260	270	280	290	300
A/Wisconsin/67/2005	RITVSTKRSQQT	VIPNIGSRPRIRNIP	SRISYWTIVKPGDILL	INSTGNLIAPRGYFKIR	SGKSSIMRSDAPIGKCN	SECITP	NGSIPNDKPFQNVNRI			
A/Brisbane/10/2007		V								
A/Hunan-Yanfeng/165/2011 CDC		V				MQ				
A/Kumamoto/36/2011 NIID		X.V.X				MQ				
A/Victoria/210/2009		XV.X				MQ				
A/Wisconsin/15/2009	A	V								
A/Perth/16/2009	S	V								
A/Taiwan/1328/2011 CDC	A									
A/Hong Kong/3969/2011	A									
A/South Australia/3/2011 AUS	A									
A/Norway/1762/2011	A									
A/Bangladesh/5071/2011 CDC	A									
A/Ghana/FS1037/2011	A									
A/Stockholm/18/2011	A									
A/Johannesburg/99/2011	A									
A/Yamagata/285/2011 NIID	A	V								
A/Johannesburg/153/2011	A	V								
A/Victoria/208/2009	A	V								
A/Alabama/05/2010	K.A	V.D	V					A		
A/Santa Fe/1431/2011	A	V	V					A		
A/Stockholm/17/2011	A	V	V					A		
A/Chile/64/2011	A	V	V					A		
A/Iowa/19/2010	A	V	V					A		
A/Hong Kong/3961/2011	A	V	V					A		
A/Hunan-Beihu/1143/2011 CNIC	A	V	V					T		
A/Stockholm/15/2011	A	V	V					T		
A/Sakai/20/2011 NIID	A	V	V					A		
A/Montana/05/2011 CDC	A	V	V					A		
A/Bolivia/405/2011 CDC	A	V	V					A		
A/Perth/10/2010 AUS	A	V	V					A		
	310	320	330	340	350	360	370	380	390	400
A/Wisconsin/67/2005	TYGACPRYVKQNTLKLAT	GMRNVPEKQTRGIFGA	IAGFIENGWEGMVDGWY	GFRHQNSEGIGQAADLKST	QAAINQINGKLNRLIGKT	NEKFHQIEKEFS				
A/Brisbane/10/2007										
A/Hunan-Yanfeng/165/2011 CDC						R		D		
A/Kumamoto/36/2011 NIID										
A/Victoria/210/2009						R		D		
A/Wisconsin/15/2009						R		D		
A/Perth/16/2009						R		D		
A/Taiwan/1328/2011 CDC	S					R		D		
A/Hong Kong/3969/2011	S					R		D		
A/South Australia/3/2011 AUS	S					R		D		
A/Norway/1762/2011	S					R		D		
A/Bangladesh/5071/2011 CDC	S					R		D		
A/Ghana/FS1037/2011						R		D		
A/Stockholm/18/2011						R		D		
A/Johannesburg/99/2011						R		D		
A/Yamagata/285/2011 NIID	S									
A/Johannesburg/153/2011						R		D		
A/Victoria/208/2009						R		D		
A/Alabama/05/2010						R		D		
A/Santa Fe/1431/2011			K			R		D		
A/Stockholm/17/2011						R		D		
A/Chile/64/2011						R		D		
A/Iowa/19/2010						R		D		
A/Hong Kong/3961/2011						R		D		
A/Hunan-Beihu/1143/2011 CNIC						R		D		
A/Stockholm/15/2011						R		D		
A/Sakai/20/2011 NIID										
A/Montana/05/2011 CDC						R		D		
A/Bolivia/405/2011 CDC						R		D		
A/Perth/10/2010 AUS						R		D		

Figure 11 continued.

HA protein alignment for A(H3N2) viruses.

	410	420	430	440	450	460	470	480	490	500
A/Wisconsin/67/2005		EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSMNKLFERTKKQLRENAEDMGNGCFKIYHKCDNACIGSIRNGTYDHDVYRDEALNNRF								
A/Brisbane/10/2007		K.....								
A/Hunan-Yanfeng/165/2011 CDC		K.....								
A/Kumamoto/36/2011 NIID		~.....								
A/Victoria/210/2009		K.....							N	
A/Wisconsin/15/2009		K.....							A	
A/Perth/16/2009		K.....								
A/Taiwan/1328/2011 CDC		K.....								
A/Hong Kong/3969/2011		K.....								
A/South Australia/3/2011 AUS		K.....								
A/Norway/1762/2011		K.....							N	
A/Bangladesh/5071/2011 CDC		K.....							N	
A/Ghana/FS1037/2011		K.....							N	
A/Stockholm/18/2011		K.....							N	
A/Johannesburg/99/2011		K.....							N	
A/Yamagata/285/2011 NIID		~.....								
A/Johannesburg/153/2011		K.....								
A/Victoria/208/2009		K.....								
A/Alabama/05/2010		K.....								
A/Santa Fe/1431/2011		K.....								
A/Stockholm/17/2011		K.....								
A/Chile/64/2011		K.....								
A/Iowa/19/2010		K.....								
A/Hong Kong/3961/2011		K.....								
A/Hunan-Beihu/1143/2011 CNIC		K.....							N	
A/Stockholm/15/2011		K.....								
A/Sakai/20/2011 NIID		~.....								
A/Montana/05/2011 CDC		K.....								
A/Bolivia/405/2011 CDC		G.....								
A/Perth/10/2010 AUS		K.....								
	510	520	530	540	550					
A/Wisconsin/67/2005		QIKGVELKSGYKDWLWISFAISCFLLCVALLGFFIMWACQKGNIRCNICI*								
A/Brisbane/10/2007		*								
A/Hunan-Yanfeng/165/2011 CDC		*								
A/Kumamoto/36/2011 NIID		~.....								
A/Victoria/210/2009		*								
A/Wisconsin/15/2009		*								
A/Perth/16/2009		*								
A/Taiwan/1328/2011 CDC		*								
A/Hong Kong/3969/2011		*								
A/South Australia/3/2011 AUS		*								
A/Norway/1762/2011		E.....								
A/Bangladesh/5071/2011 CDC		*								
A/Ghana/FS1037/2011		*								
A/Stockholm/18/2011		*								
A/Johannesburg/99/2011		*								
A/Yamagata/285/2011 NIID		~.....								
A/Johannesburg/153/2011		*								
A/Victoria/208/2009		*								
A/Alabama/05/2010		*								
A/Santa Fe/1431/2011		*								
A/Stockholm/17/2011		*								
A/Chile/64/2011		*								
A/Iowa/19/2010		*								
A/Hong Kong/3961/2011		*								
A/Hunan-Beihu/1143/2011 CNIC		*								
A/Stockholm/15/2011		*								
A/Sakai/20/2011 NIID		~.....								
A/Montana/05/2011 CDC		*								
A/Bolivia/405/2011 CDC		*								
A/Perth/10/2010 AUS		*								

Figure 12. H3-HA locations of genetic cluster defining amino acid substitutions.

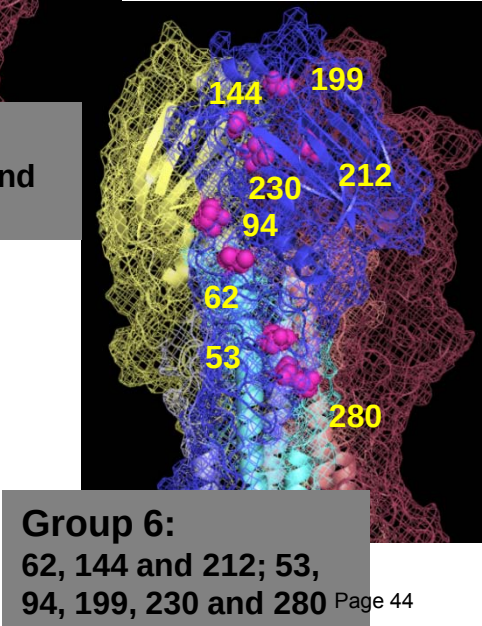
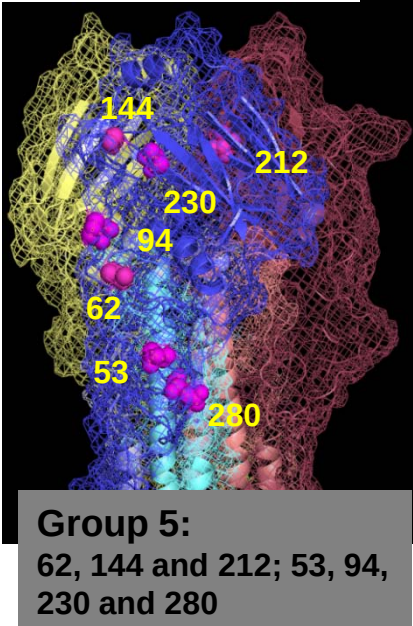
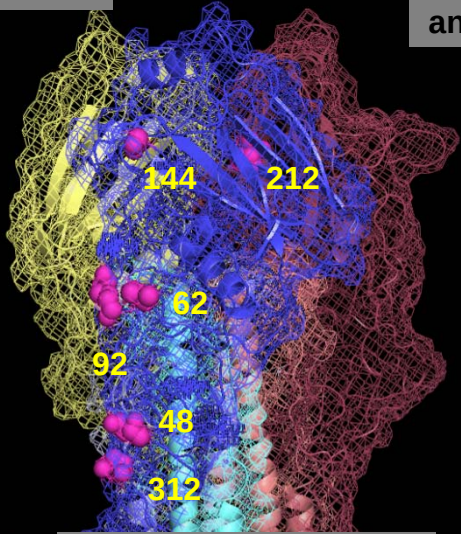
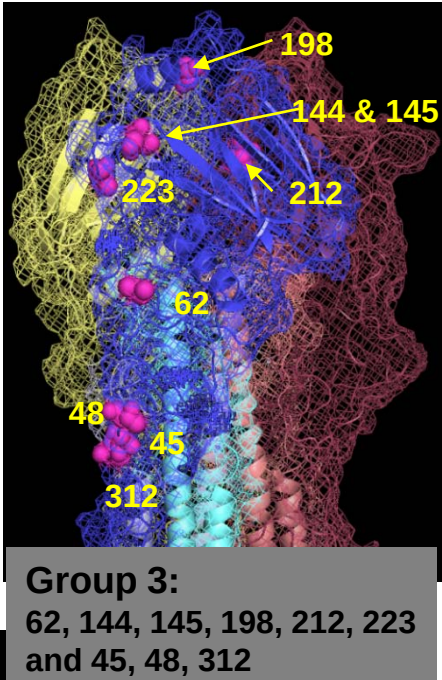
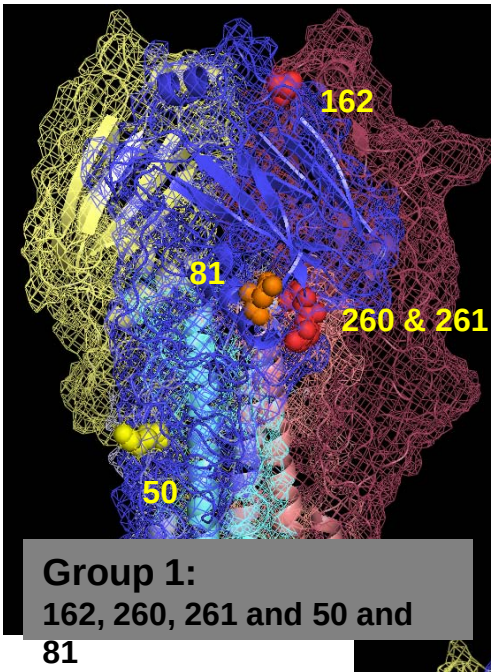
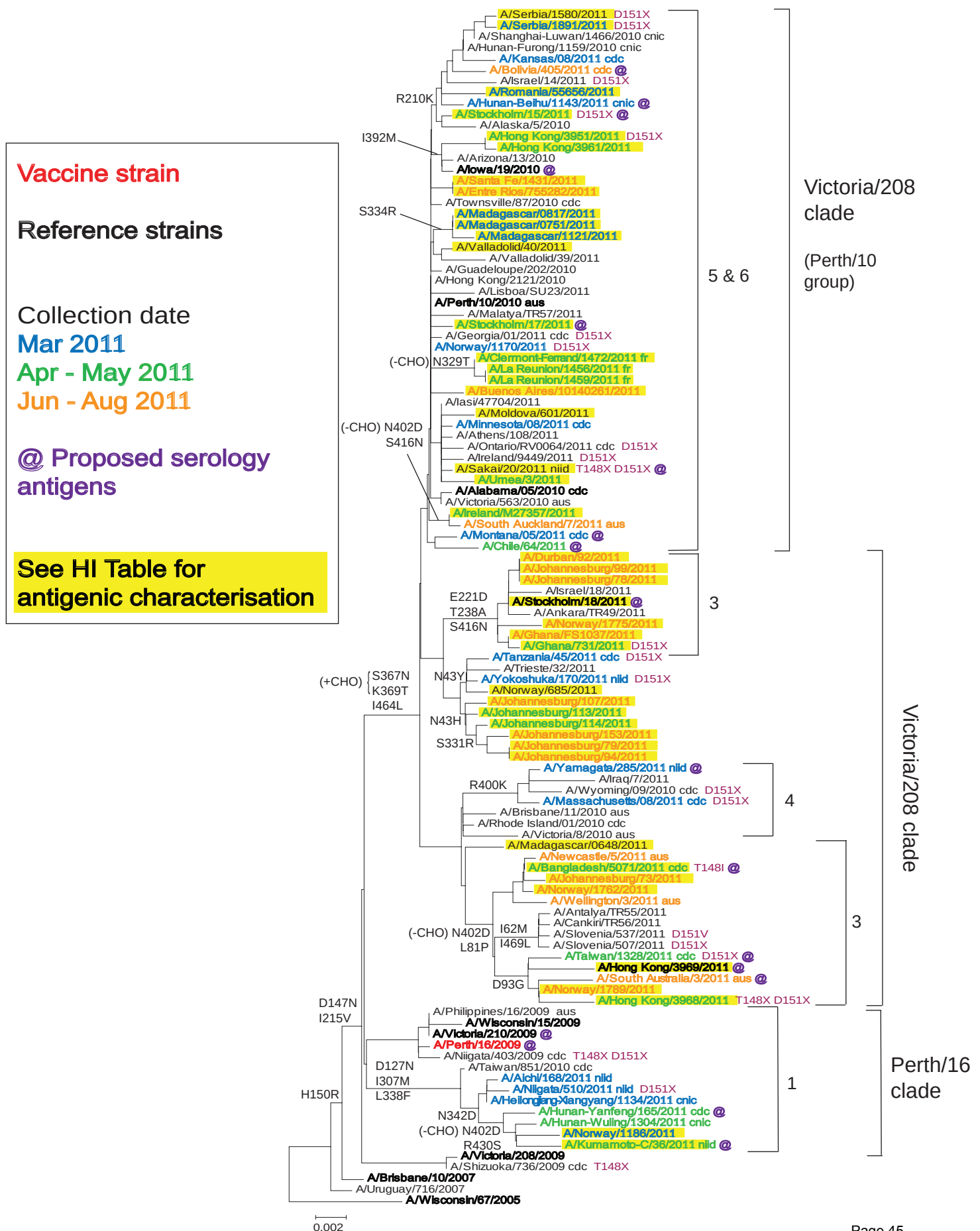


Figure 13. Phylogenetic comparison of influenza A(H3N2) NA genes



Potential N-linked glycosylation sites

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NA protein alignment for A(H3N2) viruses.

	210	220	230	240	250	260	270	280	290	300
A/Wisconsin/67/2005	ATASFIYNGRLVDSIVSWSKEILRQESCEVCINGTCTVVMTDGSASGKADTKILFIEEGKIVHTSTLSGSAQHVEECSCYPHYLGVRVCVRDNWKGNSNR									
A/Brisbane/10/2007	.	.	.	.	.	.	.	.	P.	.
A/Hunan-Yanfeng/165/2011 CDC	.	K.	V	.	.	.	.	.	P.	.
A/Kumamoto/36/2011 NIID	.	F.	V	.	V.	.	.	.	P.	.
A/Victoria/210/2009	.	.	V	.	.	.	.	.	P.	.
A/Wisconsin/15/2009	.	.	V	.	.	.	.	.	P.	.
A/Perth/16/2009	.	.	V	.	.	.	.	.	P.	.
A/Taiwan/1328/2011 CDC	.	.	V	.	.	.	.	.	P.	.
A/Hong Kong/3969/2011	.	.	V	.	N.	.	.	.	P.	.
A/South Australia/3/2011 AUS	.	.	V	.	I.	.	K.	.	P.	.
A/Norway/1762/2011	.	K.	V	.	.	.	.	.	P.	.
A/Bangladesh/5071/2011 CDC	.	.	V	.	.	.	.	.	P.	.
A/Ghana/FS1037/2011	.	.	V	.	D	.	A	.	P.	.
A/Stockholm/18/2011	.	.	V	.	D	.	A	.	P.	.
A/Johannesburg/99/2011	.	.	V	.	D	.	A	.	P.	.
A/Yamagata/285/2011 NIID	.	.	V	.	.	.	.	.	P.	.
A/Johannesburg/153/2011	.	.	V	.	.	.	.	.	P.	.
A/Victoria/208/2009	.	.	V	.	.	.	.	.	P.	.
A/Alabama/05/2010	.	.	V	.	.	.	.	.	P.	.
A/Santa Fe/1431/2011	.	.	V	.	.	.	.	.	P.	.
A/Stockholm/17/2011	.	.	V	.	.	.	.	.	P.	.
A/Chile/64/2011	.	.	V	.	.	.	.	.	P.	.
A/Iowa/19/2010	.	.	V	.	Q	.	.	.	P.	.
A/Hong Kong/3961/2011	.	.	V	.	.	.	.	.	P.	.
A/Hunan-Beihu/1143/2011 CNIC	.	K.	V	.	.	.	.	.	P.	.
A/Stockholm/15/2011	.	K.	V	.	.	.	.	.	P.	.
A/Sakai/20/2011 NIID	.	.	V	.	.	.	.	.	P.	.
A/Montana/05/2011 CDC	.	.	V	.	.	.	.	.	P.	.
A/Bolivia/405/2011 CDC	.	K.	V	.	.	.	.	.	P.	.
A/Perth/10/2010 AUS	.	.	V	.	.	.	.	.	P.	.

	310	320	330	340	350	360	370	380	390	400
A/Wisconsin/67/2005	PIVDINIKDYSIVSSYVCGLVGDTTPRRNDSSSSSHCLDPNNEEGGGHVKGWAFDDGDNDVWMGRITISEKLRSGYETFKVIEGWSNPNSKLQINRQVIHDR									
A/Brisbane/10/2007	.	H.T	.	.	.	.	S.L	.	K	.
A/Hunan-Yanfeng/165/2011 CDC	.	M.H.	.	.	F.D	.	S.L	.	K	.
A/Kumamoto/36/2011 NIID	.	M.H.	.	.	F.D	.	S.L	.	K	.
A/Victoria/210/2009	.	H.	.	.	.	.	S.L	.	K	.
A/Wisconsin/15/2009	.	H.	.	.	.	.	S.L	.	K	.
A/Perth/16/2009	.	H.	.	.	.	.	S.L	.	K	.
A/Taiwan/1328/2011 CDC	.	H.I	.	.	.	.	N.TS.L	.	K	.
A/Hong Kong/3969/2011	.	H.	.	.	.	.	N.TS.L	.	K	.
A/South Australia/3/2011 AUS	.	H.	T	.	.	.	N.TS.L	.	K	.
A/Norway/1762/2011	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Bangladesh/5071/2011 CDC	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Ghana/FS1037/2011	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Stockholm/18/2011	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Johannesburg/99/2011	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Yamagata/285/2011 NIID	.	H.	.	.	.	.	N.TS.L	.	K	K
A/Johannesburg/153/2011	.	H.	R	.	.	.	N.TS.L	.	K	.
A/Victoria/208/2009	.	H.	.	.	.	.	S.L	.	K	.
A/Alabama/05/2010	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Santa Fe/1431/2011	.	H.	R	.	.	.	N.TS.L	.	K	.
A/Stockholm/17/2011	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Chile/64/2011	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Iowa/19/2010	.	H.	.	.	.	.	N.TS.L	.	K	M.
A/Hong Kong/3961/2011	.	H.	.	G	.	.	N.TS.L	.	M	.
A/Hunan-Beihu/1143/2011 CNIC	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Stockholm/15/2011	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Sakai/20/2011 NIID	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Montana/05/2011 CDC	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Bolivia/405/2011 CDC	.	H.	.	.	.	.	N.TS.L	.	K	.
A/Perth/10/2010 AUS	.	H.	.	.	.	.	N.TS.L	.	K	.

Figure 14 continued.

NA protein alignment for A(H3N2) viruses.

	410	420	430	440	450	460	470
A/Wisconsin/67/2005	GNRS	GYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTS	SGTYGTGSWPDGADINLMPI*				
A/Brisbane/10/2007	...	...	...	...	...	...	...
A/Hunan-Yanfeng/165/2011 CDC	D	...	S	...	...	...	...
A/Kumamoto/36/2011 NIID	D	...	S	...	...	...	...
A/Victoria/210/2009	...	...	...	...	...	...	...
A/Wisconsin/15/2009	...	...	...	...	...	...	...
A/Perth/16/2009	...	...	...	...	...	...	...
A/Taiwan/1328/2011 CDC	D	...	...	...	L	...	...
A/Hong Kong/3969/2011	D	...	...	...	L	...	...
A/South Australia/3/2011 AUS	D	...	...	...	L	...	...
A/Norway/1762/2011	D	...	...	...	L	...	...
A/Bangladesh/5071/2011 CDC	D	...	...	...	L	...	...
A/Ghana/FS1037/2011	...	N	...	...	L	...	...
A/Stockholm/18/2011	...	N	...	...	L	...	...
A/Johannesburg/99/2011	...	N	...	...	L	...	...
A/Yamagata/285/2011 NIID	...	...	...	...	L	...	...
A/Johannesburg/153/2011	...	...	...	...	L	...	...
A/Victoria/208/2009	...	...	...	...	L	...	...
A/Alabama/05/2010	...	...	...	...	L	...	...
A/Santa Fe/1431/2011	...	...	...	...	L	...	...
A/Stockholm/17/2011	...	...	...	...	L	...	...
A/Chile/64/2011	...	...	...	...	L	...	...
A/Iowa/19/2010	...	...	...	...	L	...	...
A/Hong Kong/3961/2011	...	...	...	...	L	...	...
A/Hunan-Beihu/1143/2011 CNIC	...	...	...	...	L	...	...
A/Stockholm/15/2011	...	...	...	...	L	...	...
A/Sakai/20/2011 NIID	...	...	...	...	L	H	...
A/Montana/05/2011 CDC	...	...	...	...	L	...	...
A/Bolivia/405/2011 CDC	...	...	...	...	L	...	...
A/Perth/10/2010 AUS	...	...	...	...	L	...	...

Table 22. Summary of influenza B viruses (Victoria-lineage) received, collected between 2011-02-01 and present

MONTH Country	B Victoria lineage			
	Number propagated	≤2 fold*	4 fold*	≥8 fold*
FEBRUARY				
Belarus	2	2		
Czech Republic	3	3		
Ghana	6	6		
Ireland	1	1		
Latvia	2	2		
Madagascar	8	8		
Moldova	1	1		
Russia	7	7		
Serbia	3	3		
Slovakia	2	2		
Slovenia	6	6		
Spain	5	5		
Sweden	3	3		
Ukraine	3	3		
MARCH				
Belarus	1	1		
Estonia	1	1		
Ghana	3	2		1
Greece	2	2		
Ireland	1	1		
Latvia	2	2		
Madagascar	7	7		
Malta	5	5		
Norway	6	6		
Serbia	7	7		
Spain	1	1		
Sweden	9	9		
APRIL				
France	7	7		
Ghana	5	5		
Hong Kong	1	1		
Malta	1	1		
Serbia	1	1		
Sweden	2		2	
MAY				
France	0			
Ghana	3	3		
Sweden	1			1
JUNE				
Ghana	1			1
Hong Kong	2		2	
Total	121	114 94%	4 3%	3 3%

* compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 and B/Odessa/3886/2010)

Table 23. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
			B/Bris ² 60/08 Sh 524	B/Mal 2506/04 F28/05	B/England 393/08 F31/08	B/Bris 60/08 F25/10	B/Paris 1762/08 F11/09	B/HK 514/09 F13/10	B/Odessa 3886/10 F17/10
REFERENCE VIRUSES									
B/Malaysia/2506/2004	2004-12-06	E3/E3	1280	640	160	80	<	<	<
B/England/393/2008	2008-08-29	E1/E6	640	160	320	320	40	40	40
B/Brisbane/60/2008	2008-08-04	E4/E4	640	160	160	320	40	20	40
B/Paris/1762/2009	2009-02-09	C2/MDCK4	640	10	<	20	80	160	80
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK1	1280	<	<	20	80	160	80
B/Odessa/3886/2010	2010-03-19	C2/MDCK3	1280	10	<	40	80	320	160
TEST VIRUSES									
B/St. Petersburg/13/2011	2011-02-01	C1/MDCK1	1280	40	80	40	160	160	160
B/Moldova/284/2011	2011-02-02	MDCK3	640	10	<	40	160	160	160
B/Leon/34/2011	2011-02-02	MDCK1/MDCK2	1280	10	<	80	160	160	160
B/Czech Republic/35/2011	2011-02-03	MDCK2/MDCK1	640	10	10	20	80	160	80
B/Ghana/FS-11-341/2011	2011-02-03	C1/MDCK1	640	10	10	20	80	80	80
B/St. Petersburg/5/2011	2011-02-07	E2/E1	1280	80	320	320	80	40	80
B/St. Petersburg/78/2011	2011-02-07	C1/MDCK1	1280	20	80	80	160	160	160
B/Czech Republic/22/2011	2011-02-08	MDCK2/MDCK1	640	20	10	20	80	160	80
B/Czech Republic/36/2011	2011-02-08	MDCK2/MDCK1	640	20	10	40	80	160	80
B/Kaliningrad/1/2011	2011-02-08	C1/MDCK1	1280	20	80	80	160	160	160
B/Ghana/FS-11-193/2011	2011-02-10	C1/MDCK1	640	10	10	20	80	80	80
B/Ghana/FS-11-190/2011	2011-02-14	C2/MDCK1	640	10	20	40	80	160	80
B/Astrakhan/3/2011	2011-02-14	C2/MDCK1	1280	40	80	80	160	160	160
B/Ghana/FS-11-220/2011	2011-02-15	C1/MDCK1	640	20	20	40	80	160	80
B/Ghana/FS-11-222/2011	2011-02-15	C1/MDCK1	640	10	10	20	80	80	80
B/Segovia/42/2011	2011-02-18	MDCK1/MDCK1	640	<	<	80	160	640	320
B/Segovia/43/2011	2011-02-18	MDCK1/MDCK1	640	<	<	80	160	640	320
B/Ghana/FS244/2011	2011-02-20	C1/MDCK3	640	10	20	40	80	160	80
B/Latvia/2-43305/2011	2011-02-23	MDCK/MDCK1	640	10	20	40	80	160	80
B/Leon/45/2011	2011-02-23	MDCK1/MDCK1	640	<	<	40	160	320	160
B/Salamanca/44/2011	2011-02-24	MDCK1/MDCK1	640	<	<	80	160	640	320
B/Latvia/2-45256/2011	2011-02-28	MDCK1/MDCK1	640	10	20	40	80	160	80
B/Voronezh/20/2011	2011-02-28	C3/MDCK1	1280	40	40	40	160	160	160
B/Leon/46/2011	2011-03-01	MDCK1/MDCK2	640	<	<	20	160	160	160
B/Lefkada/GR8955/2011	2011-03-05	MDCK2	320	10	10	20	40	80	80
B/Latvia/3-39525p/2011	2011-03-16	MDCK1/MDCK1	640	10	20	40	80	160	80
B/Latvia/3-39620p/2011	2011-03-16	MDCK1/MDCK1	640	10	20	40	80	160	80
B/Estonia/55784/2011	2011-03-16	MDCK 2/MDCK1	1280	10	40	80	80	160	80
B/Athens/10051/2011	2011-03-16	MDCK3	640	10	10	20	80	80	80

Sequences included in the HA phylogeny

Vaccine virus

1. < = <10; 2. hyperimmune sheep serum

Table 24. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
			B/Bris ² 60/08 Sh 524	B/Mal 2506/04 F28/05	B/England 393/08 F05/11	B/Bris 60/08 F06/11	B/Paris 1762/09 F7/11	B/HK 514/09 F3/10	B/Odessa 3886/10 F17/10
REFERENCE VIRUSES									
B/Malaysia/2506/2004	2004-12-06	E3/E3	1280	320	80	160	10	<	10
B/England/393/2008	2008-08-29	E1/E6	1280	320	320	1280	80	80	80
B/Brisbane/60/2008	2008-08-04	E4/E4	640	160	320	640	80	40	80
B/Paris/1762/2009	2009-02-09	C2/MDCK4	1280	20	80	80	160	160	160
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK1	1280	20	40	80	160	160	160
B/Odessa/3886/2010	2010-03-19	C2/MDCK3	2560	20	80	160	160	320	320
TEST VIRUSES									
B/Serbia/806/2011	2011-02-02	C1/MDCK1	1280	20	40	80	160	160	320
B/Madagascar/0305/2011	2011-02-02	MDCK2/MDCK1	1280	20	80	80	160	160	320
B/Ireland/11497/2011	2011-02-03	MDCK2/MDCK1	1280	20	40	80	80	160	160
B/Madagascar/0475/2011	2011-02-04	MDCK1/MDCK1	1280	20	80	80	160	320	320
B/Ukraine/54/2011	2011-02-07	MDCK1/MDCK1	640	20	40	80	80	160	160
B/Madagascar/0499/2011	2011-02-07	MDCK1/MDCK1	1280	20	40	80	80	160	320
B/Madagascar/0560/2011	2011-02-08	MDCK1/MDCK1	1280	40	80	80	160	320	320
B/Serbia/1028/2011	2011-02-09	C1/MDCK1	640	20	40	80	80	160	160
B/Madagascar/0579/2011	2011-02-14	MDCK1/MDCK1	1280	20	80	80	160	80	160
B/Madagascar/0645/2011	2011-02-18	MDCK1/MDCK1	1280	20	40	80	160	320	320
B/Madagascar/0662/2011	2011-02-21	MDCK1/MDCK1	1280	20	40	80	160	160	320
B/Serbia/1575/2011	2011-02-24	C1/MDCK1	1280	20	40	80	160	160	320
B/Voronezh/2/2011	2011-02-28	C2/MDCK2	1280	20	40	80	160	160	320
B/Madagascar/0758/2011	2011-02-28	MDCK1/MDCK1	1280	20	40	80	160	160	320
B/Serbia/1693/2011	2011-03-01	C1/MDCK1	640	20	80	80	160	160	320
B/Serbia/1760/2011	2011-03-03	C1/MDCK1	1280	20	80	160	160	160	320
B/Serbia/1761/2011	2011-03-03	C1/MDCK1	1280	20	40	80	160	160	320
B/Madagascar/0778/2011	2011-03-04	MDCK1/MDCK1	1280	20	40	80	160	160	160
B/Malta/MV636793/2011	2011-03-05	MDCK2	1280	80	80	80	160	320	320
B/Serbia/1885/2011	2011-03-07	C1/MDCK1	640	20	40	80	80	80	160
B/Madagascar/0783/2011	2011-03-07	MDCK1/MDCK1	1280	20	40	80	160	160	320
B/Malta/MV636714/2011	2011-03-07	MDCK2	1280	40	40	80	160	160	320
B/Madagascar/1013/2011	2011-03-10	MDCK1/MDCK1	1280	20	80	80	160	160	160
B/Ireland/21653/2011	2011-03-14	MDCKP1/MDCK1	1280	10	20	40	80	160	80
B/Madagascar/1016/2011	2011-03-14	MDCK1/MDCK1	1280	20	40	80	160	160	160
B/Madagascar/1023/2011	2011-03-14	MDCK1/MDCK1	1280	20	80	80	160	320	320
B/Serbia/2133/2011	2011-03-17	C1/MDCK1	1280	20	40	80	160	160	160
B/Norway/1249/2011	2011-03-18	MDCK1/MDCK1	640	10	20	40	80	160	80
B/Madagascar/1097/2011	2011-03-21	MDCK1/MDCK1	1280	20	80	80	160	160	320
B/Madagascar/1119/2011	2011-03-21	MDCK1/MDCK1	1280	20	40	80	160	160	160
B/Norway/1248/2011	2011-03-22	MDCK1/MDCK1	640	10	20	40	80	80	80
B/Serbia/2225/2011	2011-03-22	C1/MDCK1	1280	20	80	80	160	160	320
B/Malta/MV639140/2011	2011-03-22	MDCK2	1280	20	40	80	80	80	160
B/Malta/MV639141/2011	2011-03-22	MDCK2	1280	20	40	80	160	80	160
B/Malta/MV000738/2011	2011-03-25	MDCK2	1280	40	80	160	160	80	320
B/Serbia/2324/2011	2011-03-28	C1/MDCK1	1280	20	40	80	160	160	160
B/Norway/1276/2011	2011-03-29	MDCK1/MDCK1	640	10	20	40	160	160	80
B/Malta/MV642476/2011	2011-04-02	MDCK2	1280	80	40	80	160	160	320
B/Serbia/2689/2011	2011-04-13	C1/MDCK1	1280	40	80	160	160	160	320
B/Hong Kong/591/2011	2011-04-20	MDCK2/MDCK3	640	20	40	80	80	160	160

Sequences included in the HA phylogeny

Vaccine virus

1. < = <10; 2. hyperimmune sheep serum

Table 25. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
			B/Bris ² 60/08 Sh 524	B/Mal 2506/04 F28/05	B/England 393/08 F05/11	B/Bris 60/08 F25/10	B/Paris 1762/09 F07/11	B/HK 514/09 F3/10	B/Odessa 3886/10 F17/10
REFERENCE VIRUSES									
B/Malaysia/2506/2004	2004-12-06	E3/E4	1280	640	80	80	10	<	10
B/England/393/2008	2008-08-29	E1/E6	640	160	320	320	40	40	40
B/Brisbane/60/2008	2008-08-04	E4/E4	640	160	320	640	40	40	40
B/Paris/1762/2009	2009-02-09	C2/MDCK4	1280	20	80	80	160	160	160
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK1	2560	20	80	80	160	160	320
B/Odessa/3886/2010	2010-03-19	C2/MDCK3	2560	20	80	160	160	320	320
TEST VIRUSES									
B/Minsk/939/2011	2011-02-01	MDCK2/MDCK1	640	20	80	40	80	80	80
B/Slovakia/249/2011	2011-02-02	MDCKx/MDCK1	1280	20	40	20	80	80	320
B/Slovenia/819/2011	2011-02-03	MDCK1/MDCK1	640	40	40	80	80	80	80
B/Slovakia/302/2011	2011-02-07	MDCKx/MDCK1	1280	20	40	40	80	160	160
B/Stockholm/1/2011	2011-02-09	MDCK1/MDCK1	1280	20	40	40	80	160	160
B/Stockholm/6/2011	2011-02-09	MDCK1/MDCK1	1280	20	80	40	160	320	320
B/Stockholm/3/2011	2011-02-09	MDCK1/MDCK2	1280	20	40	80	80	80	80
B/Minsk/1138/2011	2011-02-14	MDCK2/MDCK1	640	20	80	40	80	80	80
B/Slovenia/1010/2011	2011-02-15	MDCK1/MDCK1	640	40	80	80	160	80	160
B/Minsk/1559/2011	2011-03-14	MDCK2/MDCK1	640	20	10	40	80	80	80
B/Stockholm/7/2011	2011-03-14	MDCK2/MDCK1	1280	20	80	80	160	160	320
B/Stockholm/16/2011	2011-03-14	MDCK2/MDCK1	1280	10	80	80	80	80	160
B/Stockholm/8/2011	2011-03-14	MDCK2/MDCK2	640	10	20	80	80	80	160
B/Stockholm/15/2011	2011-03-14	MDCK0/MDCK1	640	10	40	40	80	160	160
B/Norway/1188/2011	2011-03-17	MDCK1/MDCK3	640	10	40	40	80	80	80
B/Norway/1227/2011	2011-03-19	MDCK1/MDCK4	1280	20	40	40	80	160	160
B/Lyon-CHU/88/2011	2011-03-21	MDCK2/MDCK1	2560	320	640	640	160	80	160
B/Ghana/832/2011	2011-03-24	SIAT1/MDCK1	1280	40	80	80	160	160	160
B/Ghana/430/2011	2011-03-25	SIAT1/MDCK1	1280	20	40	80	160	160	160
B/Lyon-CHU/1127/2011	2011-03-27	MDCK2/MDCK1	640	20	10	80	80	80	160
B/Stockholm/9/2011	2011-03-28	MDCK2/MDCK2	640	10	20	40	80	160	160
B/Stockholm/10/2011	2011-03-28	MDCK0/MDCK2	640	10	20	80	80	80	160
B/Stockholm/14/2011	2011-03-28	MDCK0/MDCK1	1280	20	80	80	160	160	320
B/Stockholm/13/2011	2011-03-28	MDCK0/MDCK1	1280	20	40	80	160	160	160
B/Stockholm/11/2011	2011-03-28	MDCK0/MDCK2	640	10	20	20	80	80	160
B/Lyon-CHU/584/2011	2011-03-31	MDCK2/MDCK1	640	20	10	80	80	80	160
B/Norway/1326/2011	2011-03-31	MDCK1/MDCK4	1280	20	80	80	160	160	160
B/Ghana/495/2011	2011-04-02	SIAT1/MDCK1	640	20	40	80	160	80	160
B/Ghana/503/2011	2011-04-05	SIAT3/MDCK2	1280	20	40	80	160	80	160
B/Ghana/516/2011	2011-04-06	SIAT1/MDCK2	1280	20	80	160	320	160	320
B/Lyon-CHU/411/2011	2011-04-06	MDCK2/MDCK1	640	20	10	40	80	80	80
B/Stockholm/4/2011	2011-04-06	MDCK1/MDCK1	2560	20	160	80	80	80	160
B/Stockholm/5/2011	2011-04-06	MDCK1/MDCK2	640	10	20	20	40	80	80
B/Lyon-CHU/767/2011	2011-04-08	MDCK2/MDCK1	640	20	10	40	80	80	80
B/Lyon-CHU/15216/2011	2011-04-12	MDCK2/MDCK1	640	20	10	40	80	80	160
B/Lyon-CHU/489/2011	2011-04-13	MDCK2/MDCK1	640	20	10	40	80	80	80
B/Ghana/577/2011	2011-04-14	SIAT2/MDCK1	1280	40	40	160	160	80	160
B/Lyon-CHU/15731/2011	2011-04-15	MDCK2/MDCK1	640	20	10	40	80	80	80
B/Lyon-CHU/16167/2011	2011-04-18	MDCK2/MDCK1	640	20	10	40	80	80	80
B/Ghana/882/2011	2011-04-19	SIAT1/MDCK1	640	20	40	80	80	80	160
B/Lyon-CHU/16432/2011	2011-04-20	MDCK2/MDCK1	640	20	20	40	160	80	160
B/Malmoe/2/2011	2011-05-04	MDCK1/MDCK1	1280	320	80	80	<	<	<
B/Ghana/735/2011	2011-05-11	SIAT1/MDCK1	1280	40	40	80	160	80	160
B/Ghana/760/2011	2011-05-18	SIAT1/MDCK1	1280	40	40	80	80	80	160
B/Ghana/816/2011	2011-05-24	SIAT2/MDCK2	1280	40	40	80	160	80	160
B/Ghana/847/2011	2011-06-03	SIAT1/MDCK1	640	40	40	40	80	40	80

Sequences included in the HA phylogeny

Vaccine virus

1. < = <10; 2. hyperimmune sheep serum

Table 26. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
			B/Bris ² 60/08 Sh 523	B/Mal 2506/04 F28/05	B/England 393/08 F05/11	B/Bris 60/08 F06/11	B/Paris 1762/09 F07/11	B/HK 514/09 F3/10	B/Odessa 3886/10 F17/10
REFERENCE VIRUSES									
B/Malaysia/2506/2004	2004-12-06	E3/E3	2560	640	80	160	10	<	10
B/England/393/2008	2008-08-29	E1/E6	2560	320	640	640	80	40	80
B/Brisbane/60/2008	2008-08-04	E4/E4	2560	160	640	1280	80	40	80
B/Paris/1762/2009	2009-02-09	C2/MDCK4	5120	40	80	160	320	160	160
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK2	5120	20	40	80	160	160	160
B/Odessa/3886/2010	2010-03-19	C2/MDCK3	5120	40	80	160	320	320	320
TEST VIRUSES									
B/Donetsk/4548/2011	2011-02-13	MDCK0/MDCK1	5120	20	40	80	160	80	160
B/Slovenia/1042/2011	2011-02-16	MDCKx/MDCK1	5120	40	40	80	160	160	160
B/Donetsk/4547/2011	2011-02-17	MDCK0/MDCK1	5120	20	40	80	160	160	160
B/Slovenia/1088/2011	2011-02-19	MDCKx/MDCK1	5120	40	80	160	320	320	320
B/Slovenia/1123/2011	2011-02-22	MDCKx/MDCK1	5120	40	80	160	320	160	160
B/Slovenia/1141/2011	2011-02-24	MDCKx/MDCK1	5120	40	80	160	160	160	160
B/Ghana/850/2011	2011-03-31	SIAT1/MDCK3	2560	20	40	40	20	20	40
B/Hong Kong/762/2011	2011-06-22	MDCK3	1280	320	160	320	40	40	80
B/Hong Kong/767/2011	2011-06-27	MDCK3	1280	320	320	320	80	40	160
Sequences included in the HA phylogeny						Vaccine virus			

1. < = <10; 2. hyperimmune sheep serum

Table 27. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			B/Bris ²	B/Mal	B/England	B/Bris	B/Paris	B/HK	B/Odessa	B/Cam
			60/08 Sh 523	2506/04 F28/05	393/08 F05/11	60/08 F06/11	1762/08 F07/11	514/09 F13/10	3886/10 F17/10	30/2011 Aus F1904
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E3/E3	1280	320	80	160	10	<	<	640
B/England/393/2008	2008-08-29	E1/E6	2560	160	320	1280	80	40	40	640
B/Brisbane/60/2008	2008-08-04	E4/E5	2560	160	640	1280	80	80	80	640
B/Paris/1762/2008	2009-02-09	C2/MDCK5	5120	160	80	160	160	160	160	<
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK2	5120	80	40	80	160	160	160	<
B/Odessa/3886/2010	2010-03-19	C2/MDCK3	5120	40	40	80	160	160	160	<
TEST VIRUSES										
B/Cambodia/30/2011	2011-01-04	E3/E1	320	80	20	80	<	<	<	320
B/Santiago/6025/2011	2011-03-25	E3/E2/E1	2560	160	320	1280	160	80	80	160
B/Nevada/03/2011	2011-02-02	E3/E1	2560	160	320	640	80	80	80	320
B/Shanghai-Jingan/1392/2011	2011-04-18	E2/E1	1280	160	160	320	40	40	40	160
Sequences included in the HA phylogeny			Vaccine virus							

1. < = <10; 2. hyperimmune sheep serum

Figure 15.

Phylogenetic comparison of influenza B (Victoria-lineage) HA genes (HA1 region)

Vaccine virus

Reference viruses

Collection date

Mar 2011

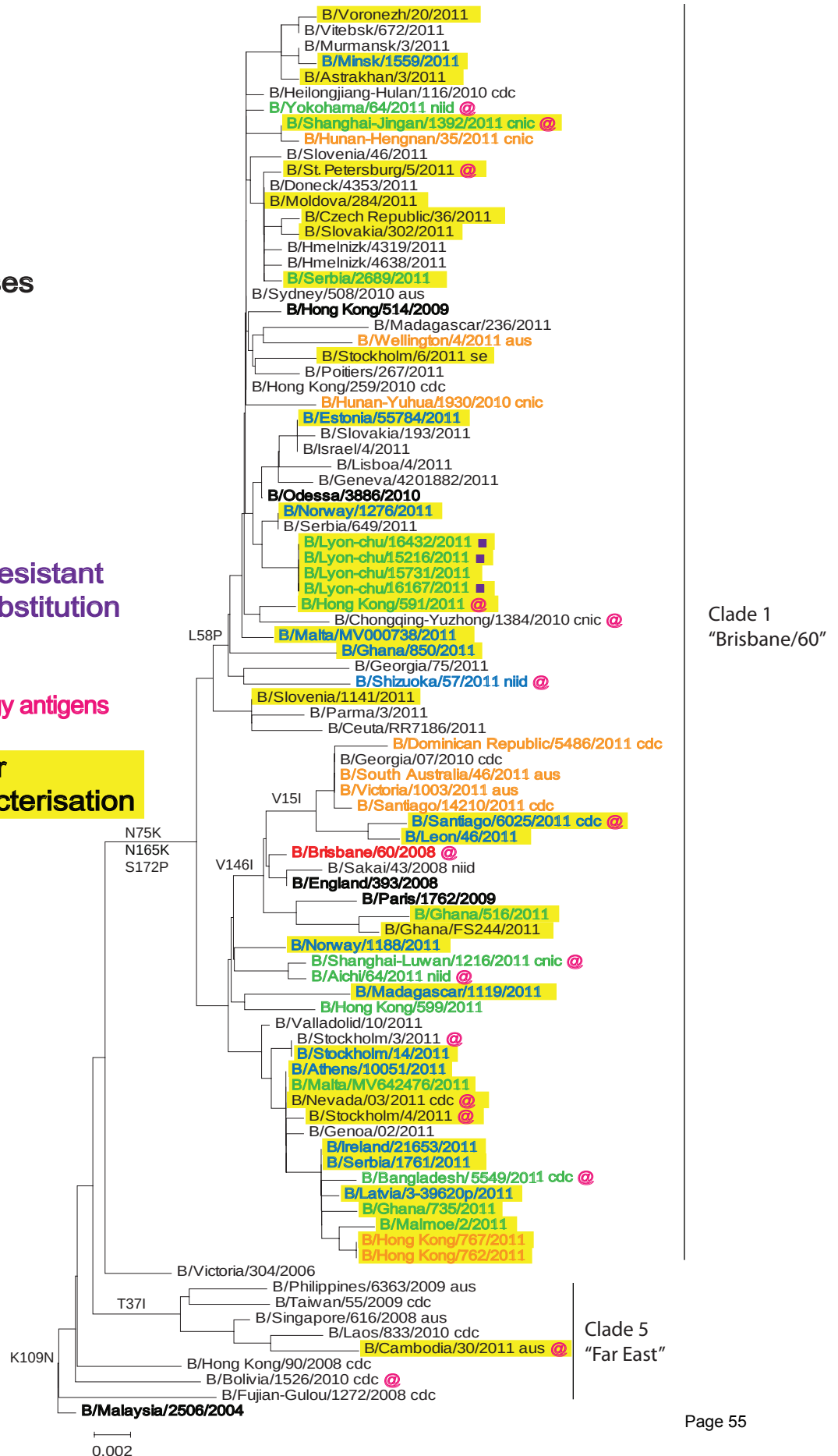
Apr - May 2011

Jun - Jul 2011

■ Oseltamivir resistant
due to I221L substitution
in NA

@ Proposed serology antigens

See HI Table for
antigenic characterisation



Potential N-linked glycosylation sites

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HA protein alignment for B-Victoria lineage viruses.

[illegible]

Figure 16 continued.

HA protein alignment for B-Victoria lineage viruses.

	410	420	430	440	450	460	470	480	490	500
B/Malaysia/2506/2004		SLSELEVKNLQRLSGAMDELHNEI	LELDEKVDLRADTISSQIELAVLLS	NEGIINSEDEHLLALERKLLKMLG	PSAVEIGNGCFETKHKCNQTCL	DRIA				
B/Bolivia/1526/2010 CDC										
B/Cambodia/30/2011 AUS										
B/Victoria/304/2006										
B/Hong Kong/767/2011										
B/Malmoë/2/2011										
B/Bangladesh/5549/2011 CDC										
B/Nevada/03/2011 CDC										
B/Stockholm/3/2011 SE										
B/Aichi/64/2011 NIID										
B/Shanghai~Luwan/1216/2011 CNIC										
B/Ghana/516/2011										
B/Paris/1762/2009			D							
B/England/393/2008										
B/Brisbane/60/2008										
B/Santiago/6025/2011 CDC										
B/Dominican Republic/5486/2011										
B/Shizuoka/57/2011 NIID										
B/Chongqing~Yuzhong/1384/2010										
B/Hong Kong/591/2011										
B/Lyon~chu/16432/2011										
B/Norway/1276/2011										
B/Odessa/3886/2010										
B/Estonia/55784/2011										
B/Hong Kong/514/2009										
B/Serbia/2689/2011										
B/St. Petersburg/5/2011										
B/Shanghai/Jingan/1392/2011 CNIC										
B/Yokohama/64/2011 NIID										
B/Minsk/1559/2011										
	510	520	530	540	550	560	570			
B/Malaysia/2506/2004	AGTTFDAGEFSLPTFFDSLNIT	AAASLNDDGLDNHT	ILLYYSTAASSLAVTLMIAIF	VVYVMSRDNVSCSICL*						
B/Bolivia/1526/2010 CDC										
B/Cambodia/30/2011 AUS										
B/Victoria/304/2006										
B/Hong Kong/767/2011										
B/Malmoë/2/2011										
B/Bangladesh/5549/2011 CDC										
B/Nevada/03/2011 CDC										
B/Stockholm/3/2011 SE										
B/Aichi/64/2011 NIID										
B/Shanghai/Luwan/1216/2011 CNIC										
B/Ghana/516/2011										
B/Paris/1762/2009										
B/England/393/2008										
B/Brisbane/60/2008										
B/Santiago/6025/2011 CDC										
B/Dominican Republic/5486/2011										
B/Shizuoka/57/2011 NIID										
B/Chongqing~Yuzhong/1384/2010										
B/Hong Kong/591/2011										
B/Lyon~chu/16432/2011										
B/Norway/1276/2011										
B/Odessa/3886/2010										
B/Estonia/55784/2011										
B/Hong Kong/514/2009										
B/Serbia/2689/2011										
B/St. Petersburg/5/2011			N							
B/Shanghai~Jingan/1392/2011 CNIC										
B/Yokohama/64/2011 NIID										
B/Minsk/1559/2011										

Figure 17. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

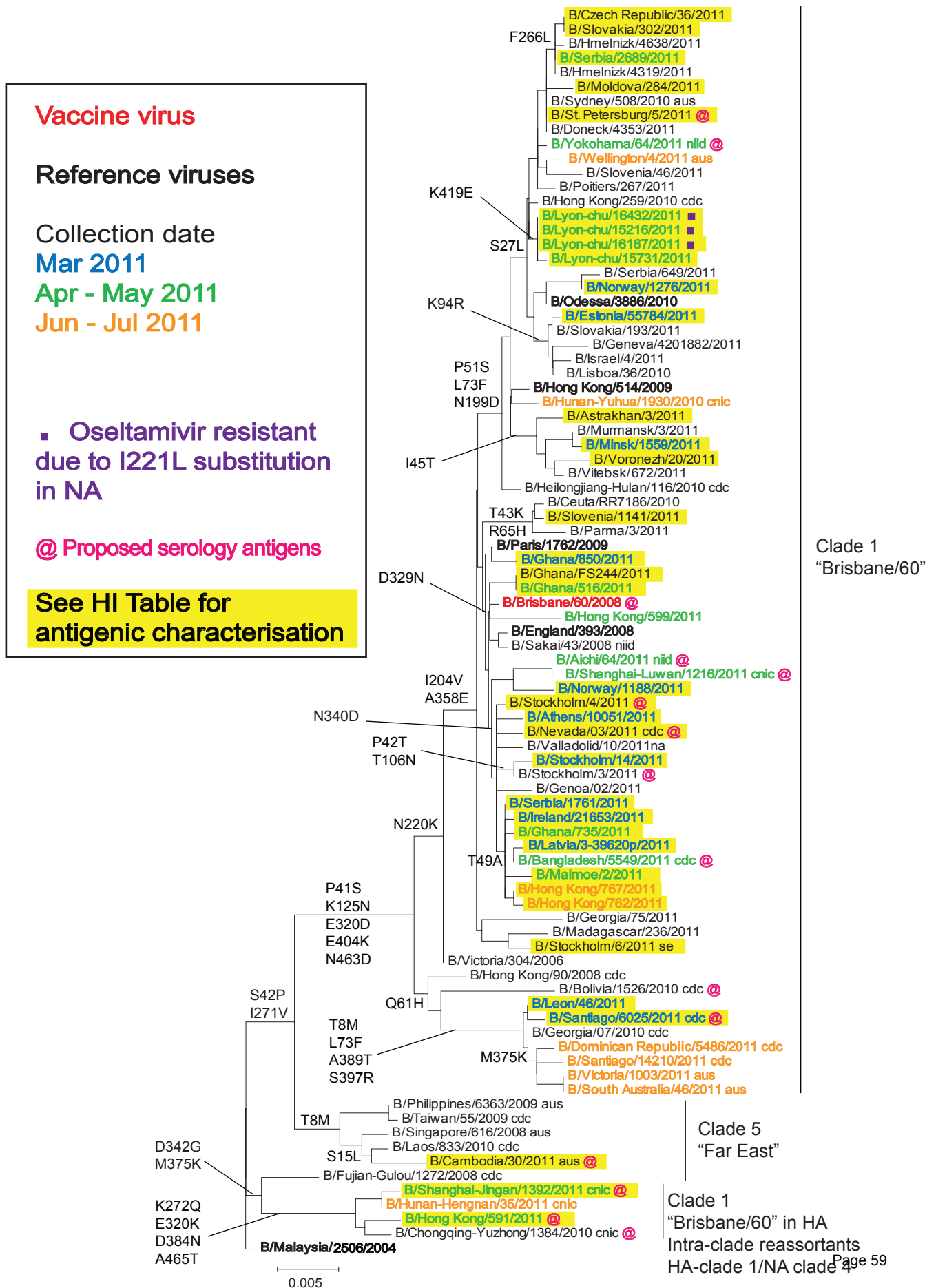


Figure 18.
NA protein alignment for B-Victoria lineage viruses.

Potential N-linked glycosylation sites

	10	20	30	40	50	60	70	80	90	100
B/Malasia/2506/2004	M	L	P	S	T	I	Q	T	L	T
B/Bolivia/1526/2010 CDC	M	L	P	S	T	I	Q	T	L	T
B/Cambodia/30/2011 AUS	M	L	P	S	T	I	Q	T	L	T
B/Victoria/304/2006	M	L	P	S	T	I	Q	T	L	T
B/Hong Kong/767/2011	M	L	P	S	T	I	Q	T	L	T
B/Malmoe/2/2011	M	L	P	S	T	I	Q	T	L	T
B/Bangladesh/5549/2011 CDC	M	L	P	S	T	I	Q	T	L	T
B/Nevada/03/2011 CDC	M	L	P	S	T	I	Q	T	L	T
B/Stockholm/3/2011	M	L	P	S	T	I	Q	T	L	T
B/Aichi/64/2011 NIID	M	L	P	S	T	I	Q	T	L	T
B/Shanghai-Luwan/1216/2011 CNIC	M	L	P	S	T	I	Q	T	L	T
B/Ghana/516/2011	M	L	P	S	T	I	Q	T	L	T
B/Paris/1762/2009	M	L	P	S	T	I	Q	T	L	T
B/England/393/2008	M	L	P	S	T	I	Q	T	L	T
B/Brisbane/60/2008	M	L	P	S	T	I	Q	T	L	T
B/Santiago/6025/2011 CDC	M	L	P	S	T	I	Q	T	L	T
B/Dominican Republic/5486/2011	M	L	P	S	T	I	Q	T	L	T
B/Chongqing-Yuzhong/1384/2010	M	L	P	S	T	I	Q	T	L	T
B/Hong Kong/591/2011	M	L	P	S	T	I	Q	T	L	T
B/Lyon-chu/16432/2011	M	L	P	S	T	I	Q	T	L	T
B/Norway/1276/2011	M	L	P	S	T	I	Q	T	L	T
B/Odessa/3886/2010	M	L	P	S	T	I	Q	T	L	T
B/Estonia/55784/2011	M	L	P	S	T	I	Q	T	L	T
B/Hong Kong/514/2009	M	L	P	S	T	I	Q	T	L	T
B/Serbia/2689/2011	M	L	P	S	T	I	Q	T	L	T
B/St. Petersburg/5/2011	M	L	P	S	T	I	Q	T	L	T
B/Shanghai-Jingan/1392/2011 CNIC	M	L	P	S	T	I	Q	T	L	T
B/Yokohama/64/2011 NIID	M	L	P	S	T	I	Q	T	L	T
B/Minsk/1559/2011	M	L	P	S	T	I	Q	T	L	T
B/Malasia/2506/2004	H	R	F	G	E	T	K	G	N	S
B/Bolivia/1526/2010 CDC	H	R	F	G	E	T	K	G	N	S
B/Cambodia/30/2011 AUS	H	R	F	G	E	T	K	G	N	S
B/Victoria/304/2006	H	R	F	G	E	T	K	G	N	S
B/Hong Kong/767/2011	H	R	F	G	E	T	K	G	N	S
B/Malmoe/2/2011	H	R	F	G	E	T	K	G	N	S
B/Bangladesh/5549/2011 CDC	H	R	F	G	E	T	K	G	N	S
B/Nevada/03/2011 CDC	H	R	F	G	E	T	K	G	N	S
B/Stockholm/3/2011	H	R	F	G	E	T	K	G	N	S
B/Aichi/64/2011 NIID	H	R	F	G	E	T	K	G	N	S
B/Shanghai-Luwan/1216/2011 CNIC	H	R	F	G	E	T	K	G	N	S
B/Ghana/516/2011	H	R	F	G	E	T	K	G	N	S
B/Paris/1762/2009	H	R	F	G	E	T	K	G	N	S
B/England/393/2008	H	R	F	G	E	T	K	G	N	S
B/Brisbane/60/2008	H	R	F	G	E	T	K	G	N	S
B/Santiago/6025/2011 CDC	H	R	F	G	E	T	K	G	N	S
B/Dominican Republic/5486/2011	H	R	F	G	E	T	K	G	N	S
B/Chongqing-Yuzhong/1384/2010	H	R	F	G	E	T	K	G	N	S
B/Hong Kong/591/2011	H	R	F	G	E	T	K	G	N	S
B/Lyon-chu/16432/2011	H	R	F	G	E	T	K	G	N	S
B/Norway/1276/2011	H	R	F	G	E	T	K	G	N	S
B/Odessa/3886/2010	H	R	F	G	E	T	K	G	N	S
B/Estonia/55784/2011	H	R	F	G	E	T	K	G	N	S
B/Hong Kong/514/2009	H	R	F	G	E	T	K	G	N	S
B/Serbia/2689/2011	H	R	F	G	E	T	K	G	N	S
B/St. Petersburg/5/2011	H	R	F	G	E	T	K	G	N	S
B/Shanghai-Jingan/1392/2011 CNIC	H	R	F	G	E	T	K	G	N	S
B/Yokohama/64/2011 NIID	H	R	F	G	E	T	K	G	N	S
B/Minsk/1559/2011	H	R	F	G	E	T	K	G	N	S

Figure 18 continued.

NA protein alignment for B-Victoria lineage viruses.

	210	220	230	240	250	260	270	280	290	300
B/Malasia/2506/2004	LLKIKYGEAYTD	TYHSYANNILRT	QESACNCIGGNCYL	MITDGSASGVSECR	FLKIREGRIIKEIF	PTGRIKHTEECT	CGFAS	NKTIECACRDNS	YTAKR	
B/Bolivia/1526/2010 CDC							V			
B/Cambodia/30/2011 AUS	F						V			
B/Victoria/304/2006		K					V			
B/Hong Kong/767/2011	V	K					V			
B/Malmoe/2/2011	V	K					V			
B/Bangladesh/5549/2011 CDC	V	K					V			
B/Nevada/03/2011 CDC	V	K					V			
B/Stockholm/3/2011	V	K					V			
B/Aichi/64/2011 NIID	V	K					V			
B/Shanghai-Luwan/1216/2011 CNIC	V	K					V			
B/Ghana/516/2011	V	K					V			
B/Paris/1762/2009	V	K					V			
B/England/393/2008	V	K					V			
B/Brisbane/60/2008	V	K					V			
B/Santiago/6025/2011 CDC							V			
B/Dominican Republic/5486/2011							V			
B/Chongqing-Yuzhong/1384/2010							Q			
B/Hong Kong/591/2011							Q			
B/Lyon-chu/16432/2011	V	KX					V			
B/Norway/1276/2011	V	K					V			
B/Odessa/3886/2010	V	K					V			
B/Estonia/55784/2011	V	K					V			
B/Hong Kong/514/2009	V	K					V			
B/Serbia/2689/2011	V	K					L	V		
B/St. Petersburg/5/2011	V	K					V			
B/Shanghai-Jingan/1392/2011 CNIC					M		TQ			
B/Yokohama/64/2011 NIID	V	K					V			
B/Minsk/1559/2011	V	K					V			
	310	320	330	340	350	360	370	380	390	400
B/Malasia/2506/2004	PFVKLVNVEDTAE	IRLMCTETYLDT	PRPDGSGITG	PCESNGDKGSGG	IGKGFVHQ	RMASKIGRWYS	RTMSKTKRMGM	GLYVKYDGP	WADSDALAF	SGVM
B/Bolivia/1526/2010 CDC		D		N			E			
B/Cambodia/30/2011 AUS							E			
B/Victoria/304/2006		D					E			
B/Hong Kong/767/2011		D	N	I	D	E	E			
B/Malmoe/2/2011		D	N		D	E	E			
B/Bangladesh/5549/2011 CDC		D	N		D	E	E			
B/Nevada/03/2011 CDC		D	N		D	E	E			
B/Stockholm/3/2011		D	N		D	E	E			
B/Aichi/64/2011 NIID		D	N		D	E	E		T	
B/Shanghai-Luwan/1216/2011 CNIC		D	N		D	E	E		T	
B/Ghana/516/2011		D	N			E	E		V	
B/Paris/1762/2009		D	N			E	E			
B/England/393/2008		D	N			E	E			
B/Brisbane/60/2008		D	N			E	E			
B/Santiago/6025/2011 CDC		D					E		T	R
B/Dominican Republic/5486/2011				N			E	K	T	R
B/Chongqing-Yuzhong/1384/2010		K		D	G		E	K	N	V
B/Hong Kong/591/2011		K	E		G		E	K	N	V
B/Lyon-chu/16432/2011		D				E	E			
B/Norway/1276/2011		D				E	E			
B/Odessa/3886/2010		D				E	E			
B/Estonia/55784/2011		D				E	E			
B/Hong Kong/514/2009		D				E	E			
B/Serbia/2689/2011		D				E	E			
B/St. Petersburg/5/2011		D				E	E			
B/Shanghai-Jingan/1392/2011 CNIC		K			G		E	K	N	
B/Yokohama/64/2011 NIID		D				E	E			
B/Minsk/1559/2011		D				E	E			

Figure 18 continued.

NA protein alignment for B-Victoria lineage viruses.

	410	420	430	440	450	460
B/Malasia/2506/2004	VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKETWHS	AATAIYCLMGSGQLLWDTVTG	VN	MAL	*	
B/Bolivia/1526/2010 CDC	..K.	..	..	..	..	D..*
B/Cambodia/30/2011 AUS	..K.	..	..	..	..	D..*
B/Victoria/304/2006	..K.	..	..	..	..	D..*
B/Hong Kong/767/2011	..K.	..	..	..	..	D..*
B/Malmoe/2/2011	..K.	..	..	..	..	D..*
B/Bangladesh/5549/2011 CDC	..K.	..	..	..	..	D..*
B/Nevada/03/2011 CDC	..K.	..	..	..	..	D..*
B/Stockholm/3/2011	..K.	..	..	..	..	D..*
B/Aichi/64/2011 NIID	..K.	..	..	..	..	D..*
B/Shanghai-Luwan/1216/2011 CNIC	..K.	..	..	..	..	D..*
B/Ghana/516/2011	..K.	..	..	..	..	D..*
B/Paris/1762/2009	..K.	..	..	..	..	D..*
B/England/393/2008	..K.	..	..	..	..	D..*
B/Brisbane/60/2008	..K.	..	..	..	..	D..*
B/Santiago/6025/2011 CDC	..K.	..	..	..	..	D..*
B/Dominican Republic/5486/2011	..K.	..	..	..	..	D..*
B/Chongqing-Yuzhong/1384/2010	..	..	..	..	..	.T.*
B/Hong Kong/591/2011	..	..	..	..	..	.T.*
B/Lyon-chu/16432/2011	..K.	..	E.	..	..	D..*
B/Norway/1276/2011	..K.	..	..	..	..	D..*
B/Odessa/3886/2010	..K.	..	..	..	..	D..*
B/Estonia/55784/2011	..K.	..	..	..	..	D..*
B/Hong Kong/514/2009	..K.	..	..	..	..	D..*
B/Serbia/2689/2011	..K.	..	..	..	..	D..*
B/St. Petersburg/5/2011	..K.	..	..	..	..	D..*
B/Shanghai-Jingan/1392/2011 CNIC	..	..	..	..	..	.T.*
B/Yokohama/64/2011 NIID	..K.	..	..	..	..	D..*
B/Minsk/1559/2011	..K.	..	..	I.	..	D..*

Table 28. Summary of influenza B viruses (Yamagata-lineage) received, collected between 2011-02-01 and present

MONTH Country	B Yamagata lineage			
	Number propagated	≤2 fold*	4 fold*	≥8 fold*
FEBRUARY				
Estonia	1	1		
Latvia	1	1		
Sweden	1	1		
Turkey	2	1	1	
MARCH				
Estonia	5	4	1	
Greece	1	1		
Norway	0			
Serbia	3	3		
Sweden	2	2		
JUNE				
Hong Kong	2	1	1	
Total	18	15 83%	3 17%	0

* Reduction in HI titre with ferret antiserum raised against B/Bangladesh/3333/2007 (egg grown virus)

Table 29. Antigenic analyses of influenza B viruses (Yamagata lineage)

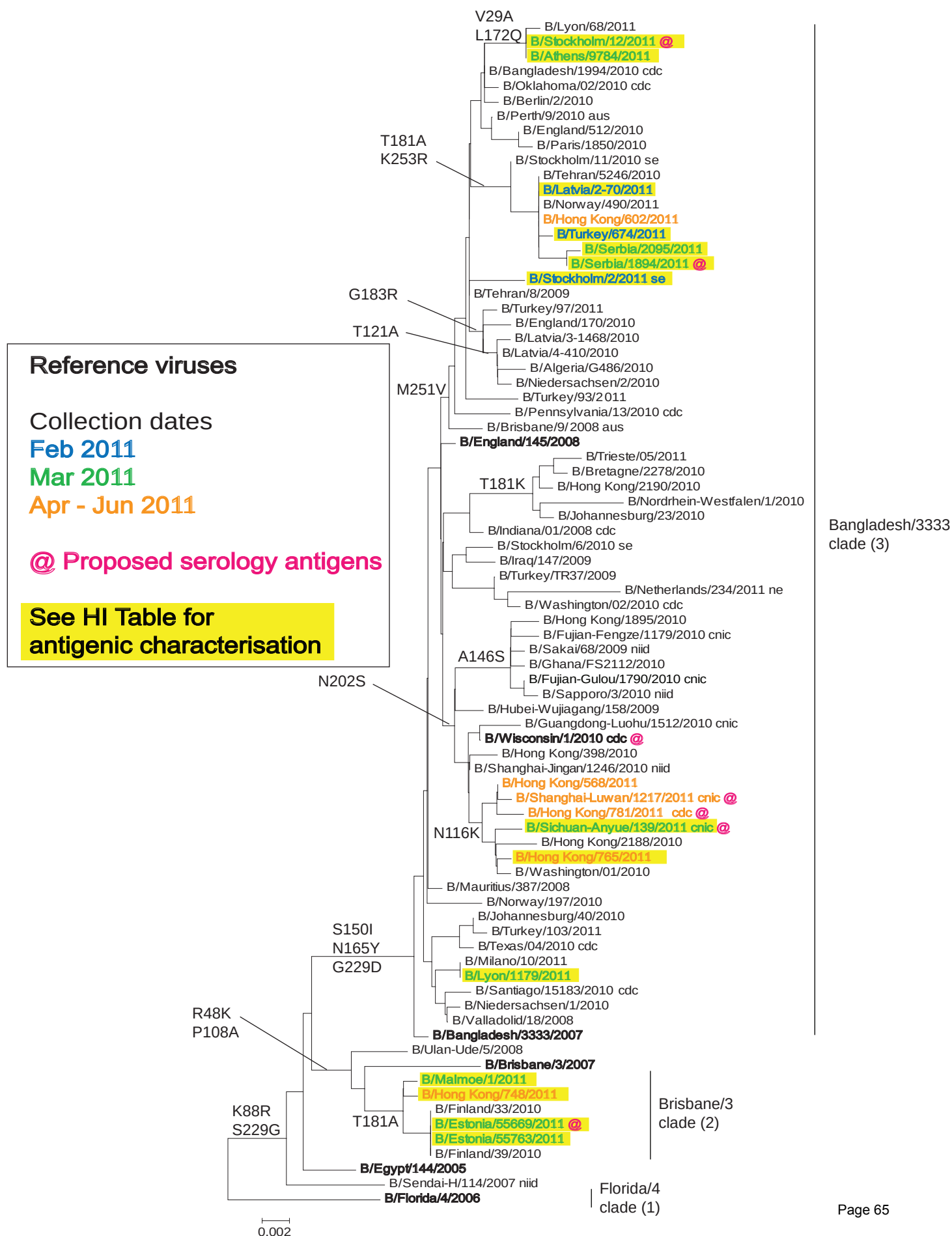
Viruses	Collection date	Passage History	Haemagglutination inhibition titre						
			B/FI ³ 4/06 SH479	Post infection ferret sera					
				B/Eg ¹ 144/05 F7/05	B/FI ¹ 4/06 F20/07	B/Bris ² 3/07 F24/07	B/Eng ² 145/08 F12/11	B/Bang ² 3333/07 F25/08	B/Wis ² 1/10 F26/10
REFERENCE VIRUSES									
B/Egypt/144/2005	2005-05-01	E3/E6	2560	640	1280	640	160	160	160
B/Florida/4/2006	2006-12-15	E3/E4	5120	640	1280	640	320	160	320
B/Brisbane/3/2007	2007-09-03	E2/E4	2560	640	1280	1280	160	320	320
B/England/145/2008		Ex/E5	640	80	160	40	160	40	20
B/Bangladesh/3333/2007	2007-08-07	E3/E4	2560	320	1280	160	160	320	320
B/Wisconsin/1/2010	2010-02-20	E3/E2	1280	160	640	160	80	160	320
TEST VIRUSES									
B/Latvia/2-70/2011	2011-02-01	MDCK1/MDCK1	5120	160	320	<	80	160	160
B/Stockholm/2/2011	2011-02-09	MDCK1/MDCK1	5120	1280	1280	640	320	640	320
B/Turkey/682/2011	2011-02-17	C2/MDCK1	5120	1280	640	<	1280	320	320
B/Turkey/674/2011	2011-02-17	C2/MDCK1	1280	320	160	<	640	80	160
B/Estonia/54563/2011	2011-02-17	MDCK1/MDCK1	2560	80	80	<	<	160	80
B/Lyon/1179/2011	2011-03-01	MDCK2/MDCK1	2560	160	320	40	160	80	80
B/Serbia/1894/2011	2011-03-08	C1/MDCK1	5120	640	1280	320	640	320	320
B/Athens/9784/2011	2011-03-13	MDCK2	5120	640	1280	640	320	320	160
B/Serbia/2017/2011	2011-03-14	C1/MDCK1	5120	1280	1280	640	640	640	320
B/Estonia/55669/2011	2011-03-14	MDCK1/MDCK1	2560	40	80	<	<	80	80
B/Estonia/55760/2011	2011-03-15	MDCK1/MDCK1	2560	160	160	<	80	320	160
B/Estonia/55758/2011	2011-03-15	MDCK1/MDCK1	2560	80	160	<	80	320	160
B/Estonia/55763/2011	2011-03-15	MDCK1/MDCK1	2560	160	160	40	80	640	320
B/Serbia/2095/2011	2011-03-15	C1/MDCK1	5120	640	640	160	320	320	160
B/Estonia/55786/2011	2011-03-16	MDCK1/MDCK1	2560	80	80	<	<	160	160
B/Malmoe/1/2011	2011-03-25	MDCK2/MDCK3	5120	640	640	320	160	160	160
B/Stockholm/12/2011	2011-03-28	MDCK0/MDCK1	5120	1280	1280	320	320	320	160
B/Sichuan-Anyue/139/2011	2011-04-26	E3	1280	160	640	320	80	320	640
B/Hong Kong/748/2011	2011-06-17	MDCK2/MDCK1	2560	320	320	160	80	80	80
B/Hong Kong/765/2011	2011-06-23	MDCK2/MDCK1	5120	640	640	320	160	160	160

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Sequences included in the HA phylogeny

Figure 19.

Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes (HA1 region)



Potential N-linked glycosylation sites

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Figure 20 continued.
HA protein alignment for B-Yamagata lineage viruses.

	210	220	230	240	250	260	270	280	290	300
B/Florida/4/2006 CDC	KNLYGDSNPQKFTSSANGVTTHYVSQIGSFDPQTEDGGLPQSGRIVVDYMMQKPGKTGTIVYQRGVLLPQKVWCASGRSKVIKGS	LP	LI	GE	AD	CL	HE	KY	G	
B/Egypt/144/2005			G							
B/Estonia/55669/2011			G							
B/Hong Kong/748/2011			G							
B/Brisbane/3/2007 AUS			G							
B/Bangladesh/3333/2007			D							
B/Lyon/1179/2011		A	D							
B/Johannesburg/40/2010			D							
B/Hong Kong/765/2011	S		D				I			
B/Sichuan-Anyue/139/2011 CNIC	S		D							
B/Hong Kong/781/2011	S		D							
B/Shanghai-Luwan/1217/2011 CNIC	S		D							
B/Wisconsin/1/2010 CDC	S		D							
B/Sapporo/3/2010 NIID	S		D							
B/Ghana/FS2112/2010	S		D							
B/Netherlands/234/2011 NE		A	D							
B/Stockholm/6/2010 SE			D							
B/Johannesburg/23/2010			D							
B/Trieste/05/2011			D							
B/England/145/2008			D							
B/Pennsylvania/13/2010 CDC			D			V				
B/Turkey/93/2011			D			V				
B/Niedersachsen/2/2010			D			V				
B/Latvia/31468/2010			D			V				
B/Serbia/1894/2011			D			V	R			
B/Hong Kong/602/2011			D			V	R			
B/Paris/1850/2010			D			V				
B/Perth/9/2010 AUS			D			V				
B/Athens/9784/2011			D			V				
B/Stockholm/12/2011			D			V				

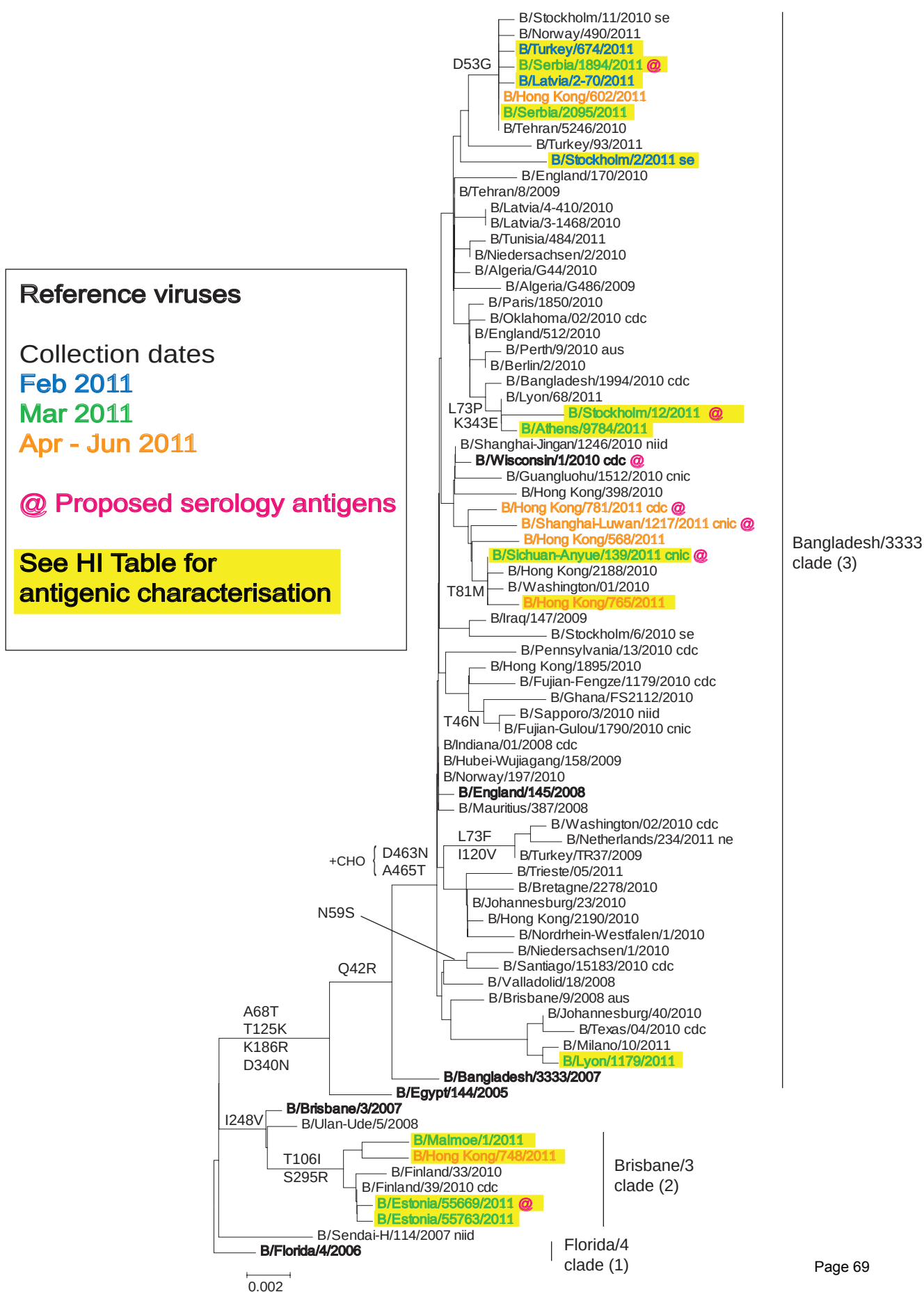
	310	320	330	340	350	360	370	380	390	400
B/Florida/4/2006 CDC	GLNKS	KPY	YTGE	HAKA	IGNC	PIW	KT	PLK	L	ANGT
B/Egypt/144/2005										X
B/Estonia/55669/2011										
B/Hong Kong/748/2011										
B/Brisbane/3/2007 AUS										
B/Bangladesh/3333/2007										
B/Lyon/1179/2011										
B/Johannesburg/40/2010										
B/Hong Kong/765/2011										
B/Sichuan-Anyue/139/2011 CNIC										
B/Hong Kong/781/2011										
B/Shanghai-Luwan/1217/2011 CNIC										
B/Wisconsin/1/2010 CDC										
B/Sapporo/3/2010 NIID										
B/Ghana/FS2112/2010										
B/Netherlands/234/2011 NE										
B/Stockholm/6/2010 SE										
B/Johannesburg/23/2010										
B/Trieste/05/2011										
B/England/145/2008										
B/Pennsylvania/13/2010 CDC										X
B/Turkey/93/2011										
B/Niedersachsen/2/2010										
B/Latvia/31468/2010										
B/Serbia/1894/2011										
B/Hong Kong/602/2011										
B/Paris/1850/2010										
B/Perth/9/2010 AUS										
B/Athens/9784/2011										X
B/Stockholm/12/2011										

Figure 20 continued.
HA protein alignment for B-Yamagata lineage viruses.

	410	420	430	440	450	460	470	480	490	500
B/Florida/4/2006 CDC	L	S	E	L	V	K	N	L	Q	R
B/Egypt/144/2005	L	S	E	L	V	K	N	L	Q	R
B/Estonia/55669/2011	L	S	E	L	V	K	N	L	Q	R
B/Hong Kong/748/2011	L	S	E	L	V	K	N	L	Q	R
B/Brisbane/3/2007 AUS	L	S	E	L	V	K	N	L	Q	R
B/Bangladesh/3333/2007	L	S	E	L	V	K	N	L	Q	R
B/Lyon/1179/2011	L	S	E	L	V	K	N	L	Q	R
B/Johannesburg/40/2010	L	S	E	L	V	K	N	L	Q	R
B/Hong Kong/765/2011	L	S	E	L	V	K	N	L	Q	R
B/Sichuan-Anyue/139/2011 CNIC	L	S	E	L	V	K	N	L	Q	R
B/Hong Kong/781/2011	L	S	E	L	V	K	N	L	Q	R
B/Shanghai-Luwan/1217/2011 CNIC	L	S	E	L	V	K	N	L	Q	R
B/Wisconsin/1/2010 CDC	L	S	E	L	V	K	N	L	Q	R
B/Sapporo/3/2010 NIID	L	S	E	L	V	K	N	L	Q	R
B/Ghana/FS2112/2010	L	S	E	L	V	K	N	L	Q	R
B/Netherlands/234/2011 NE	L	S	E	L	V	K	N	L	Q	R
B/Stockholm/6/2010 SE	L	S	E	L	V	K	N	L	Q	R
B/Johannesburg/23/2010	L	S	E	L	V	K	N	L	Q	R
B/Trieste/05/2011	L	S	E	L	V	K	N	L	Q	R
B/England/145/2008	L	S	E	L	V	K	N	L	Q	R
B/Pennsylvania/13/2010 CDC	L	S	E	L	V	K	N	L	Q	R
B/Turkey/93/2011	L	S	E	L	V	K	N	L	Q	R
B/Niedersachsen/2/2010	L	S	E	L	V	K	N	L	Q	R
B/Latvia/31468/2010	L	S	E	L	V	K	N	L	Q	R
B/Serbia/1894/2011	L	S	E	L	V	K	N	L	Q	R
B/Hong Kong/602/2011	L	S	E	L	V	K	N	L	Q	R
B/Paris/1850/2010	L	S	E	L	V	K	N	L	Q	R
B/Perth/9/2010 AUS	L	S	E	L	V	K	N	L	Q	R
B/Athens/9784/2011	L	S	E	L	V	K	N	L	Q	R
B/Stockholm/12/2011	L	S	E	L	V	K	N	L	Q	R

	510	520	530	540	550	560	570
B/Florida/4/2006 CDC	G	T	F	N	A	G	E
B/Egypt/144/2005	G	T	F	N	A	G	E
B/Estonia/55669/2011	G	T	F	N	A	G	E
B/Hong Kong/748/2011	G	T	F	N	A	G	E
B/Brisbane/3/2007 AUS	G	T	F	N	A	G	E
B/Bangladesh/3333/2007	G	T	F	N	A	G	E
B/Lyon/1179/2011	G	T	F	N	A	G	E
B/Johannesburg/40/2010	G	T	F	N	A	G	E
B/Hong Kong/765/2011	G	T	F	N	A	G	E
B/Sichuan-Anyue/139/2011 CNIC	G	T	F	N	A	G	E
B/Hong Kong/781/2011	G	T	F	N	A	G	E
B/Shanghai-Luwan/1217/2011 CNIC	G	T	F	N	A	G	E
B/Wisconsin/1/2010 CDC	G	T	F	N	A	G	E
B/Sapporo/3/2010 NIID	G	T	F	N	A	G	E
B/Ghana/FS2112/2010	G	T	F	N	A	G	E
B/Netherlands/234/2011 NE	G	T	F	N	A	G	E
B/Stockholm/6/2010 SE	G	T	F	N	A	G	E
B/Johannesburg/23/2010	G	T	F	N	A	G	E
B/Trieste/05/2011	G	T	F	N	A	G	E
B/England/145/2008	G	T	F	N	A	G	E
B/Pennsylvania/13/2010 CDC	G	T	F	N	A	G	E
B/Turkey/93/2011	G	T	F	N	A	G	E
B/Niedersachsen/2/2010	G	T	F	N	A	G	E
B/Latvia/31468/2010	G	T	F	N	A	G	E
B/Serbia/1894/2011	G	T	F	N	A	G	E
B/Hong Kong/602/2011	G	T	F	N	A	G	E
B/Paris/1850/2010	G	T	F	N	A	G	E
B/Perth/9/2010 AUS	G	T	F	N	A	G	E
B/Athens/9784/2011	G	T	F	N	A	G	E
B/Stockholm/12/2011	G	T	F	N	A	G	E

Figure 21. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes



Potential N-linked glycosylation sites

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Figure 22 continued.
NA protein alignment for B-Yamagata lineage viruses.

	210	220	230	240	250	260	270	280	290	300
B/Florida/4/2006	LLKIKYGEAYTD	TYHSYAKN	ILRTQESACN	IGGDCYLMI	TGDPASGISE	CRFLKIRE	GRIIKEIF	PTGRVKHTE	ECTCGFAS	NKTIECACRDNSYTAKR
B/Egypt/144/2005										
B/Estonia/55669/2011					V					R
B/Hong Kong/748/2011					V					R
B/Brisbane/3/2007					V					
B/Bangladesh/3333/2007										
B/Lyon/1179/2011										
B/Johannesburg/40/2010										
B/Hong Kong/765/2011										
B/Sichuan-Anyue/139/2011 CNIC										
B/Hong Kong/781/2011 CDC										
B/Shanghai-Luwan/1217/2011 CNIC										
B/Wisconsin/1/2010 CDC										
B/Sapporo/3/2010 NIID										
B/Ghana/FS2112/2010										
B/Netherlands/234/2011 NE										
B/Stockholm/6/2010 SE										
B/Johannesburg/23/2010										
B/Trieste/05/2011										
B/England/145/2008										
B/Pennsylvania/13/2010 CDC										
B/Turkey/93/2011										
B/Niedersachsen/2/2010										
B/Latvia/31468/2010										
B/Serbia/1894/2011										
B/Hong Kong/602/2011										
B/Paris/1850/2010										
B/Perth/9/2010 AUS										
B/Athens/9784/2011										
B/Stockholm/12/2011										
	310	320	330	340	350	360	370	380	390	400
B/Florida/4/2006	PFVKLN	VETDTAE	IRLMCTET	YLTTPFP	NDGSITG	PCESDGD	GKSGG	IKGGFV	HQRMASK	IGRWYSRTMSKTKRMGMGLYVKYDGD
B/Egypt/144/2005										PWTDSEALALSGVM
B/Estonia/55669/2011										
B/Hong Kong/748/2011										
B/Brisbane/3/2007										
B/Bangladesh/3333/2007										
B/Lyon/1179/2011										
B/Johannesburg/40/2010										
B/Hong Kong/765/2011										
B/Sichuan-Anyue/139/2011 CNIC										
B/Hong Kong/781/2011 CDC										
B/Shanghai-Luwan/1217/2011 CNIC										
B/Wisconsin/1/2010 CDC										
B/Sapporo/3/2010 NIID										
B/Ghana/FS2112/2010										
B/Netherlands/234/2011 NE										
B/Stockholm/6/2010 SE										
B/Johannesburg/23/2010										
B/Trieste/05/2011										
B/England/145/2008										
B/Pennsylvania/13/2010 CDC										
B/Turkey/93/2011										
B/Niedersachsen/2/2010										
B/Latvia/31468/2010										
B/Serbia/1894/2011										
B/Hong Kong/602/2011										
B/Paris/1850/2010										
B/Perth/9/2010 AUS										
B/Athens/9784/2011										
B/Stockholm/12/2011										

Figure 22 continued.

NA protein alignment for B-Yamagata lineage viruses.

	410	420	430	440	450	460
B/Florida/4/2006	VSME	EPGW	YSFG	FEIK	DKKCD	VPCIGIEMVHDGGKTTWHS
B/Egypt/144/2005						
B/Estonia/55669/2011						
B/Hong Kong/748/2011						
B/Brisbane/3/2007						
B/Bangladesh/3333/2007						
B/Lyon/1179/2011						N.T.*
B/Johannesburg/40/2010						N.T.*
B/Hong Kong/765/2011						N.T.*
B/Sichuan-Anyue/139/2011 CNIC						N.T.*
B/Hong Kong/781/2011 CDC						N.T.*
B/Shanghai-Luwan/1217/2011 CNIC				K.....		N.T.*
B/Wisconsin/1/2010 CDC						N.T.*
B/Sapporo/3/2010 NIID						N.T.*
B/Ghana/FS2112/2010						N.T.*
B/Netherlands/234/2011 NE						N.T.*
B/Stockholm/6/2010 SE						N.T.*
B/Johannesburg/23/2010						N.T.*
B/Trieste/05/2011						N.T.*
B/England/145/2008						N.T.*
B/Pennsylvania/13/2010 CDC						N.T.*
B/Turkey/93/2011		P.....			V.....	N.T.*
B/Niedersachsen/2/2010						N.T.*
B/Latvia/31468/2010						N.T.*
B/Serbia/1894/2011						N.T.*
B/Hong Kong/602/2011						N.T.*
B/Paris/1850/2010						N.T.*
B/Perth/9/2010 AUS						N.T.*
B/Athens/9784/2011						N.T.*
B/Stockholm/12/2011						N.T.*

Table 30. Antiviral testing of influenza viruses collected 2011-02-01 until present*

Influenza Type	Number tested	Number resistant to oseltamivir	Number resistant to zanamivir
H1N1pdm	97	0	0
H3N2	39	0	0
B ¹	110	3	3
Total	246	3	3

***Total number of viruses tested since 2011-02-01**

Influenza Type	Number tested	Number resistant to oseltamivir	Number resistant to zanamivir
H1N1pdm	598	2	0
H3N2	75	0	0
B ¹	373	3	3
Total	1046	5	3

1. The 3 resistant B viruses were recovered from the same French patient at different times

Table 31. Genes sequenced between 2011-02-01 and present

	Gene		
	HA	NA	M
A(H1N1)pdm	208	201	18
A(H3N2)	141	142	11
B	169	159	0
Total = 1049	518	502	29